

Control of Gene Expression

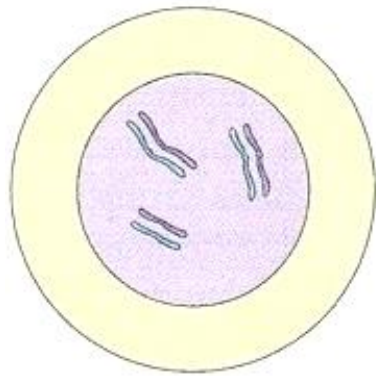
(Albert's textbook Chapter 6&7)

2026/04/23

Topics

- The structure and function of RNA
- RNA transcription
- RNA processing
- Transcription regulators
- Cell diversity and cell memory
- Post-transcriptional regulations
- Non-coding RNAs

One Genome, Many Cell Types

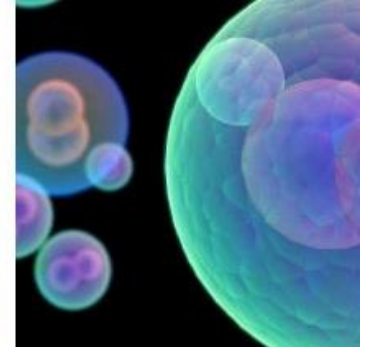
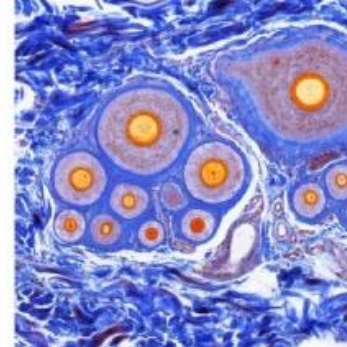
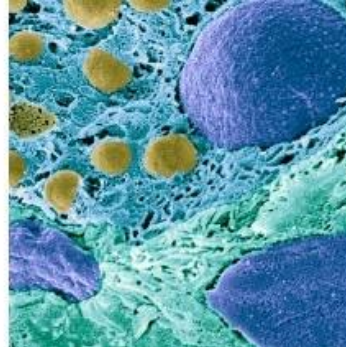
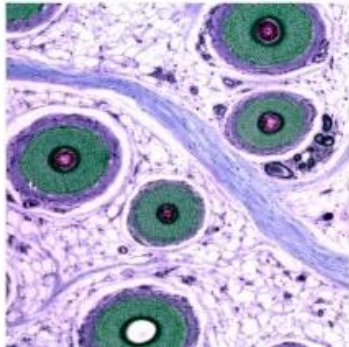


One Genome



Spatiotemporal
Gene Expression
and Regulation

Diverse Cell Types



FROM DNA TO PROTEIN: GENE EXPRESSION

RNA Molecule

Transcription: DNA to RNA

Translation: RNA to Protein

How Cells Read the Genome: From DNA to Protein

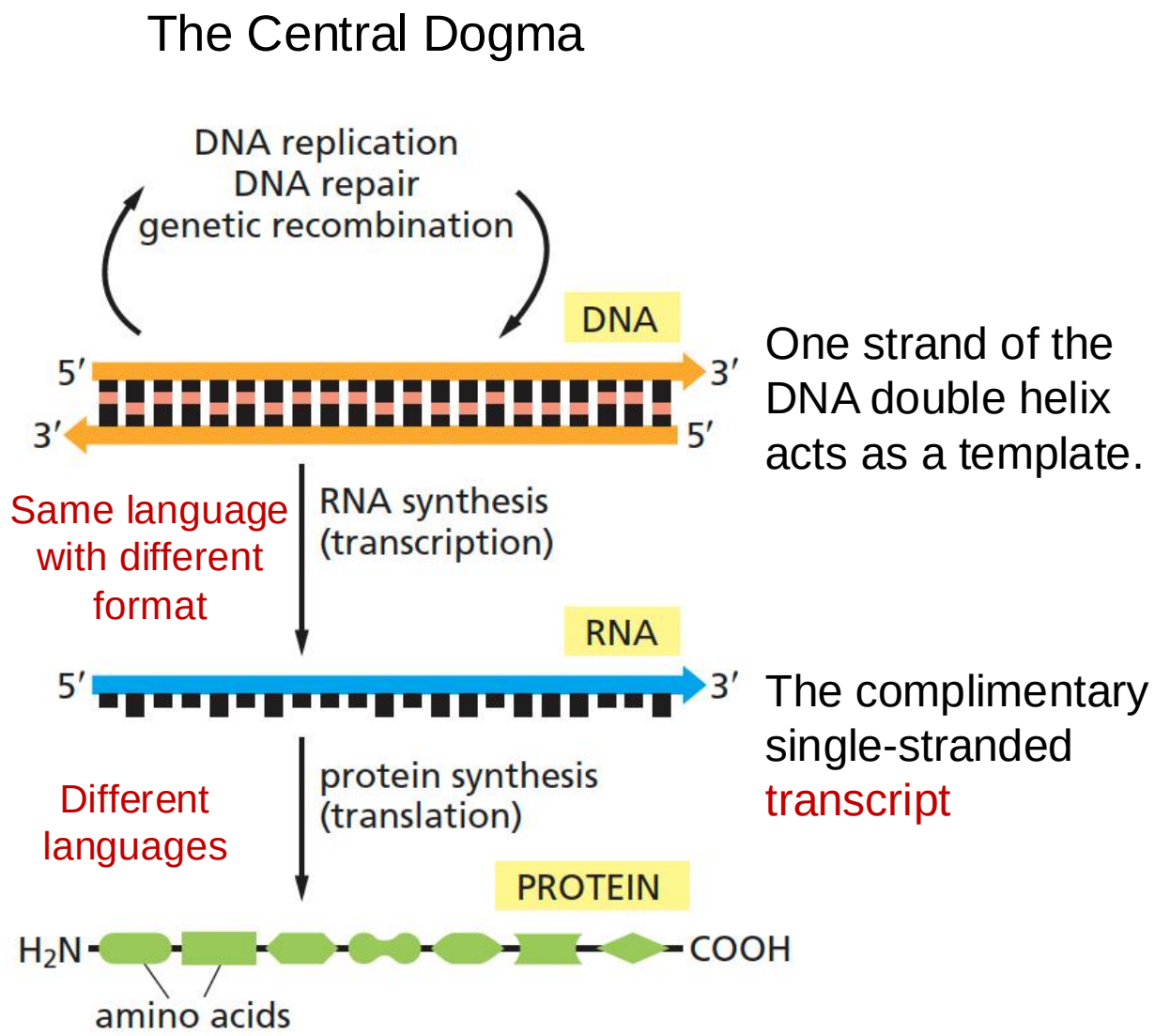


Figure 6-1, 6-3 *Molecular Biology of the Cell* (© Garland Science 2015)

RNA Molecules Differ from DNA Chemically

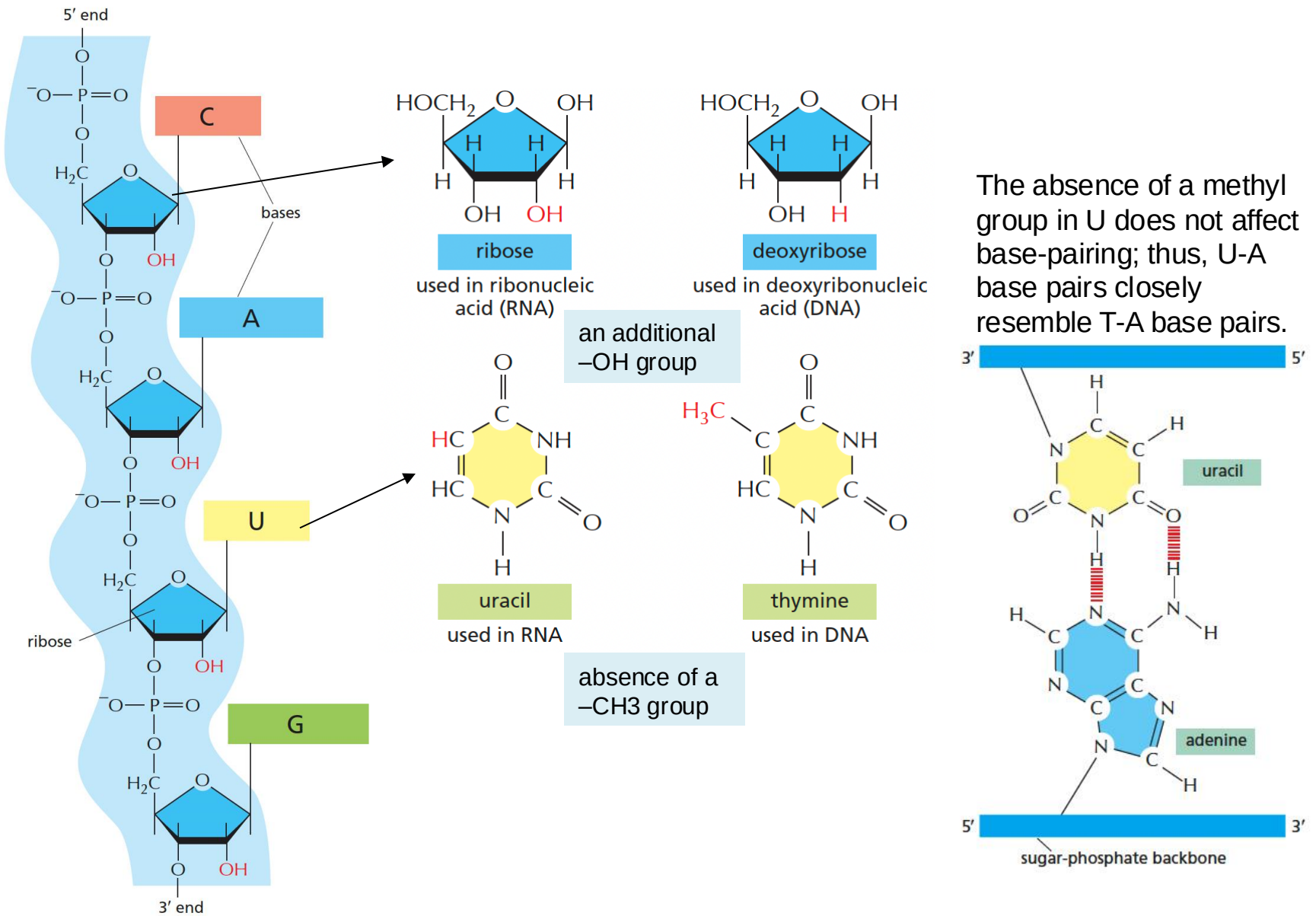
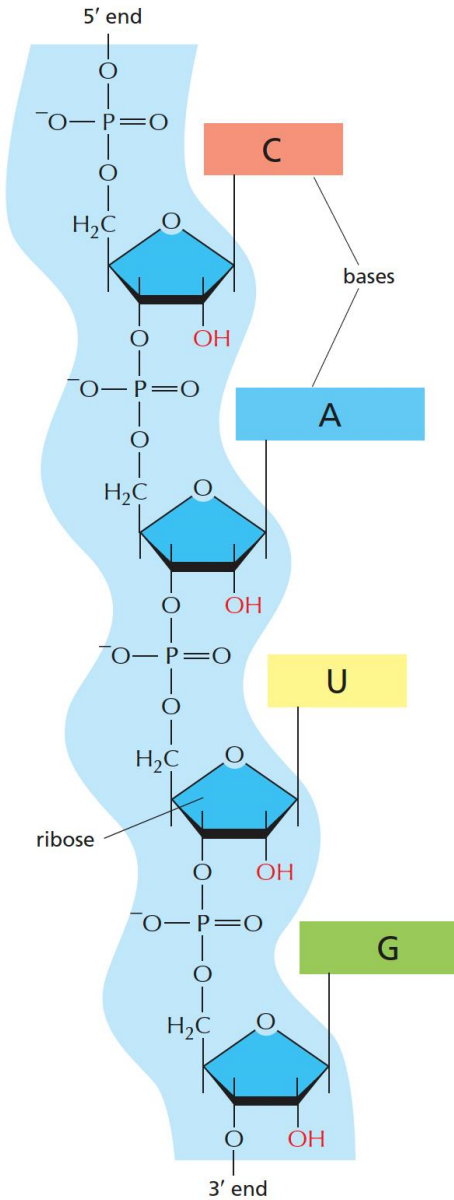
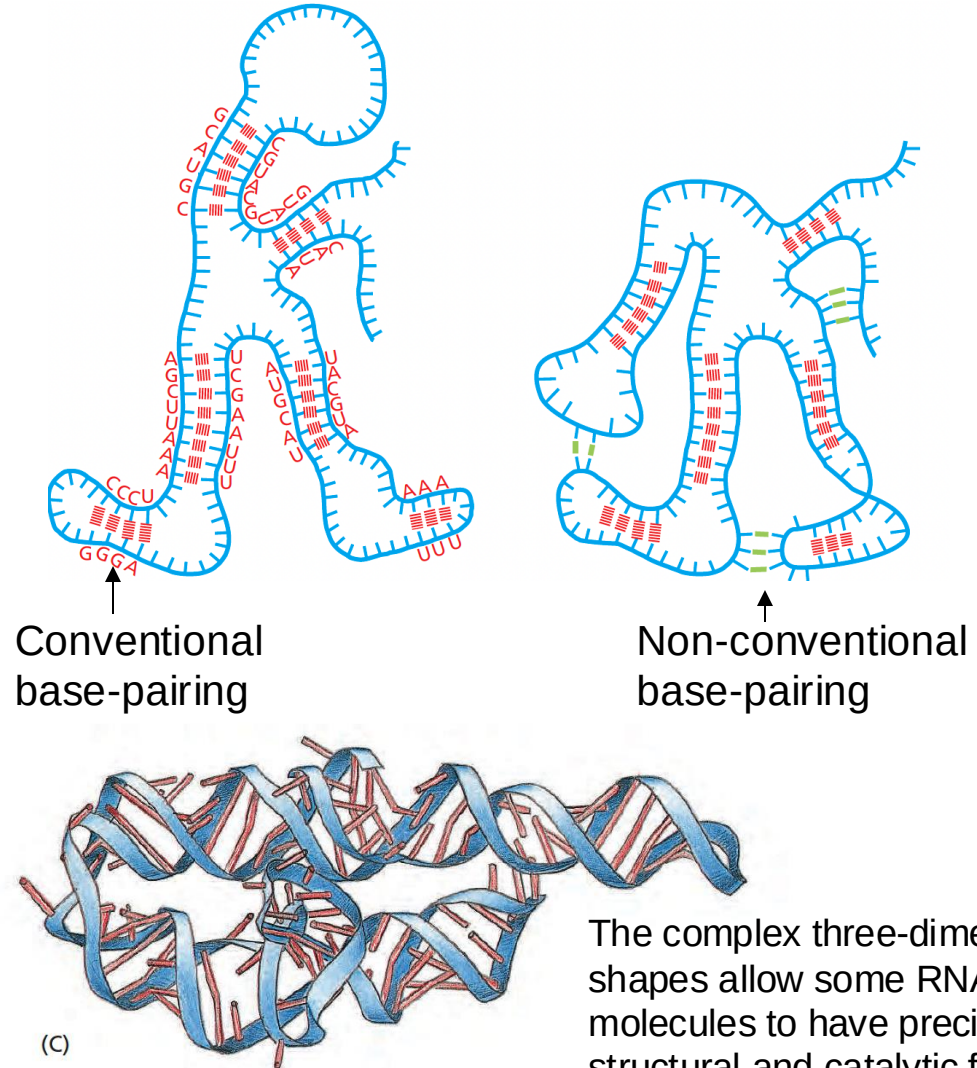


Figure 6-4, 6-5, 6-6 *Molecular Biology of the Cell* (© Garland Science 2015)

RNA Is Single-stranded and Can Fold Into Specific Structures

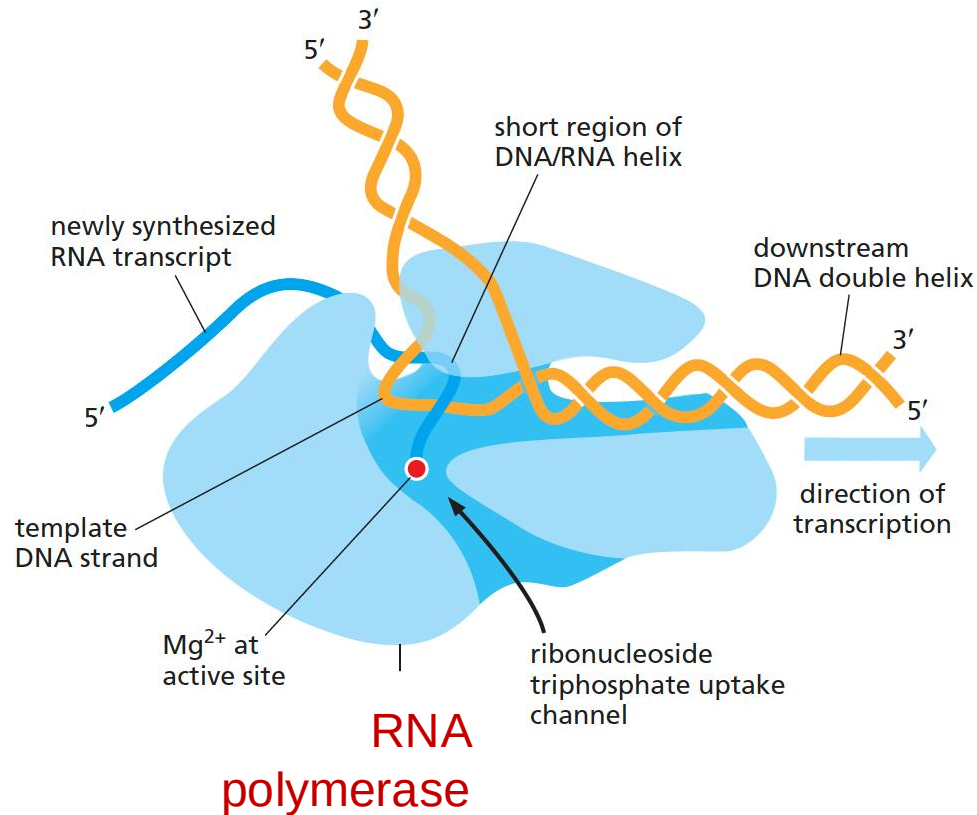


Slight chemical differences but
dramatical structure differences.



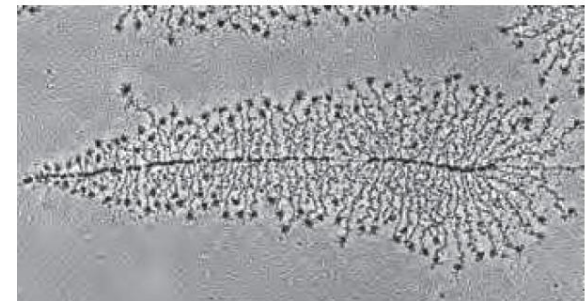
The complex three-dimensional
shapes allow some RNA
molecules to have precise
structural and catalytic functions.

The Process of RNA Transcription

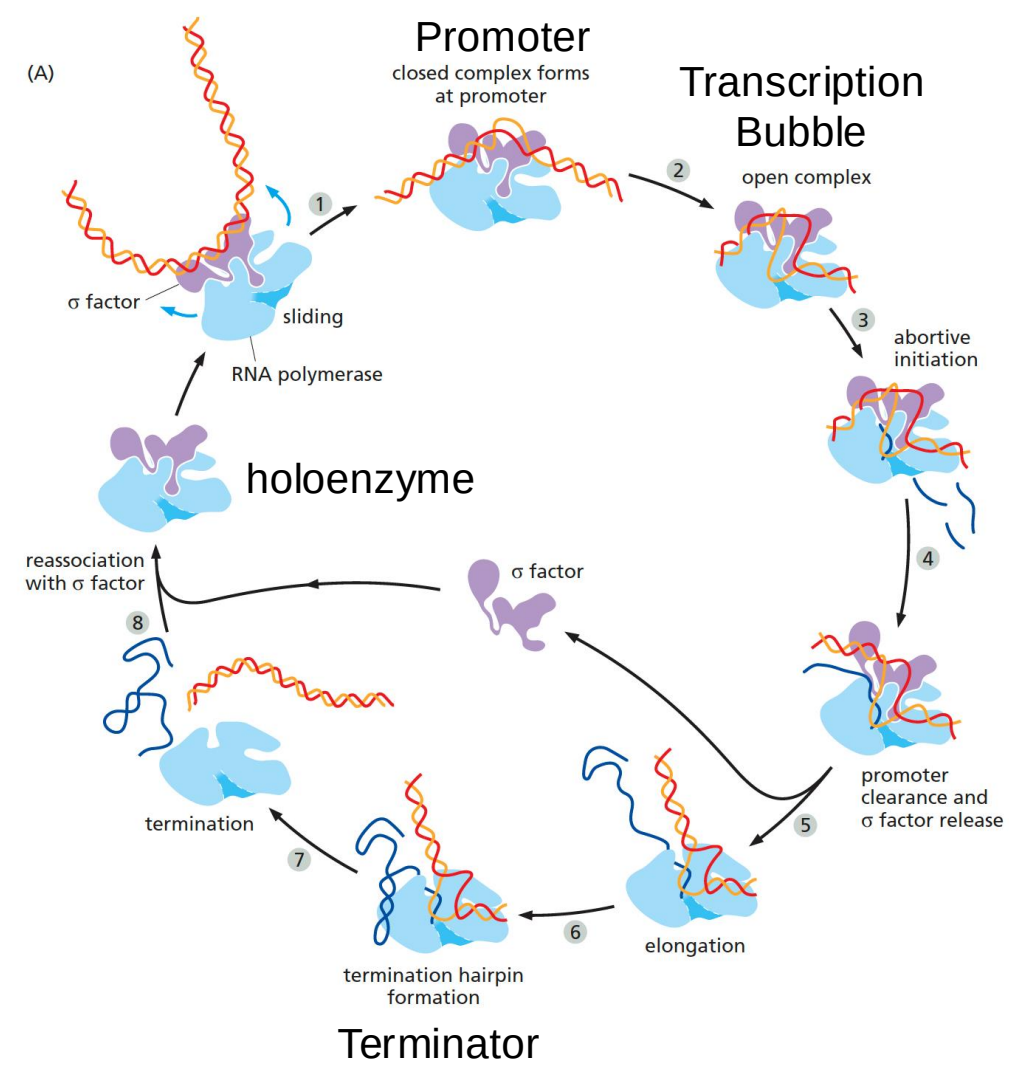


- Released immediately from the DNA template.
- Only copied from a region of the DNA therefore much shorter than DNA.
- Many molecules of RNA are simultaneously transcribed from a gene.

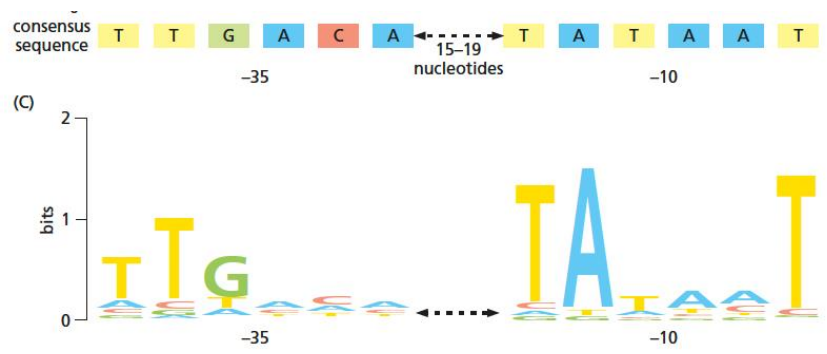
RNA Polymerases Carry Out Transcription



The Transcription Cycle of Bacterial RNA Polymerase



Consensus nucleotide sequence and sequence logo for the major class of *E. coli* promoters.



Promoter sequences are asymmetric and the polymerase can synthesize RNA only in the 5'-to-3' direction, so the promoter orientation specifies the strand to be used as a template.

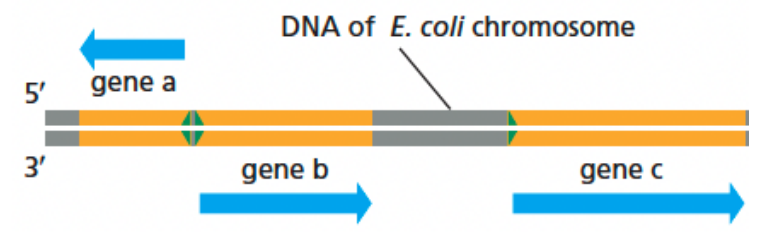
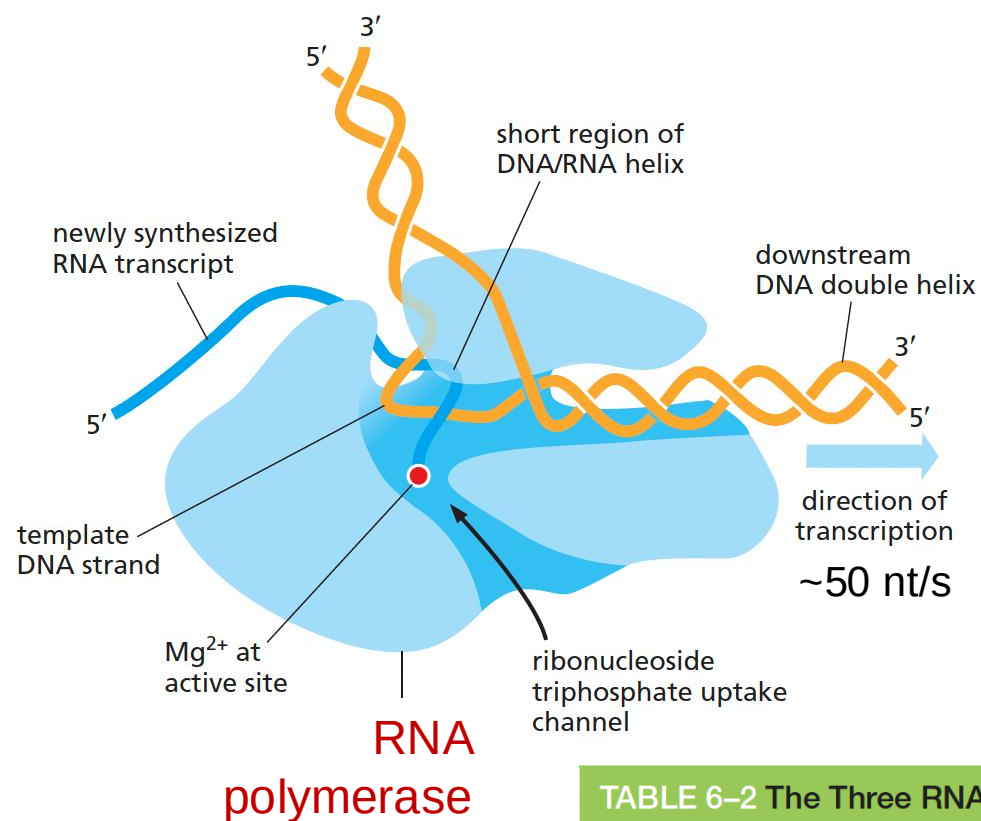
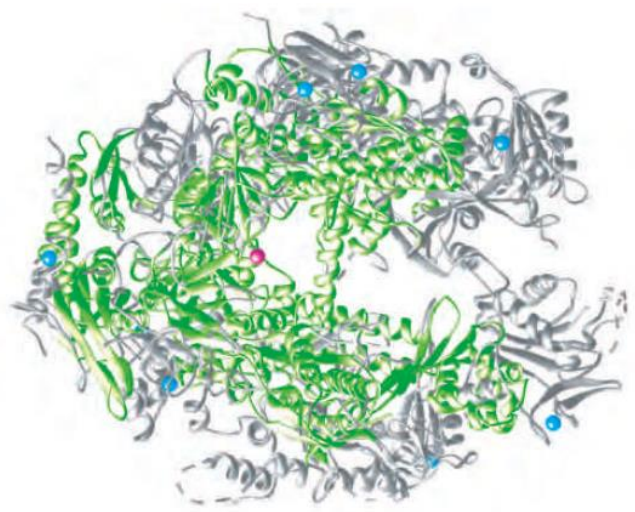


Figure 6-11,12,13 *Molecular Biology of the Cell* (© Garland Science 2015)

Eukaryotic cells Have Three RNA Polymerases



Structural similarity between a bacterial RNA polymerase and a eukaryotic RNA polymerase II.

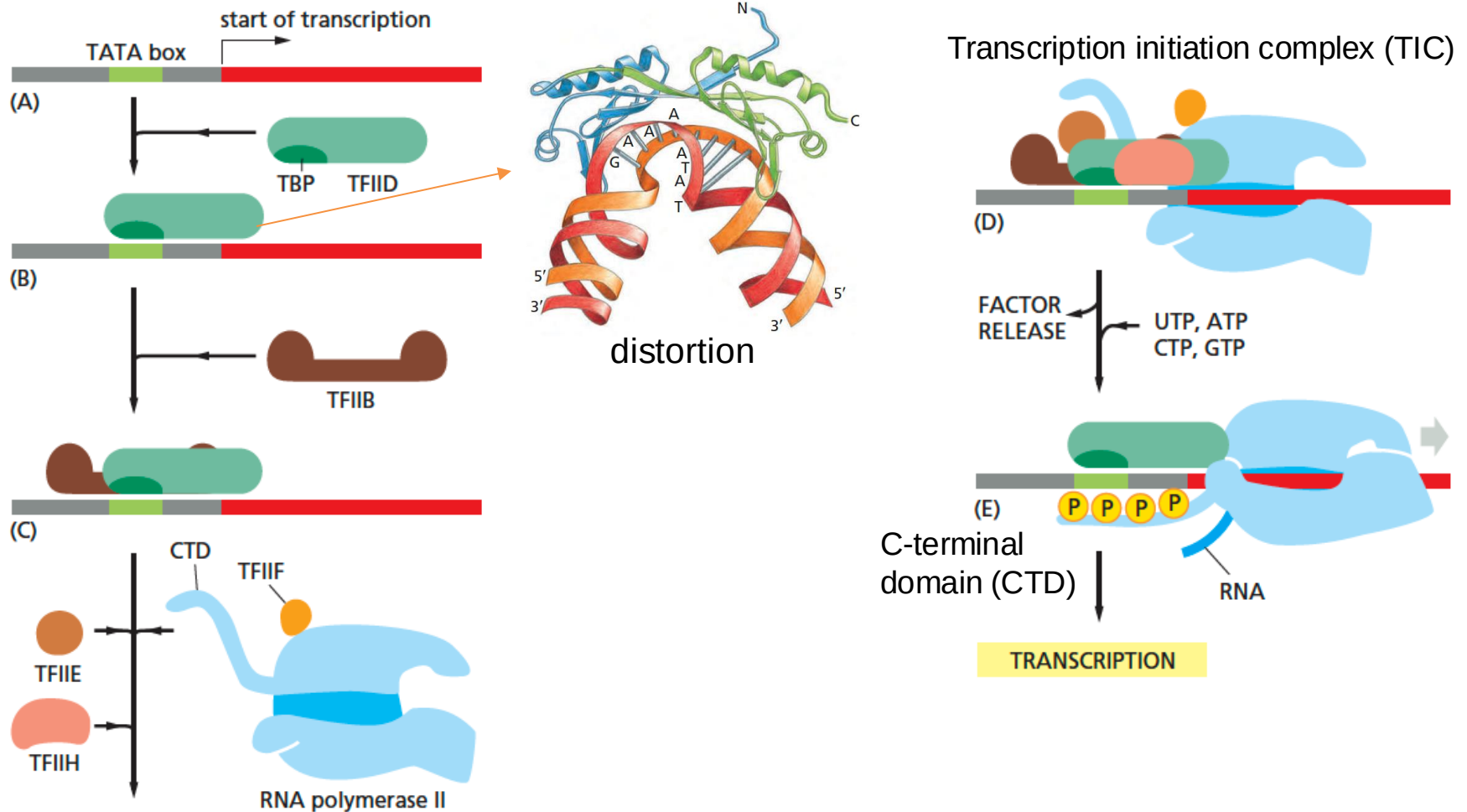


structurally similar

TABLE 6-2 The Three RNA Polymerases in Eukaryotic Cells	
Type of polymerase	Genes transcribed
RNA polymerase I	5.8S, 18S, and 28S rRNA genes
RNA polymerase II	All protein-coding genes, plus snoRNA genes, miRNA genes, siRNA genes, lncRNA genes, and most snRNA genes
RNA polymerase III	tRNA genes, 5S rRNA genes, some snRNA genes, and genes for other small RNAs
The rRNAs were named according to their “S” values, which refer to their rate of sedimentation in an ultracentrifuge. The larger the S value, the larger the rRNA.	

Figure 6-9, 6-14 Molecular Biology of the Cell (© Garland Science 2015)

RNA Polymerase II Requires a Set of General Transcription Factors



- “general” because they are needed at nearly all promoters used by RNA polymerase II.
- TFII stands for “transcription factor for polymerase II.
- Carry out functions equivalent to those of the sigma factor in bacteria.

Figure 6-15, 6-17 *Molecular Biology of the Cell* (© Garland Science 2015)

Eukaryotic Transcription Initiation Requires More Proteins

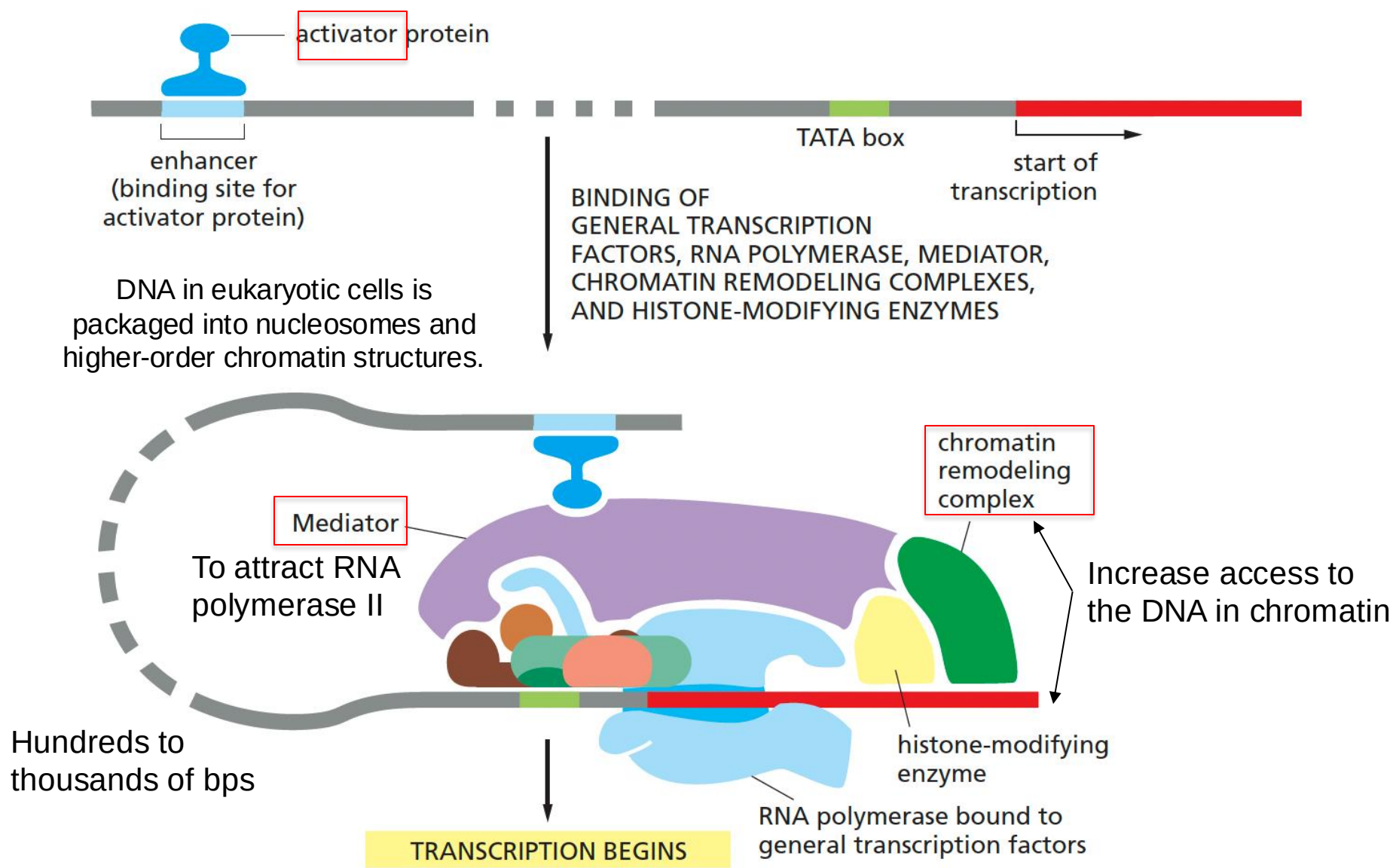
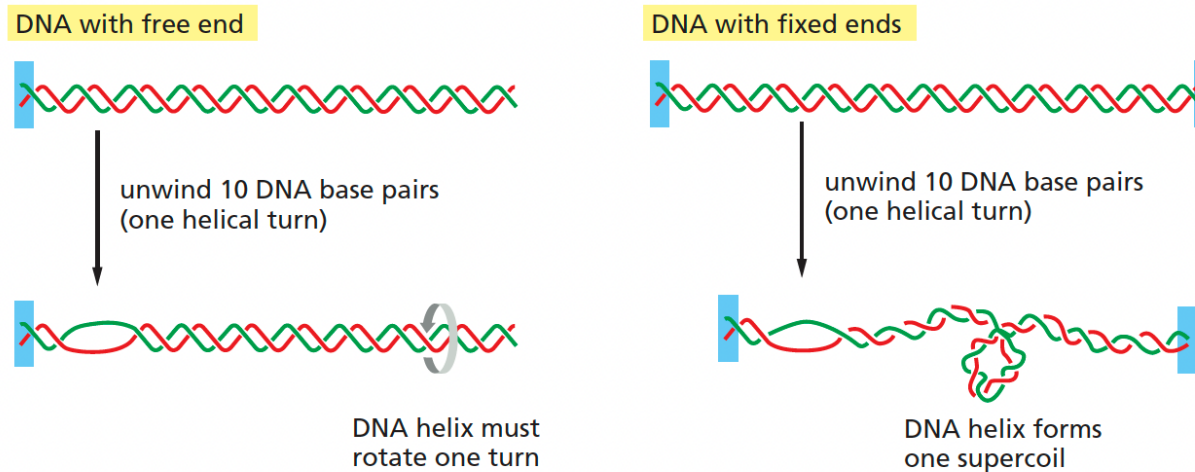
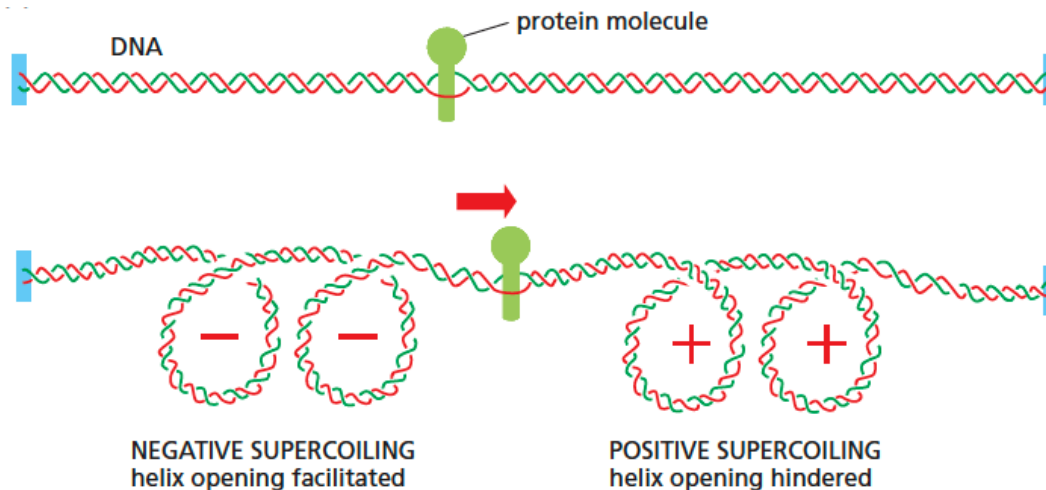


Figure 6-18 *Molecular Biology of the Cell* (© Garland Science 2015)

Transcription Creates Superhelical Tension



In eukaryotes, DNA topoisomerase enzymes rapidly remove this superhelical tension.



RNA polymerase creates superhelical tension as it moves along DNA.

Cells Produce Different Categories of RNA Molecules

TABLE 6–1 Principal Types of RNAs Produced in Cells

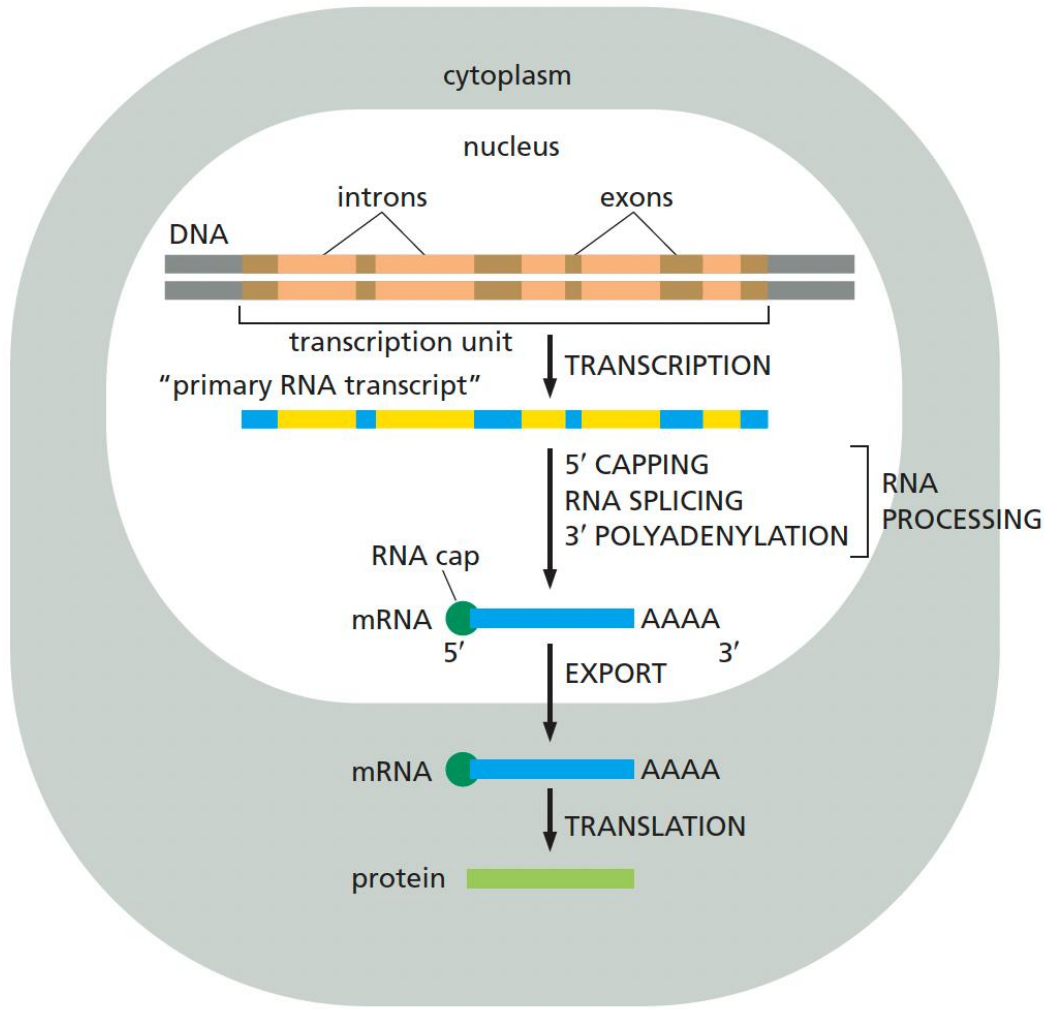
Type of RNA	Function
mRNAs	Messenger RNAs, code for proteins
rRNAs	Ribosomal RNAs, form the basic structure of the ribosome and catalyze protein synthesis
tRNAs	Transfer RNAs, central to protein synthesis as adaptors between mRNA and amino acids
snRNAs	Small nuclear RNAs, function in a variety of nuclear processes, including the splicing of pre-mRNA
snoRNAs	Small nucleolar RNAs, help to process and chemically modify rRNAs
miRNAs	MicroRNAs, regulate gene expression by blocking translation of specific mRNAs and cause their degradation
siRNAs	Small interfering RNAs, turn off gene expression by directing the degradation of selective mRNAs and the establishment of compact chromatin structures
piRNAs	Piwi-interacting RNAs, bind to piwi proteins and protect the germ line from transposable elements
lncRNAs	Long noncoding RNAs, many of which serve as scaffolds; they regulate diverse cell processes, including X-chromosome inactivation

noncoding RNAs

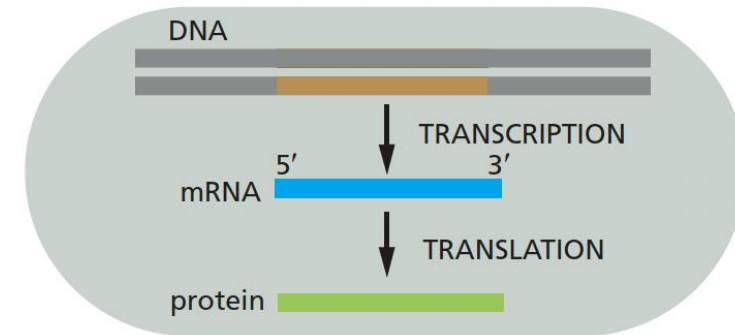
- Most of the RNA in cells is rRNA.
- mRNA comprises only 3–5% of the total RNA in a typical mammalian cell.
- On average only 10–15 molecules of each mRNA species in each cell.

Transcription In Eukaryotic Cells Is More Complex

(A) EUKARYOTES



(B) PROKARYOTES

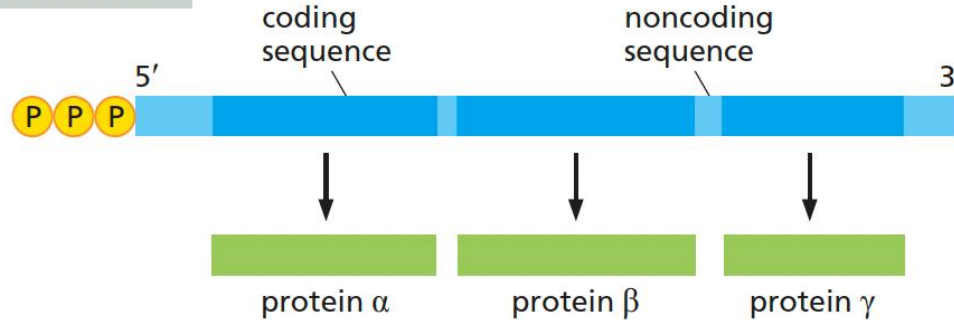


- mRNA molecule contains both coding (exon) and noncoding (intron) sequences;
- mRNA molecules need to be processed in multiple ways;
- Mature mRNA need to be transported from the nucleus to the cytoplasm.

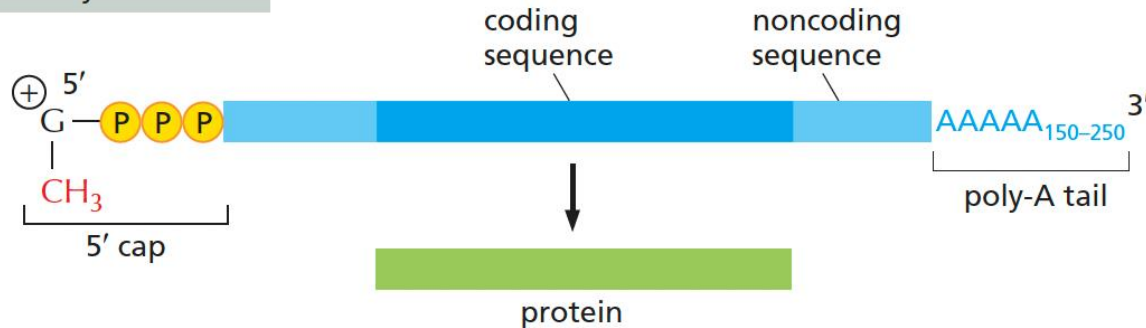
Transcription In Eukaryotic Cells Is More Complex

RNA Processing

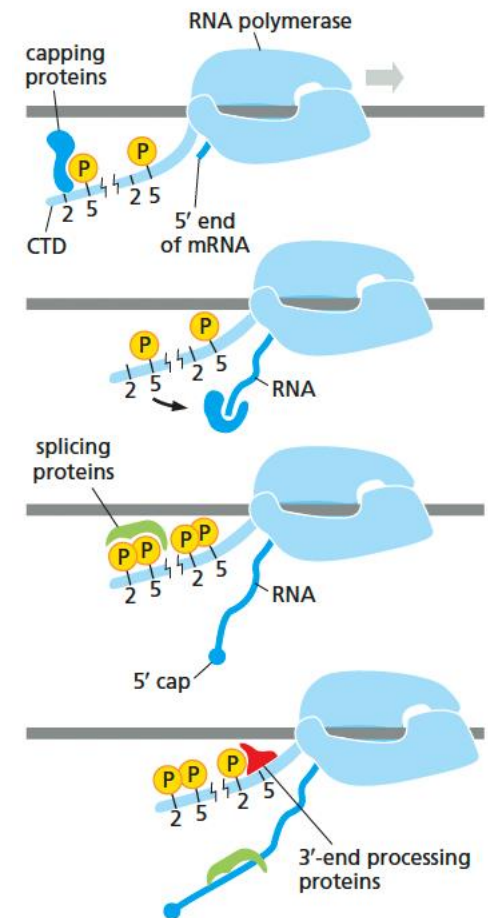
prokaryotic mRNA



eukaryotic mRNA



Transcription elongation in eukaryotes is tightly coupled to RNA processing.



RNA Capping Is the First Modification of Eukaryotic Pre-mRNAs

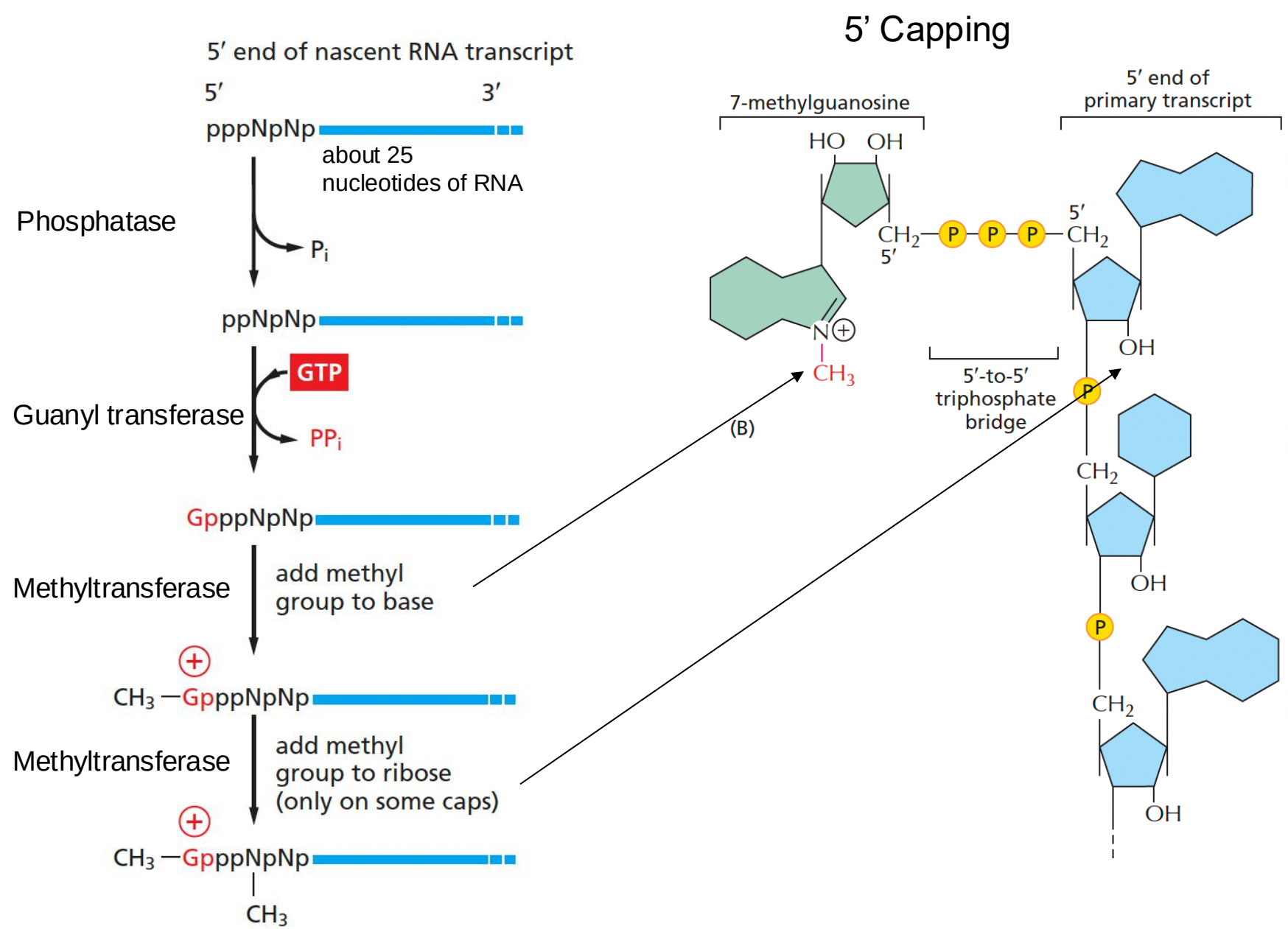
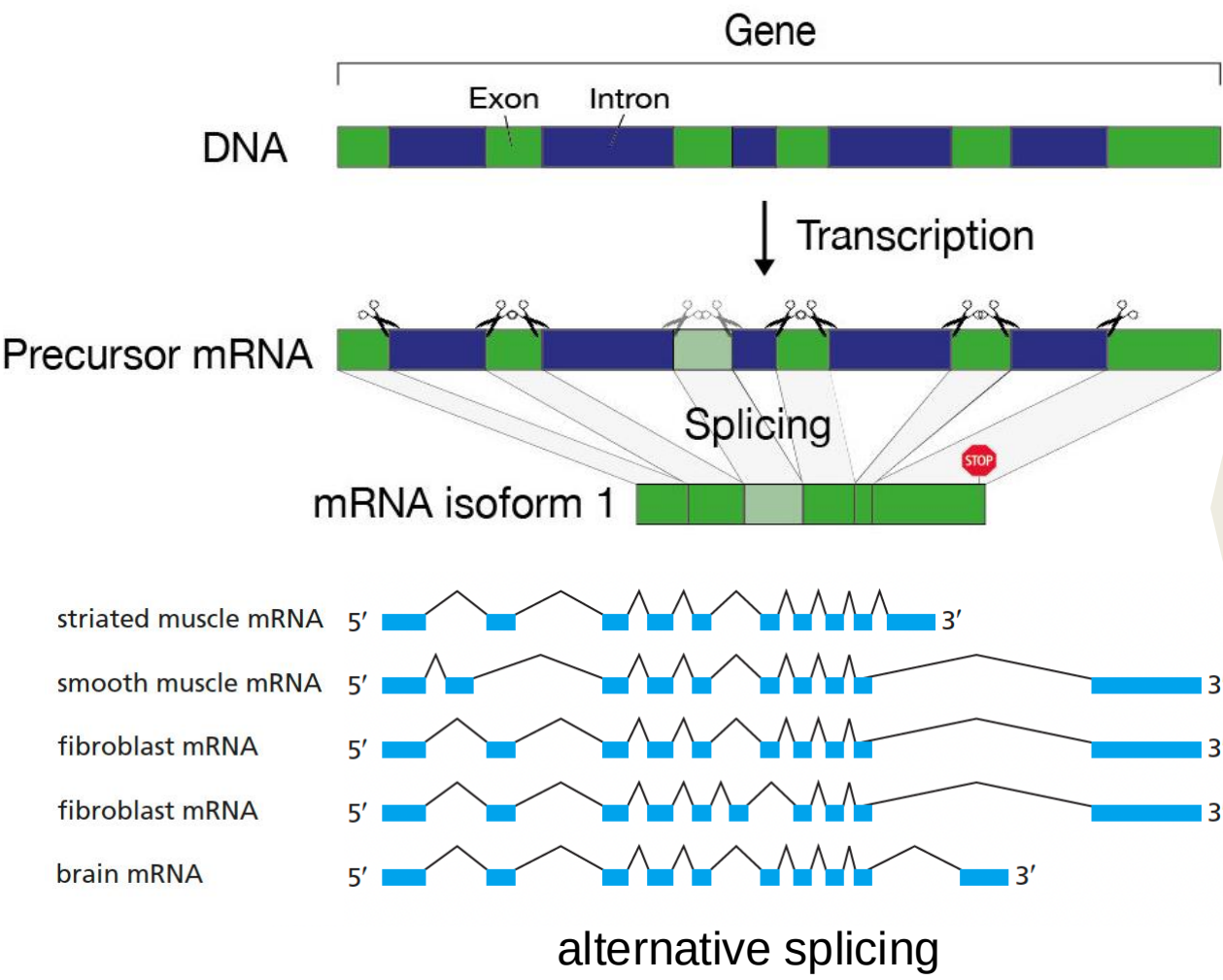


Figure 6-21B, 6-23 *Molecular Biology of the Cell* (© Garland Science 2015)

RNA Splicing Removes Intron Sequences from Pre-mRNAs



The pre-mRNA splicing reaction

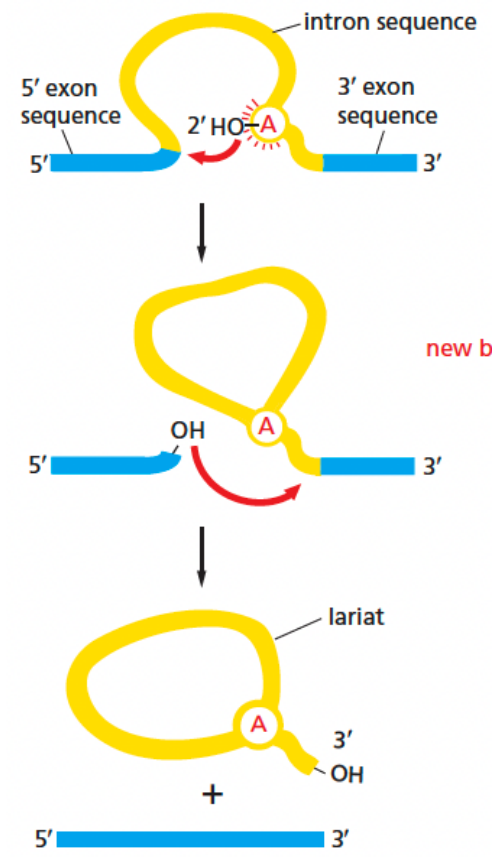
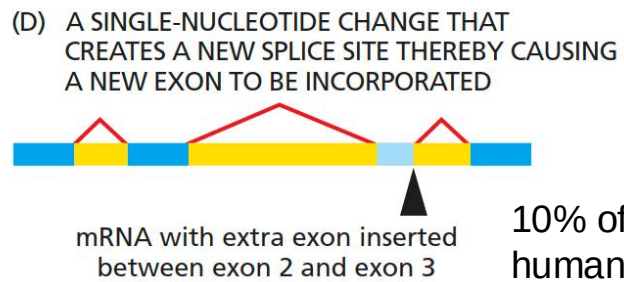
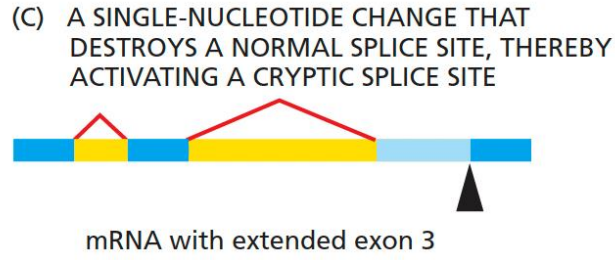
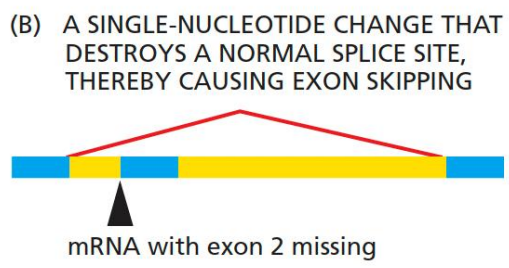
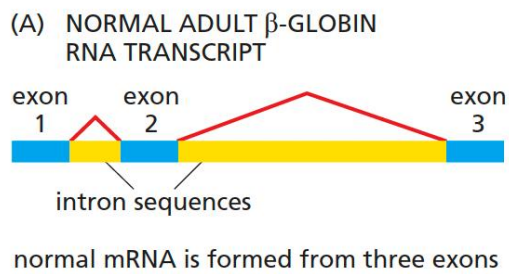
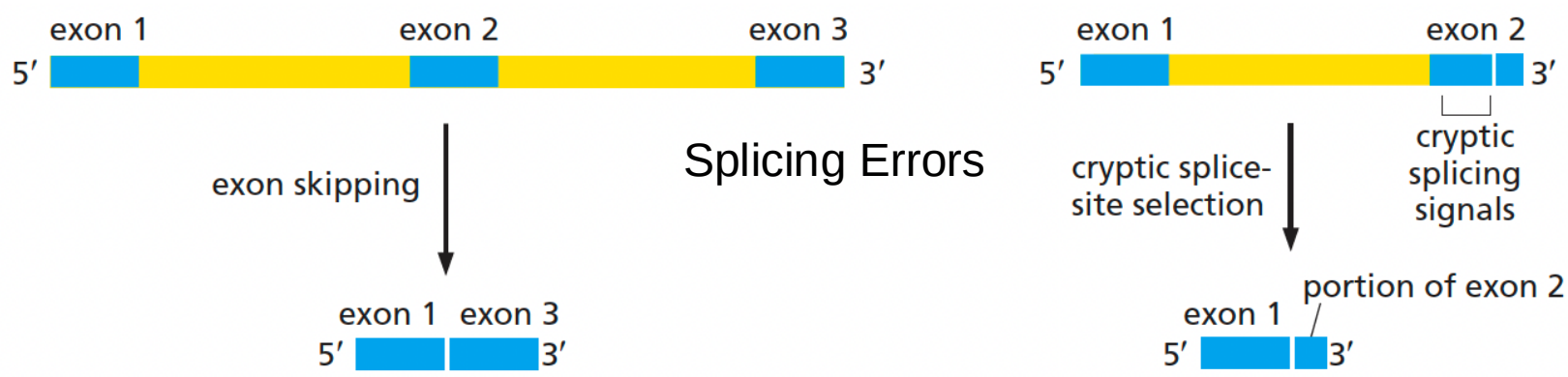


Figure 6-25, 6-26 *Molecular Biology of the Cell* (© Garland Science 2015)

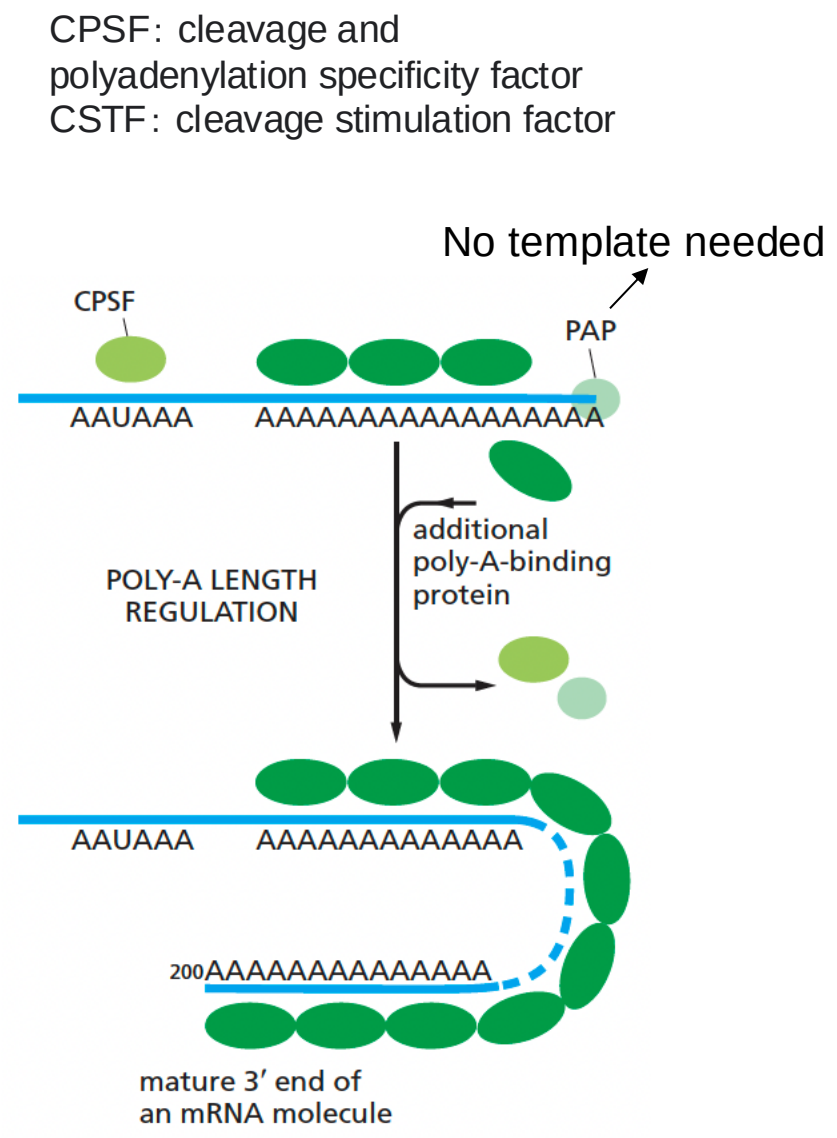
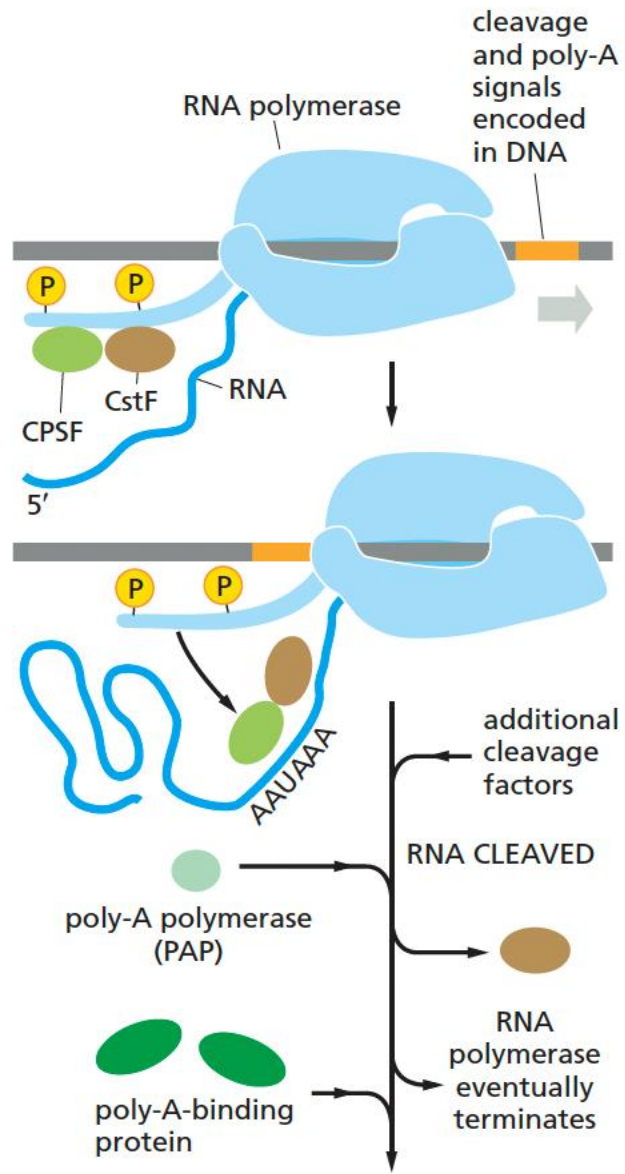
RNA Splicing Shows Remarkable Plasticity



10% of point mutations in human diseases produce aberrant splicing of the gene containing the mutation

Figure 6-30, 6-33 *Molecular Biology of the Cell* (© Garland Science 2015)

RNA-Processing Enzymes Generate the 3' End of Eukaryotic mRNAs



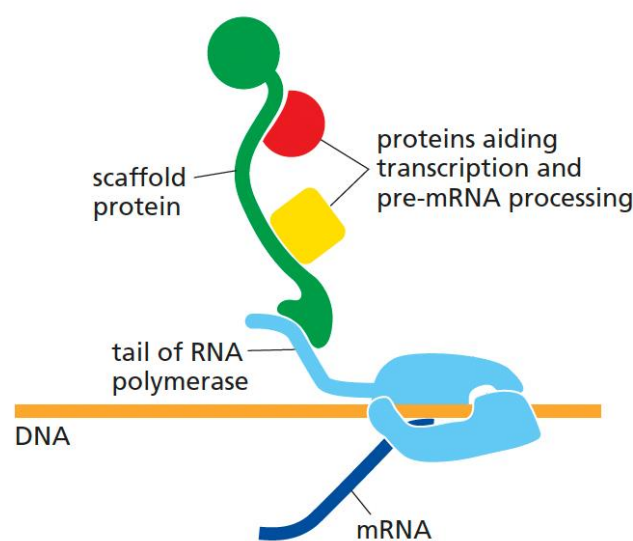
Variable length around 200 nt

Figure 6-35 Molecular Biology of the Cell (© Garland Science 2015)

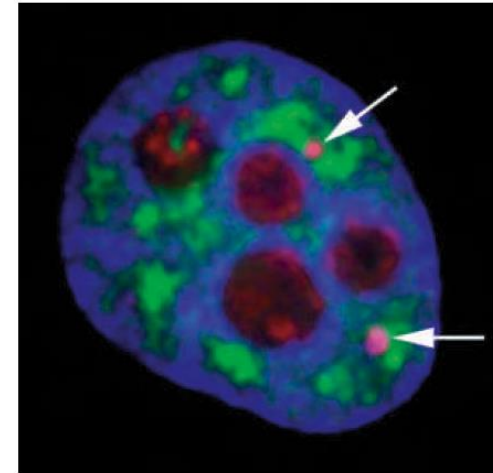
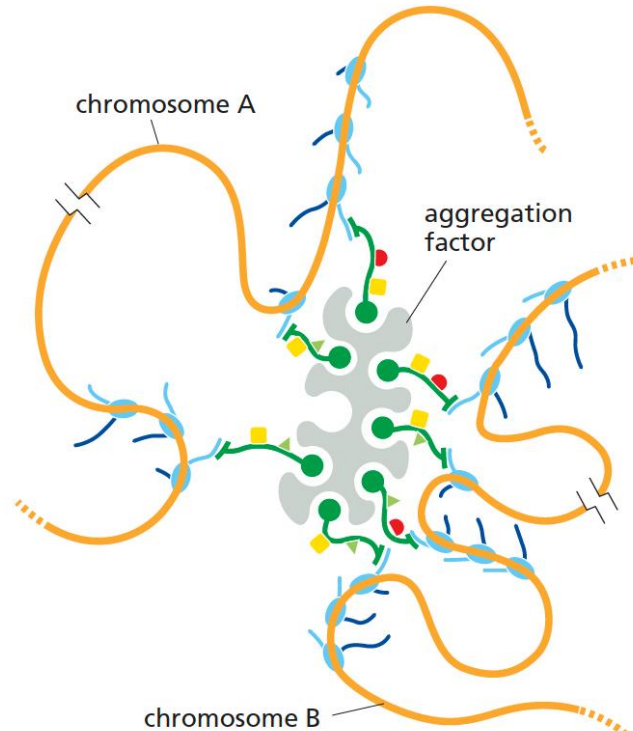
The Nucleus Contains a Variety of Subnuclear Aggregates

The main function of these aggregates is to bring components together at high concentration in order to speed up their assembly.

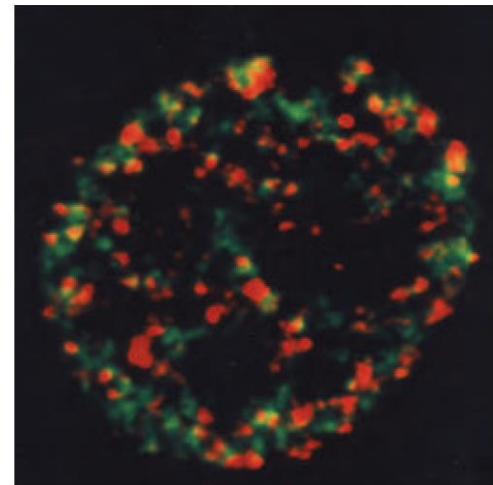
“Transcription Factory”



(A)



10 μm



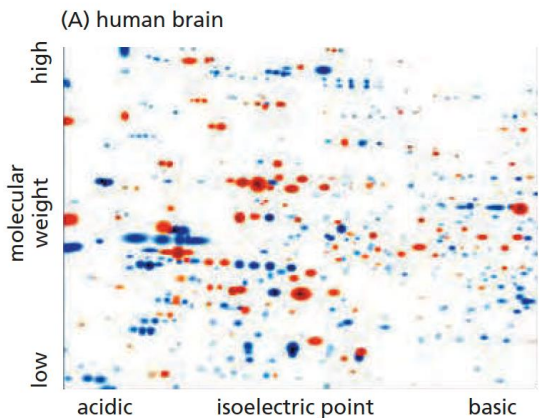
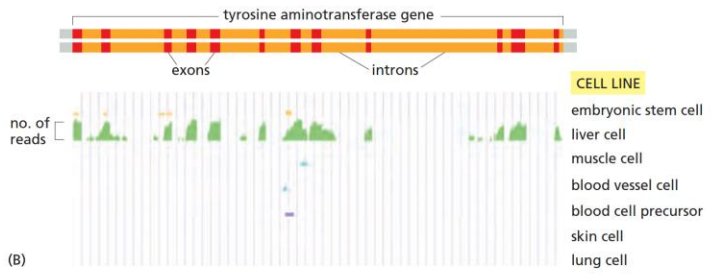
(C)

2 μm

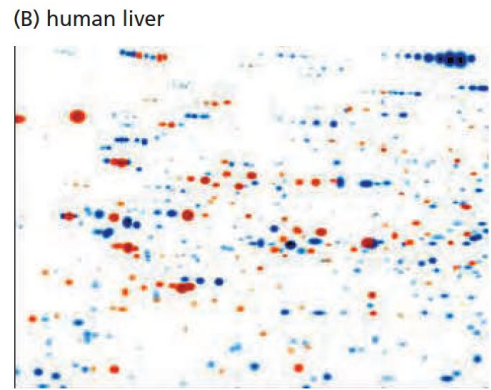
Green: newly synthesized DNA
Red: newly synthesized RNA

Different Cell Types Synthesize Different Sets of RNAs and Proteins

RNA

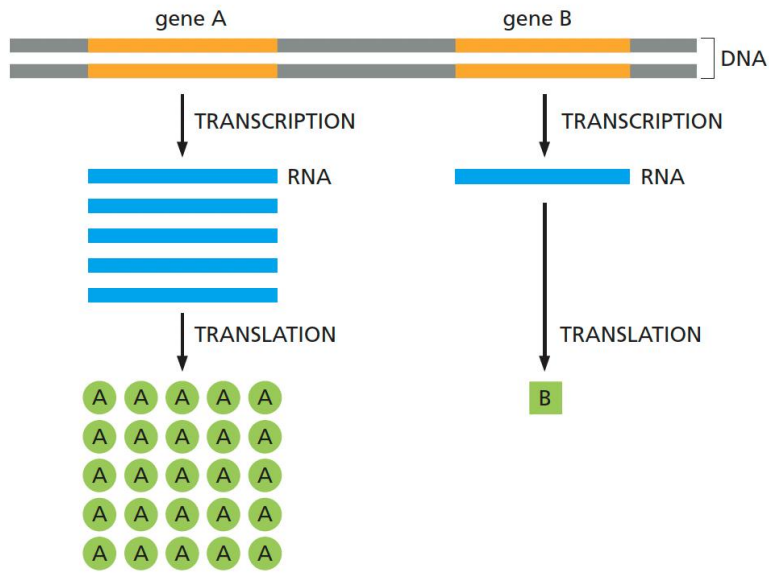


Protein



Red are common
Blue are specific

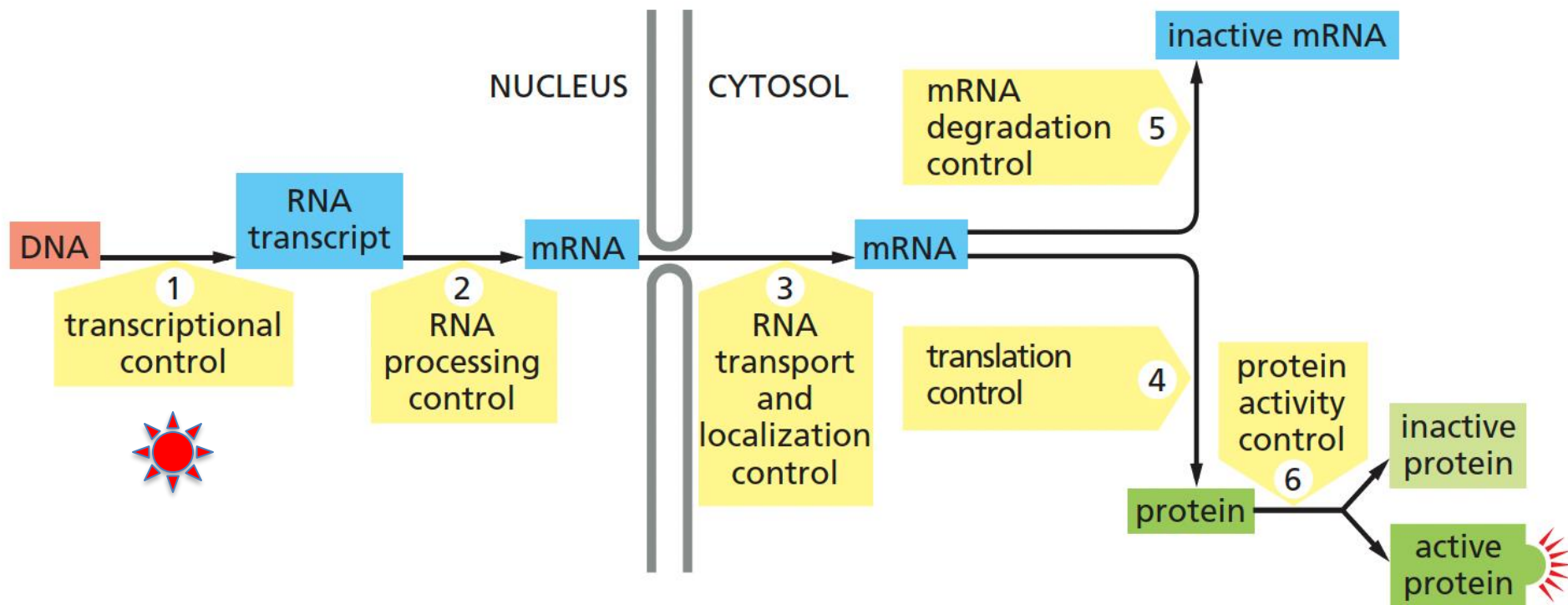
A cell can change (or regulate) the expression of each of its genes by controlling the production of its RNA.



Fascinating challenges:

- Variations between organisms
- Products beyond proteins
- Chaotic genome organizations
- Multiple-levels regulation

There Are Many Steps From DNA to Protein



- Cells can change the levels of their proteins according to their needs.
- Gene expression can be regulated at many of the steps in the pathway from DNA to RNA to protein.
- Any or all of the steps in this figure could be regulated for each individual protein.

CONTROL OF TRANSCRIPTION BY SEQUENCE-SPECIFIC DNA-BINDING PROTEINS

Transcription Regulators
cis-regulatory Sequences

Transcription Regulators Contain Structural Motifs That Can Read DNA Sequences

Each transcription regulator makes a series of contacts with the DNA, involving hydrogen bonds, ionic bonds, and hydrophobic interactions.

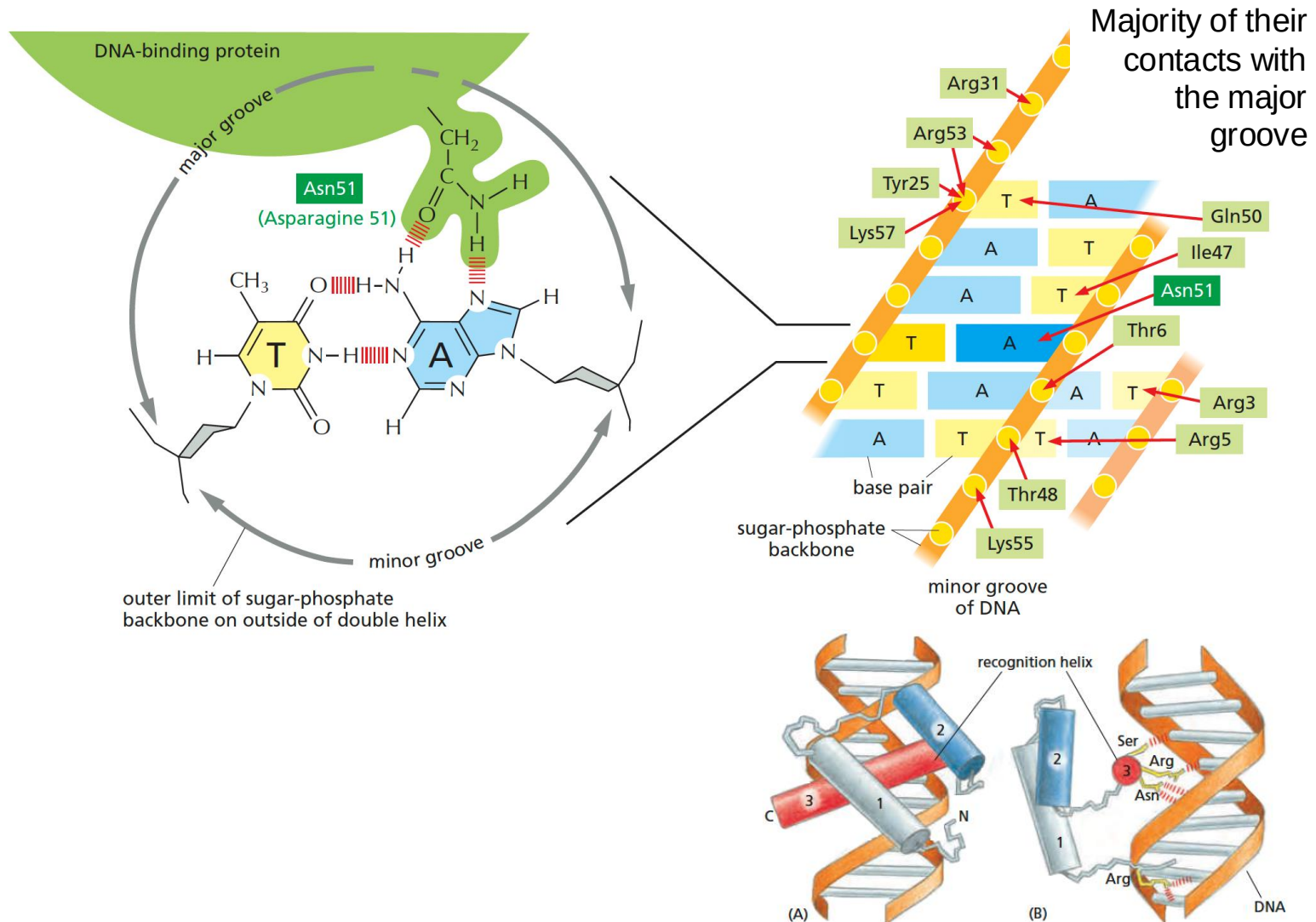


Figure 7-8 *Molecular Biology of the Cell* (© Garland Science 2015)

Transcription Regulators And *cis*-regulatory Sequences

CRS Logo: A collection of closely related DNA sequences



(A) Nanog *cis*-regulatory sequence

transcription regulator



(B) *cis*-regulatory sequence in genome

Dimerization of transcription regulators increases their affinity and specificity for DNA.



dimer

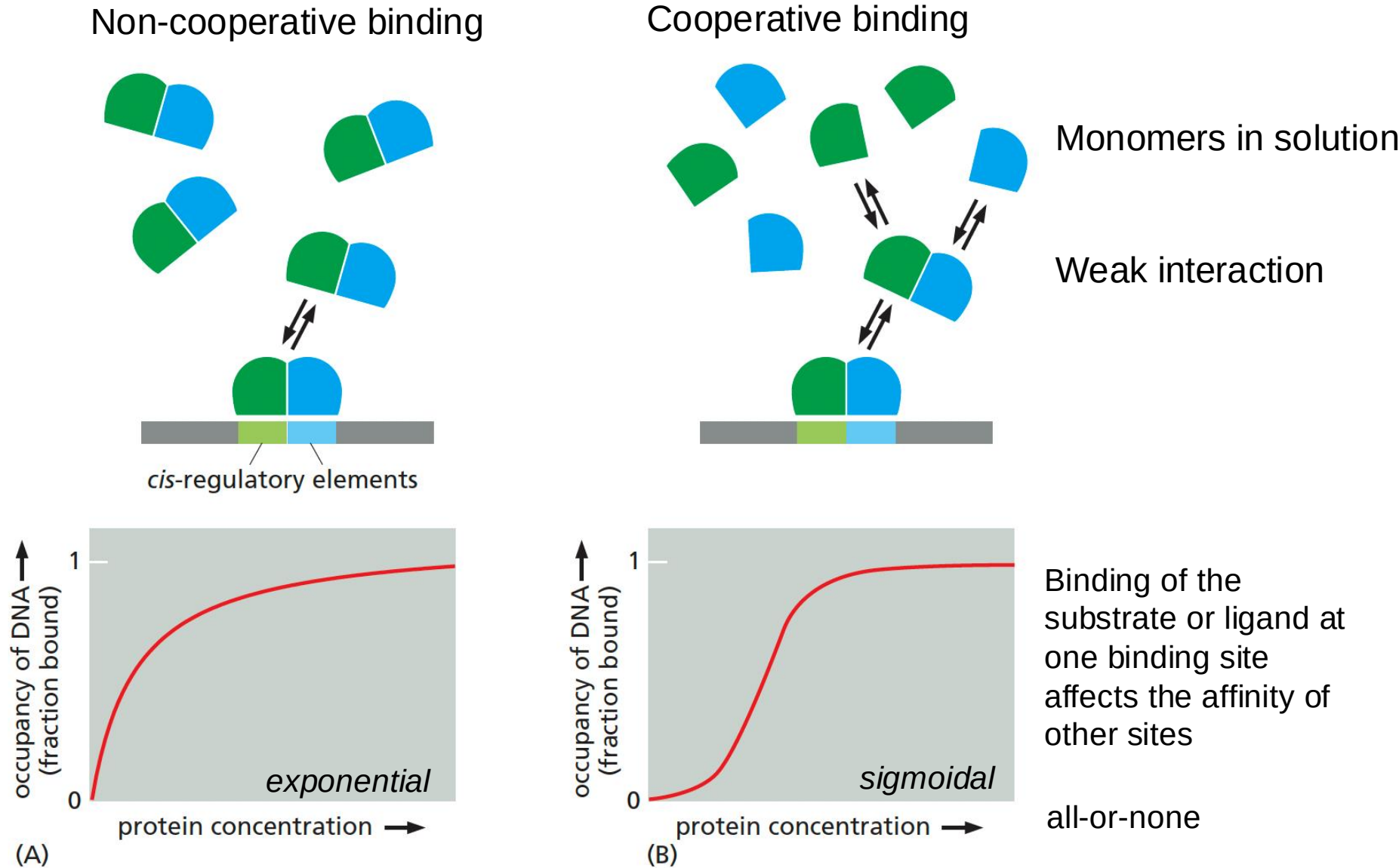


dimer



heterodimer

Transcription Regulators Bind Cooperatively to DNA



Nucleosome Structure Promotes Cooperative Binding of Transcription Regulators

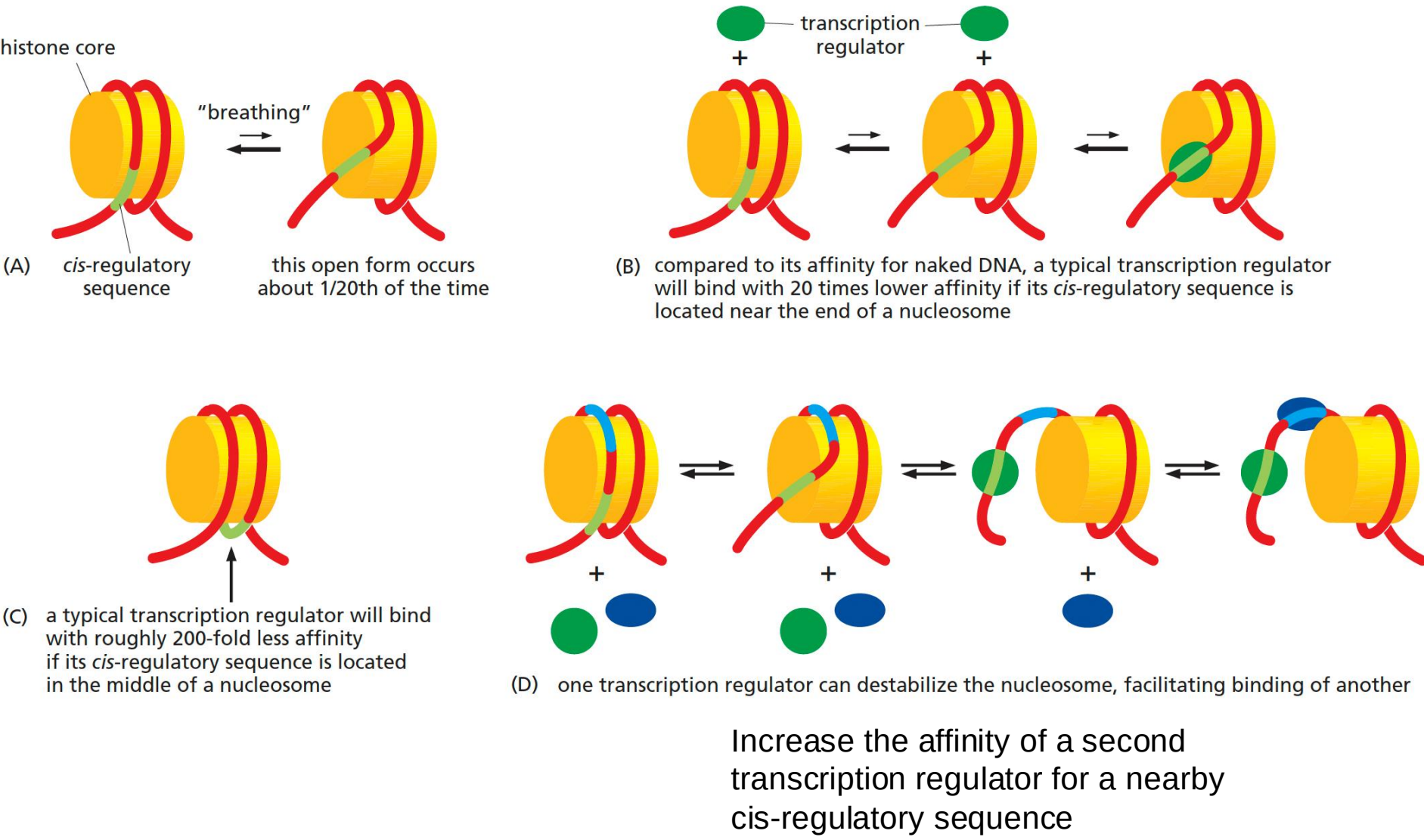
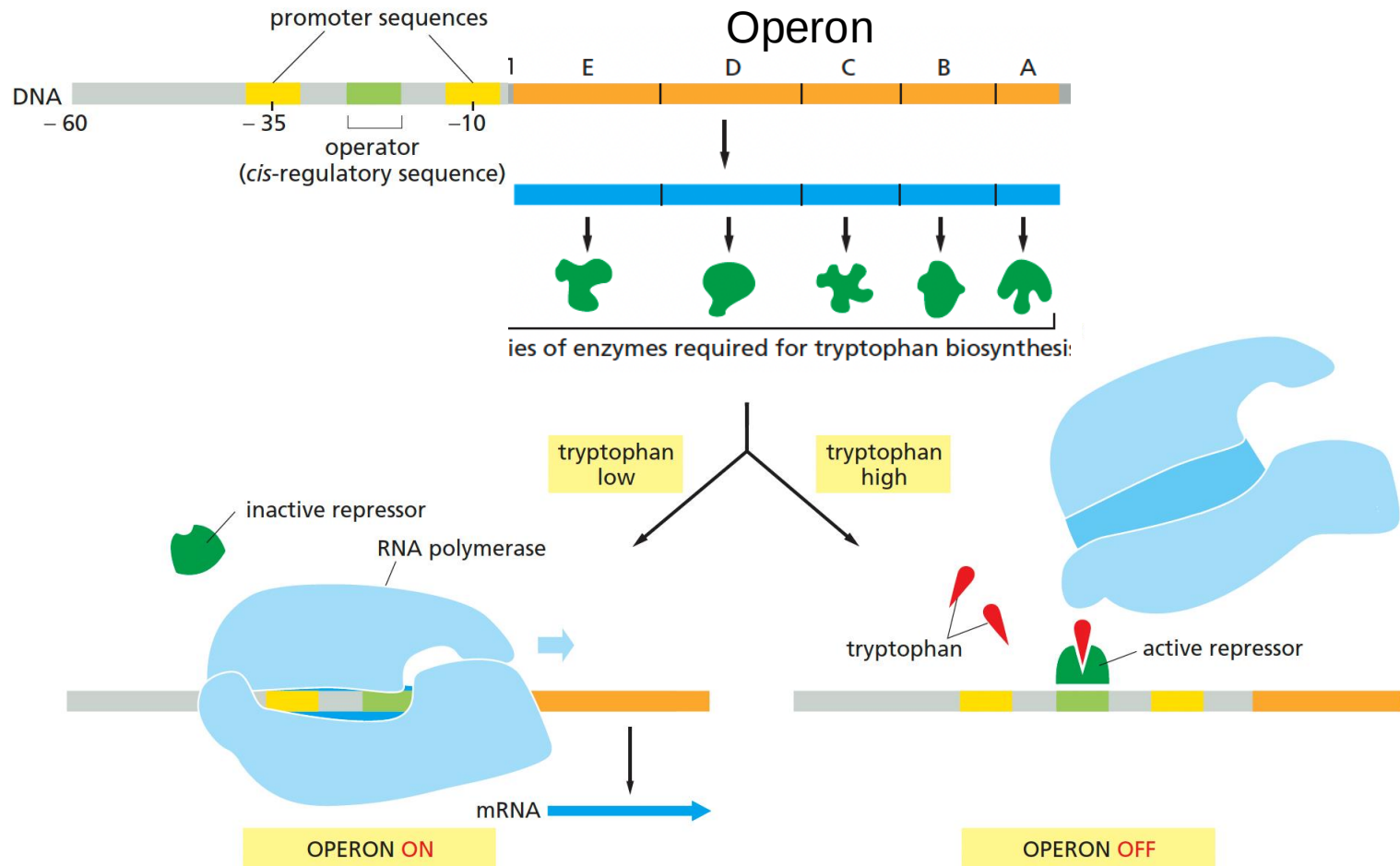


Figure 7-11 Molecular Biology of the Cell (© Garland Science 2015)

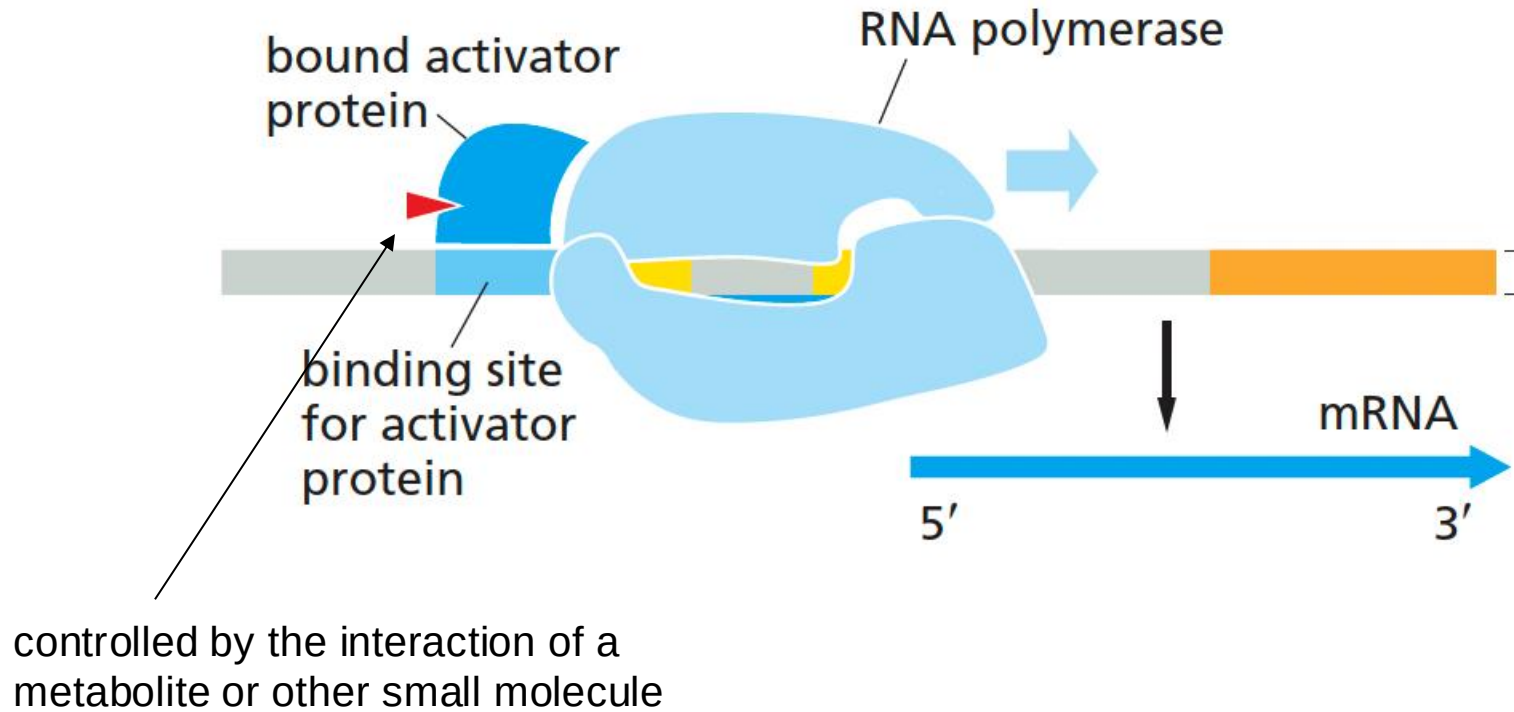
The Tryptophan Repressor Switches Genes Off

A cluster of bacterial genes can be transcribed from a single promoter.

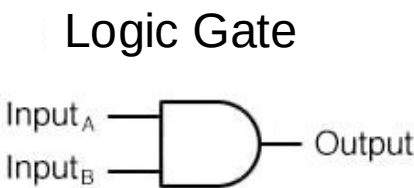


The bacterium can respond very rapidly to a rise or fall in tryptophan concentration.

Repressors Turn Genes Off and Activators Turn Them On



An Activator and a Repressor Control the Lac Operon



A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

Only in this case

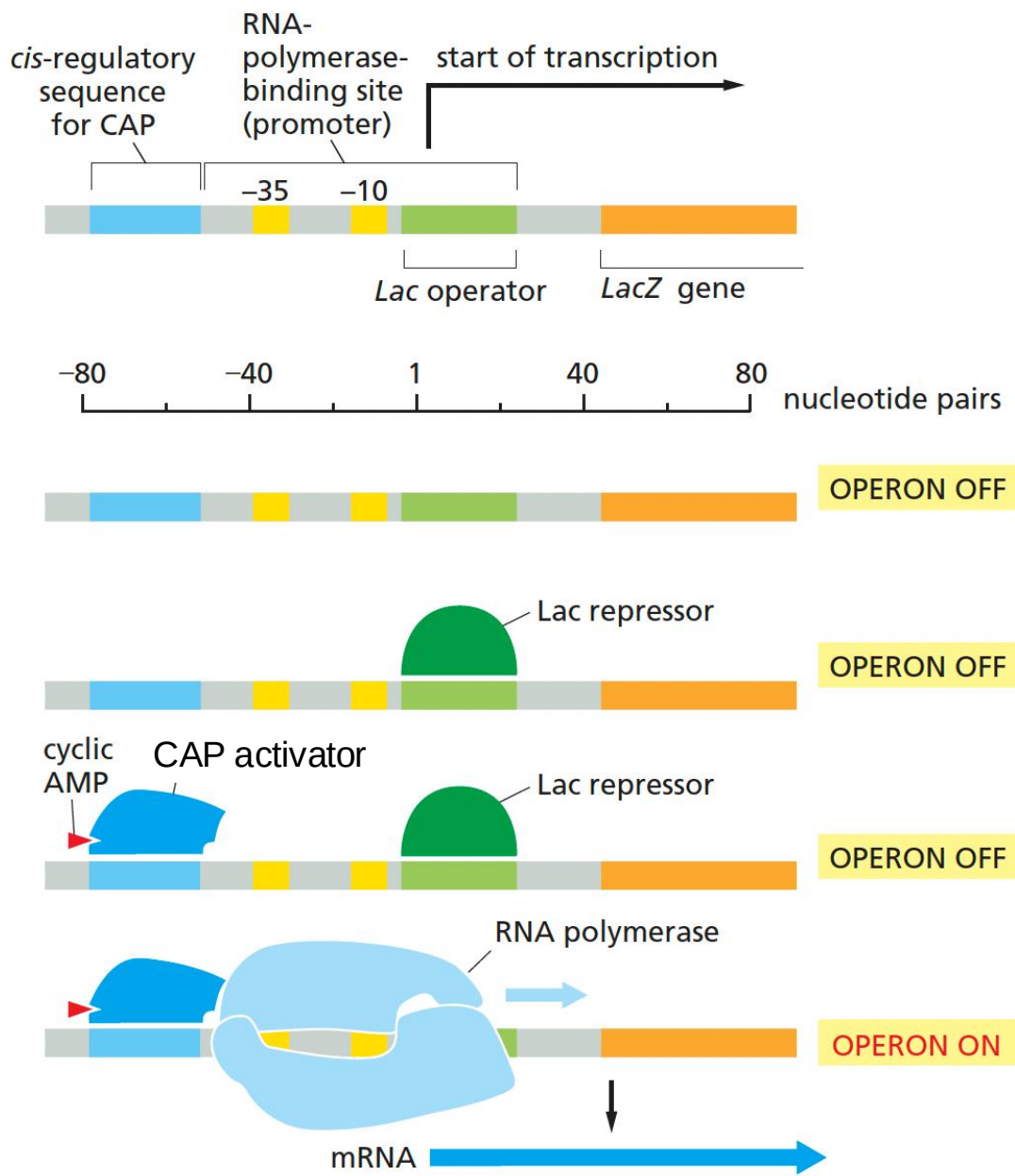
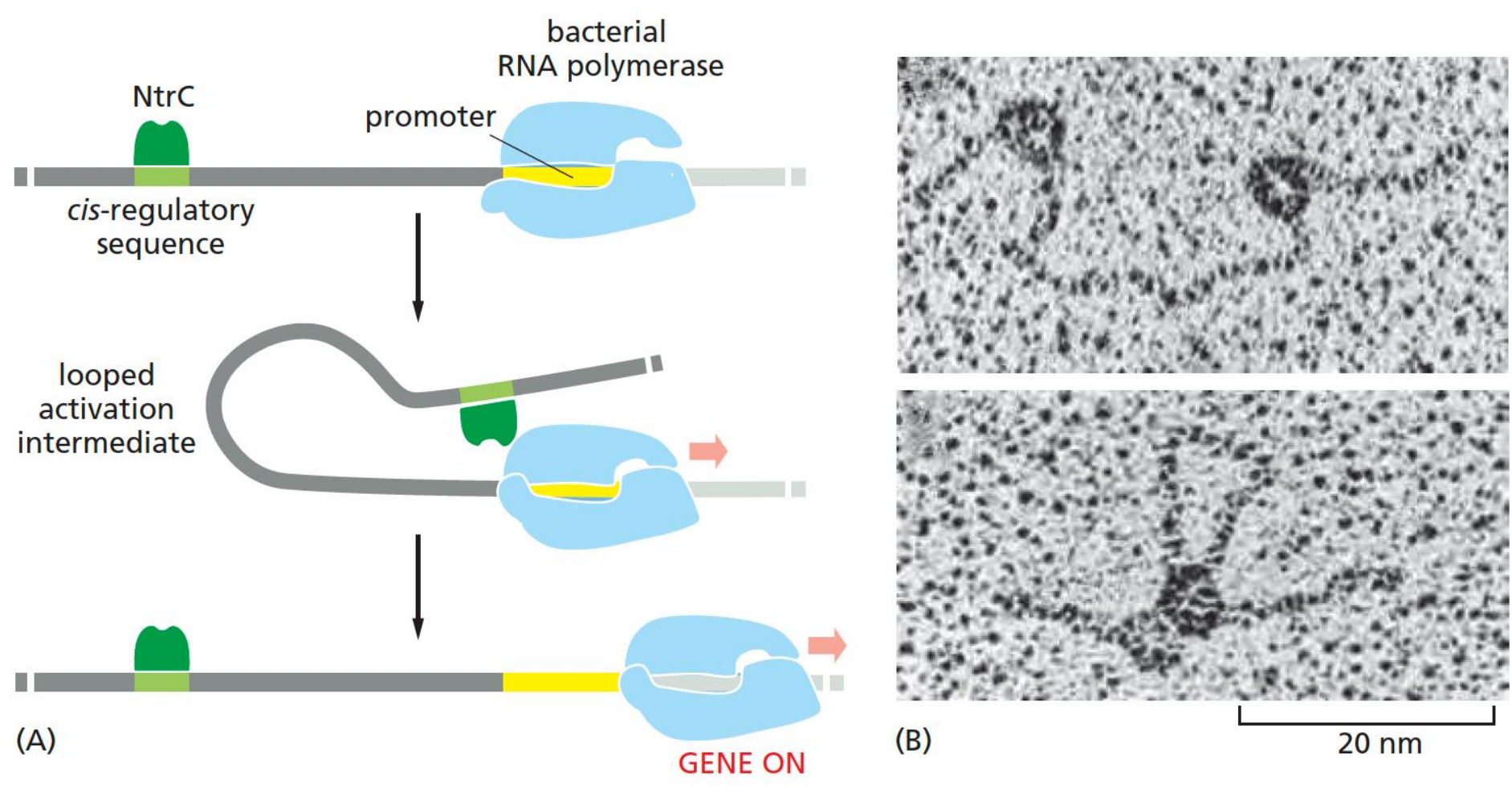


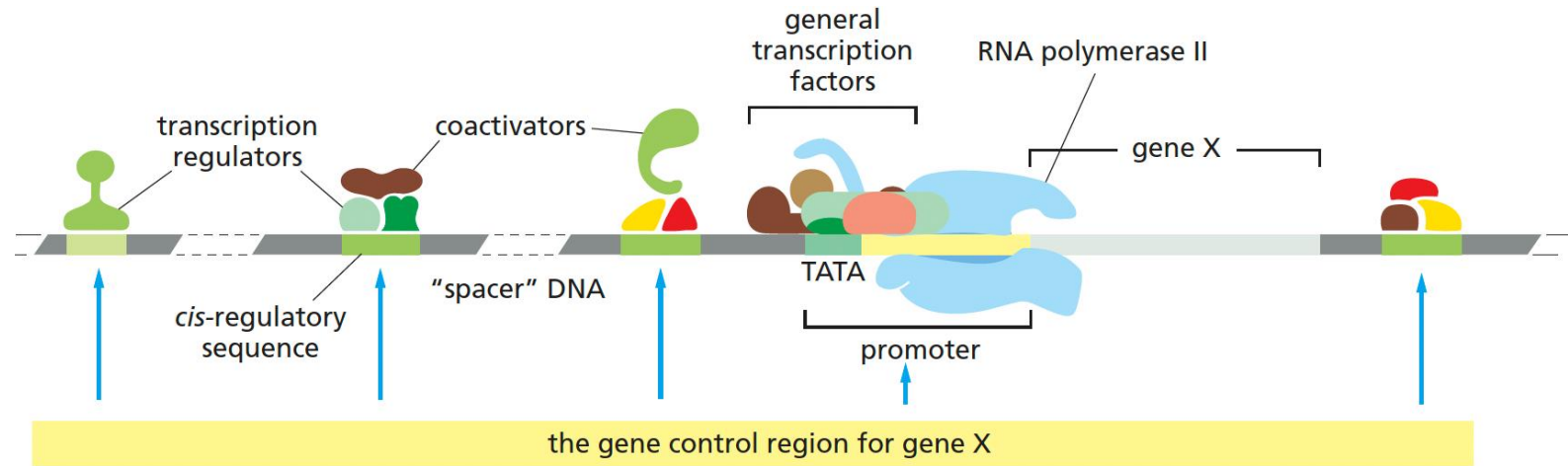
Figure 7-15 Molecular Biology of the Cell (© Garland Science 2015)

DNA Looping Can Occur During Bacterial Gene Regulation

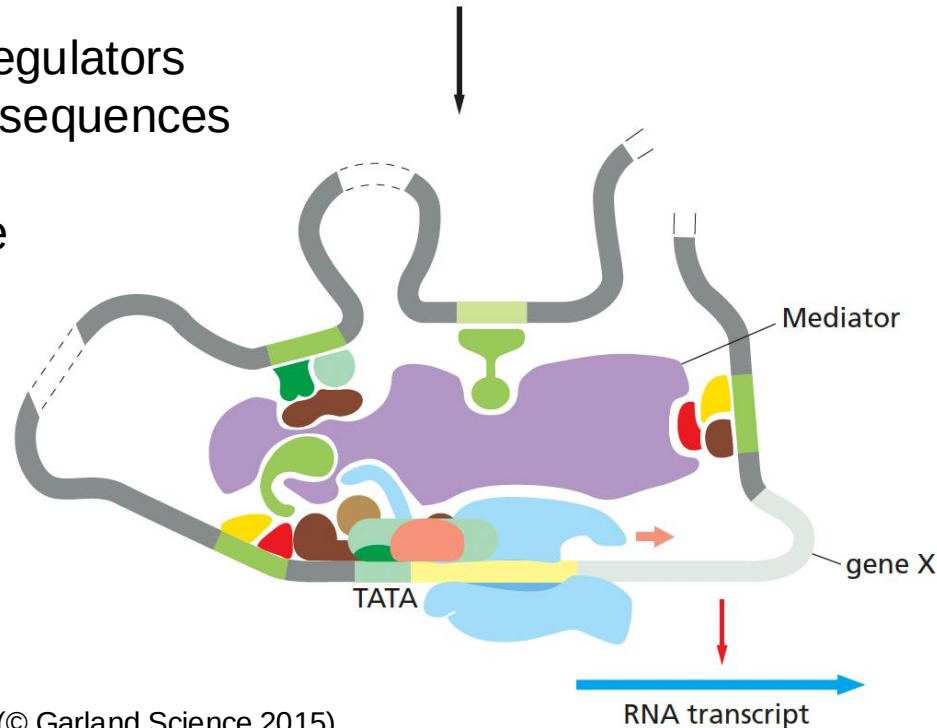


DNA looping occurs in the regulation of nearly every eukaryotic gene.

A Eukaryotic Gene Control Region Consists of a Promoter Plus Many *cis*-Regulatory Sequences



Multiple transcription regulators
Multiple *cis*-regulatory sequences
More DNA looping
Chromosome structure



Eukaryotic Transcription Regulators Work in Groups

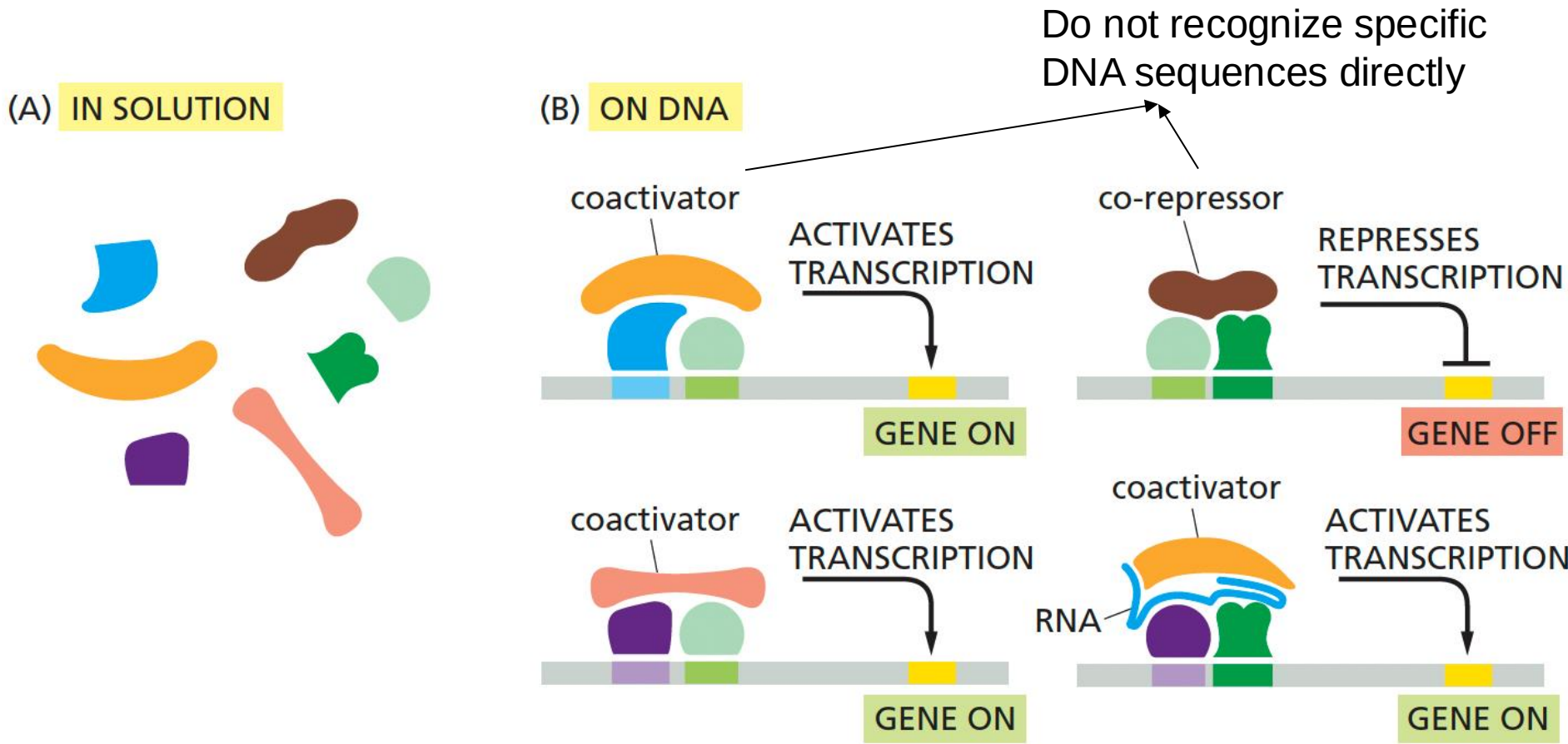


Figure 7-18 *Molecular Biology of the Cell* (© Garland Science 2015)

Eukaryotic Transcription Activators Direct the Modification of Local Chromatin Structure

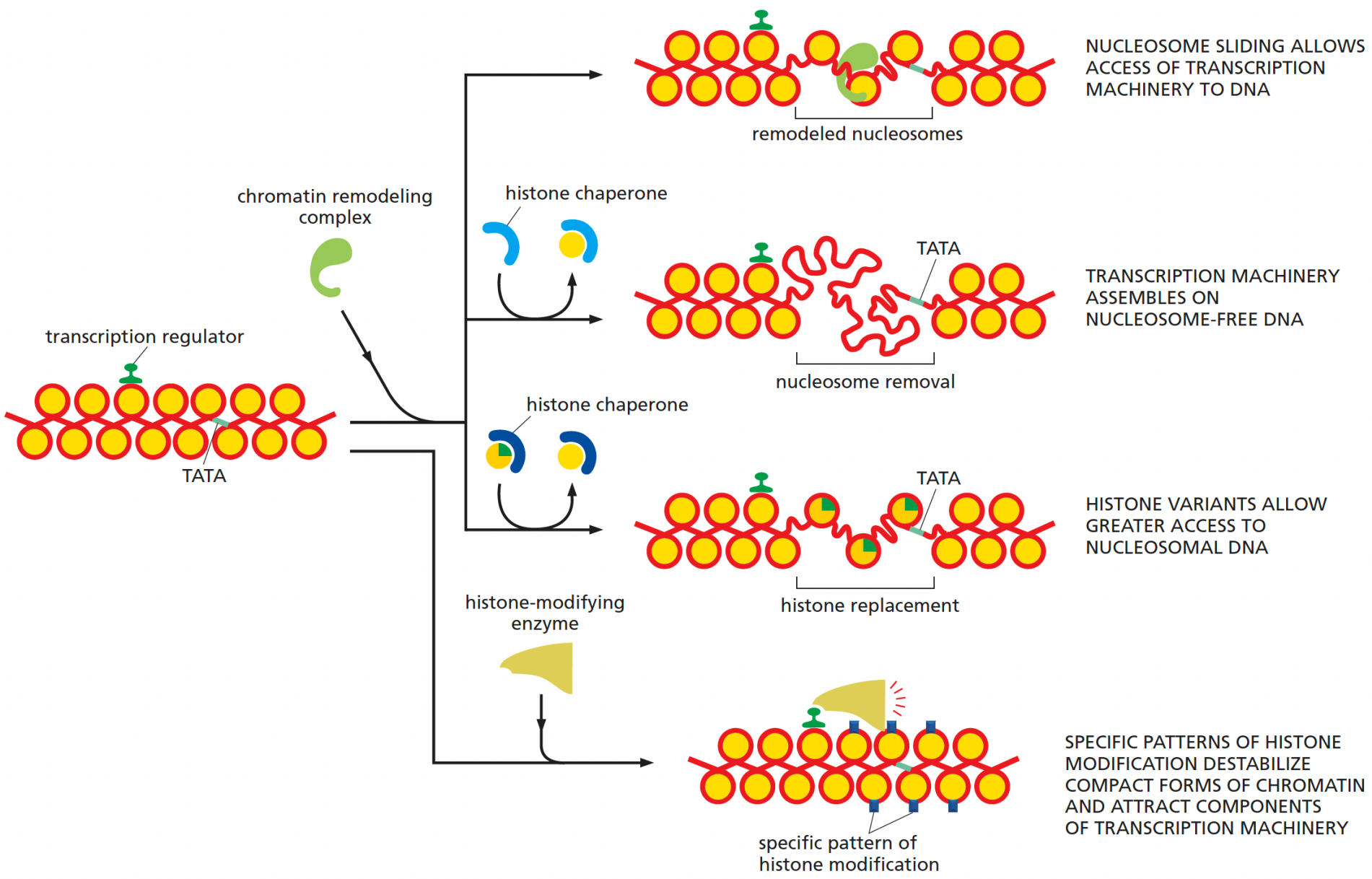
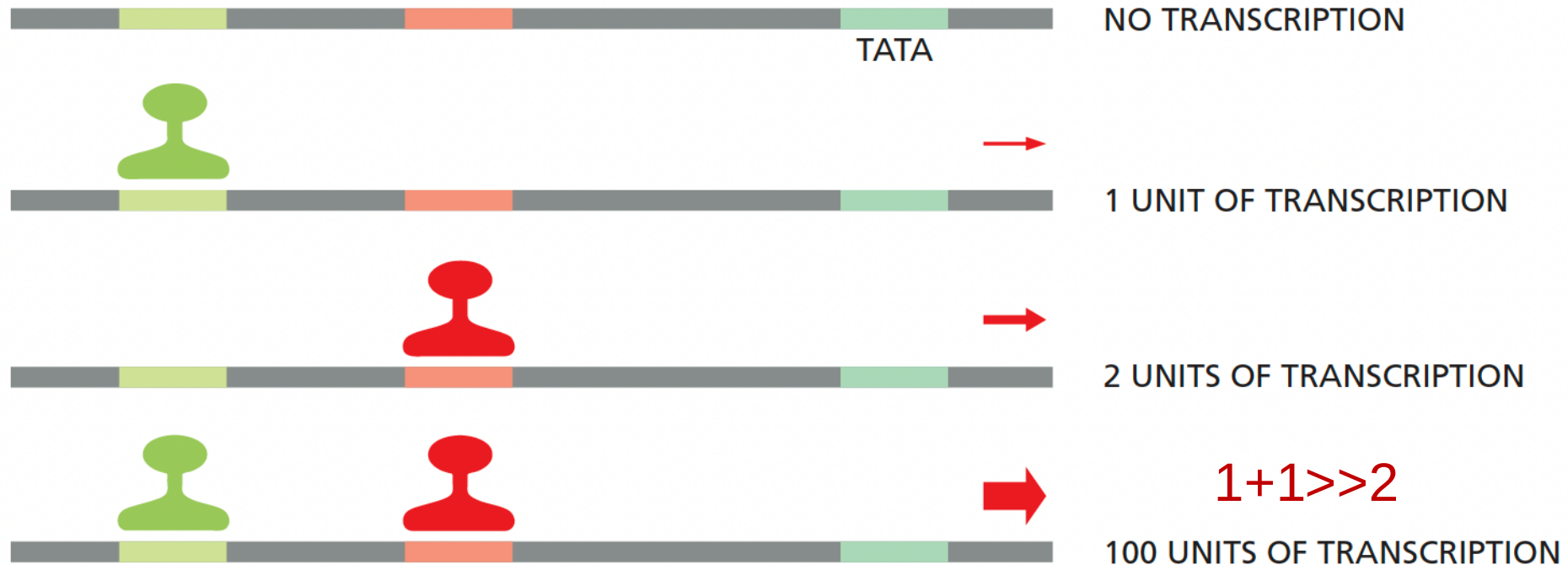


Figure 7-19 *Molecular Biology of the Cell* (© Garland Science 2015)

Transcription Activators Work Synergistically

Transcriptional synergy: a greater than additive effect of multiple activators working together.



Eukaryotic Transcription Repressors Can Inhibit Transcription in Several Ways

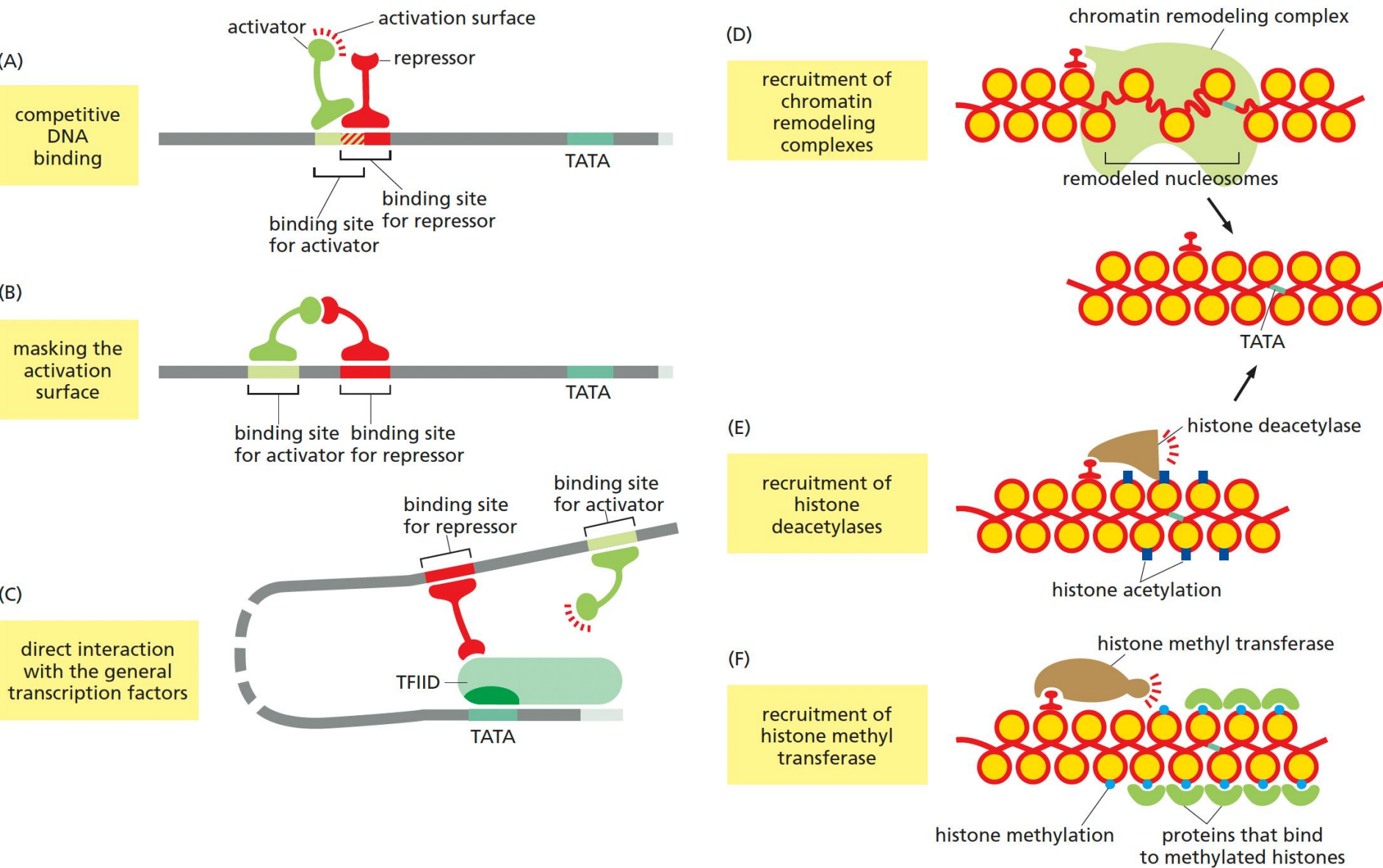
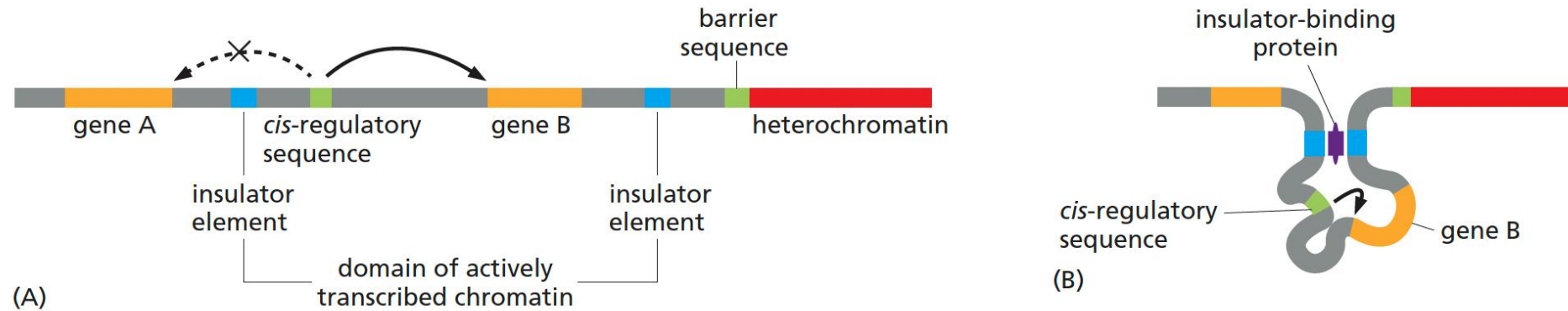
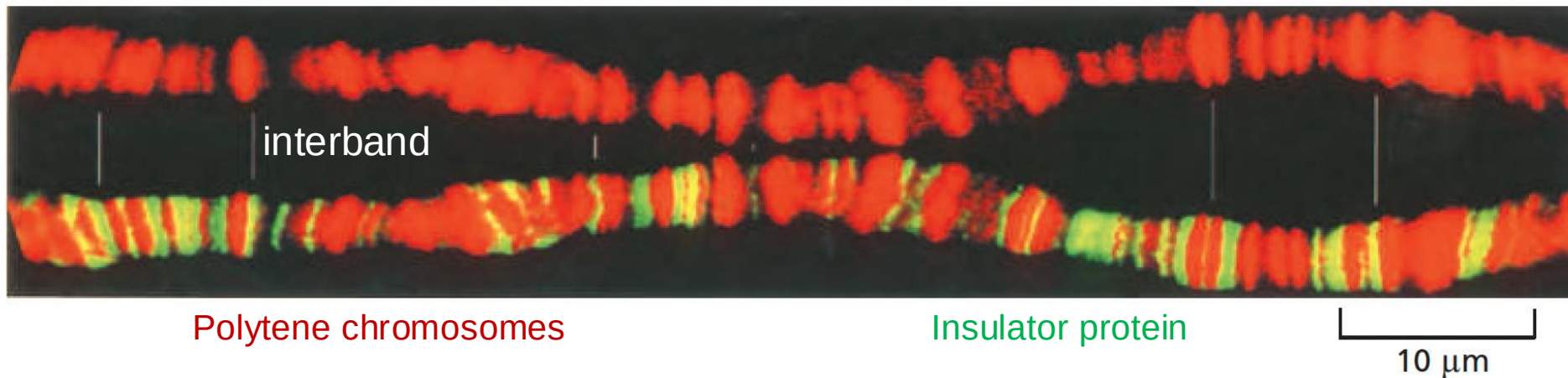


Figure 7-23 *Molecular Biology of the Cell* (© Garland Science 2015)

Insulator DNA Sequences Prevent Eukaryotic Transcription Regulators from Influencing Distant Genes



Localization of a *Drosophila* insulator-binding protein on polytene chromosomes



MOLECULAR GENETIC MECHANISMS THAT CREATE AND MAINTAIN SPECIALIZED CELL TYPES

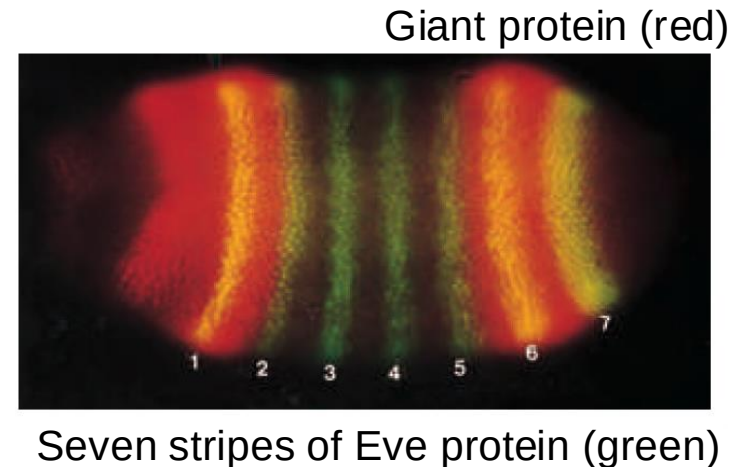
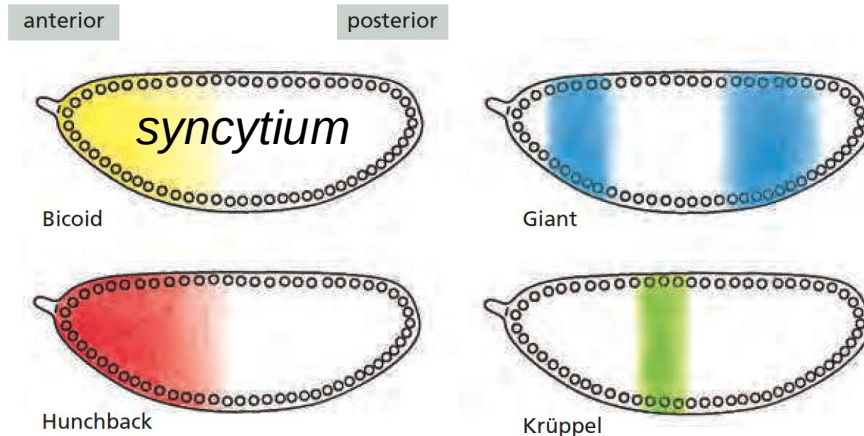
Cell Diversity

Cell Memory

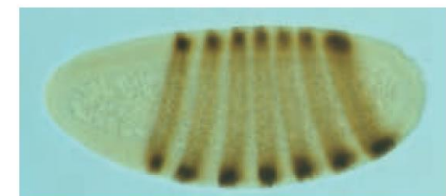
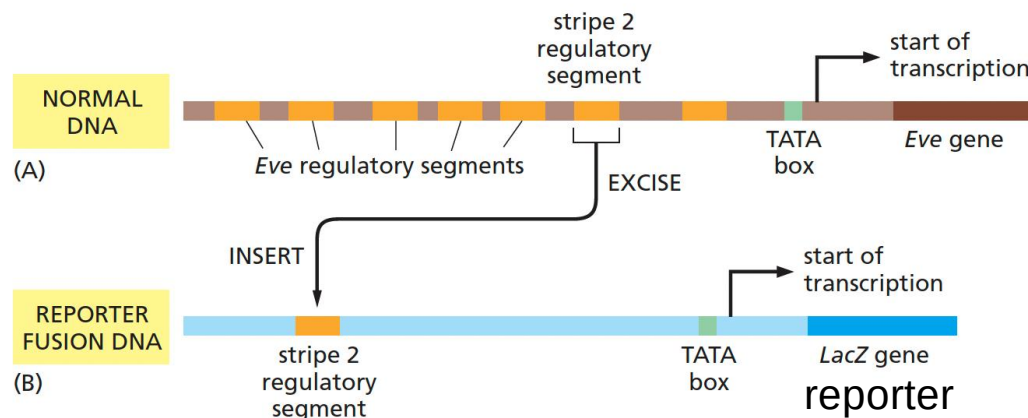
Transcription Regulation Circuits

Complex Genetic Switches That Regulate *Drosophila* Development Are Built Up from Smaller Molecules

The non-uniform distribution of transcription regulators in an early *Drosophila* embryo



The modular construction of the *Eve* gene regulatory region



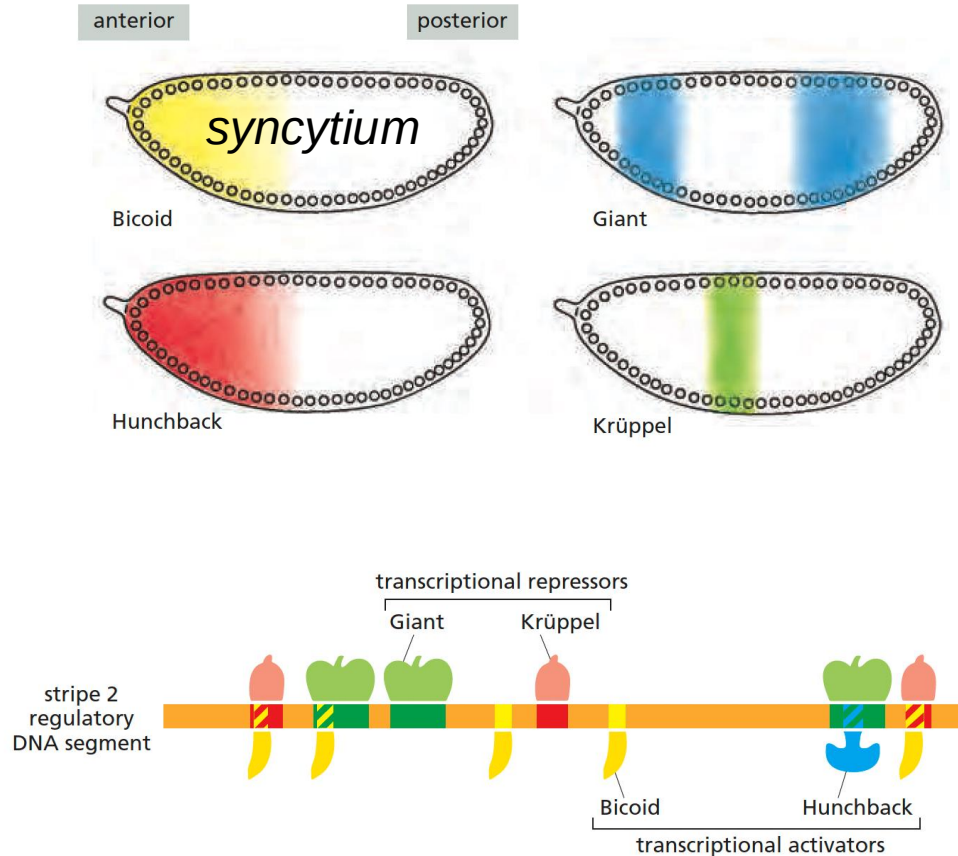
(C)



(D)

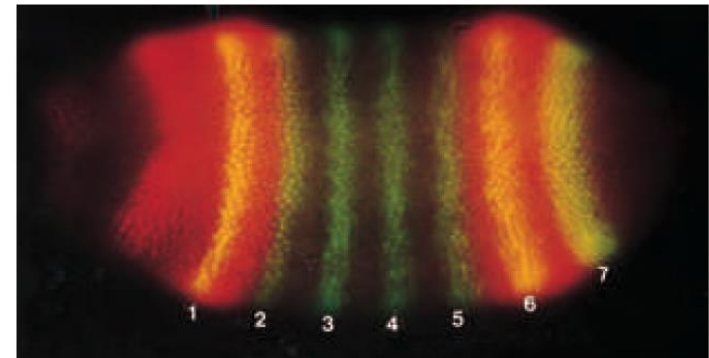
The Drosophila Eve Gene Is Regulated by Combinatorial Controls

The non-uniform distribution of transcription regulators in an early Drosophila embryo

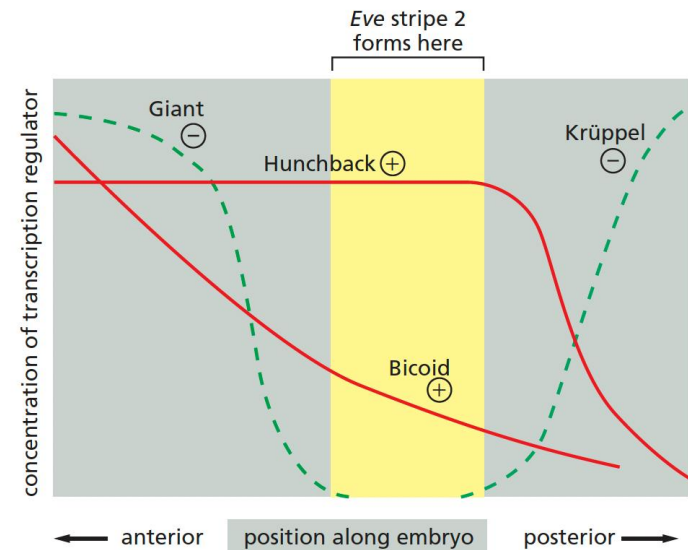


Only in those nuclei located where the levels of both Bicoid and Hunchback are high and both Krüppel and Giant are absent.

Giant protein (red)



Seven stripes of Eve protein (green)



positional information

Transcription Regulators Are Controlled in Eukaryotic Cells

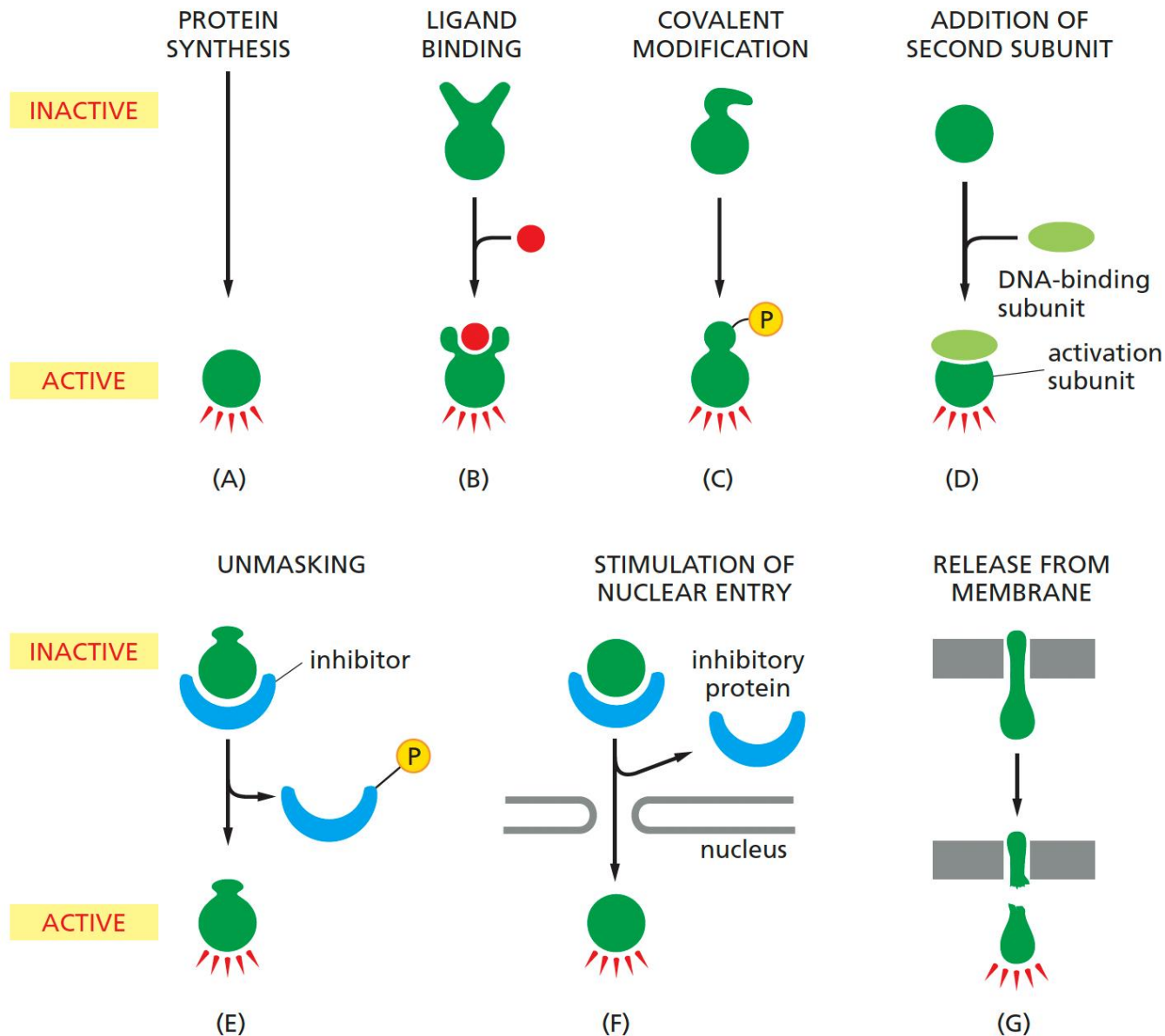
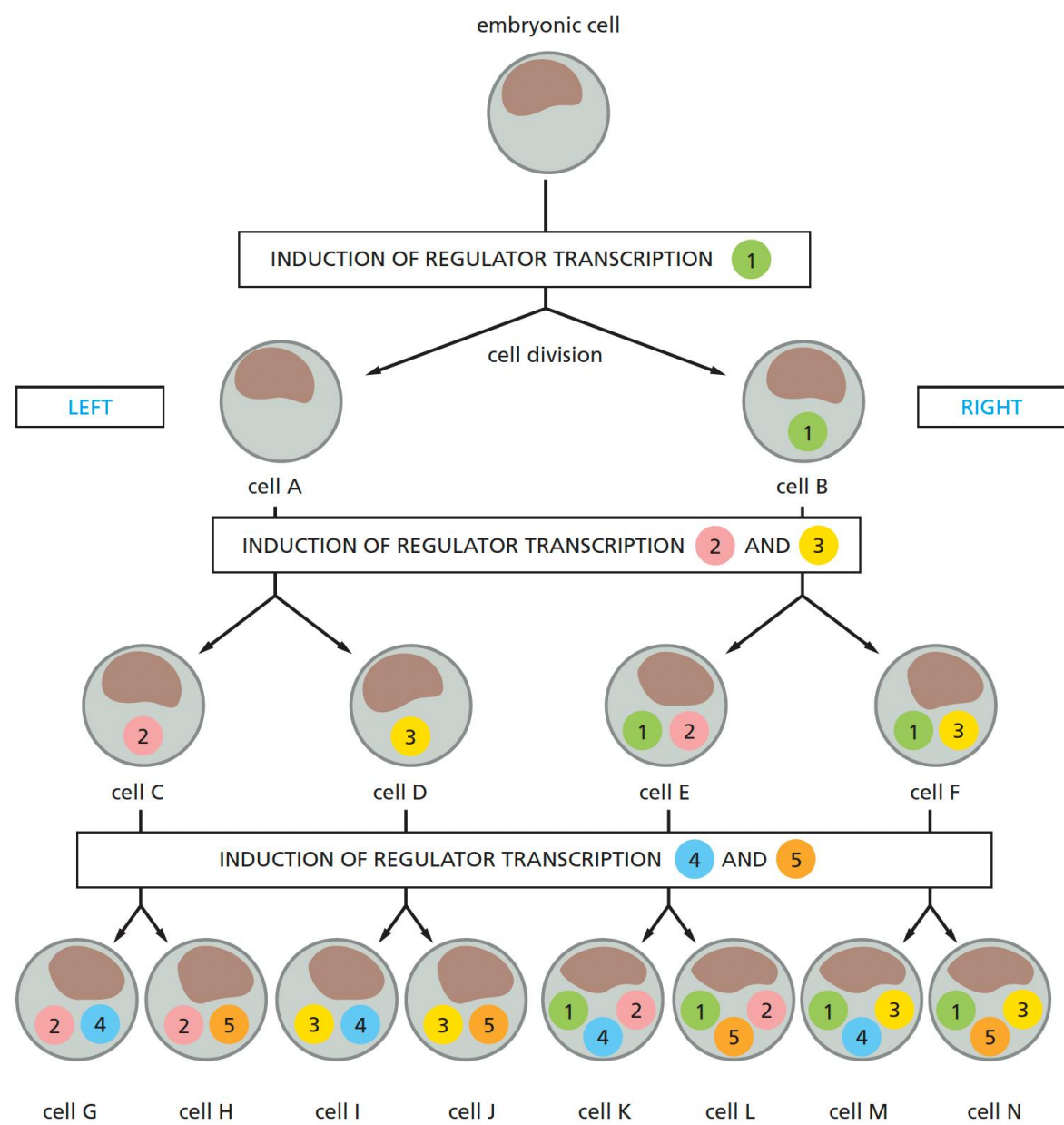


Figure 7-32 *Molecular Biology of the Cell* (© Garland Science 2015)

Combinatorial Gene Control Creates Many Different Cell Types

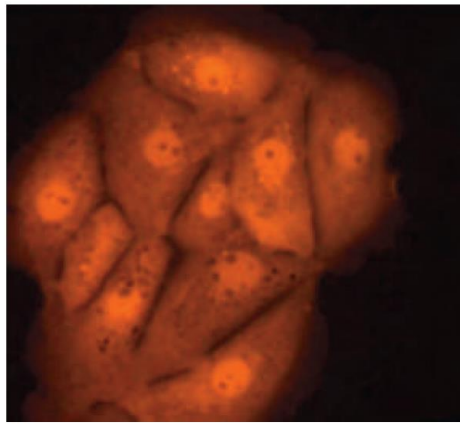


The context matters.
The history matters.

Combinatorial gene control causes the effect of adding a new transcription regulator to a cell to depend on that cell's past history, since it is this history that determines which transcription regulators are already present.

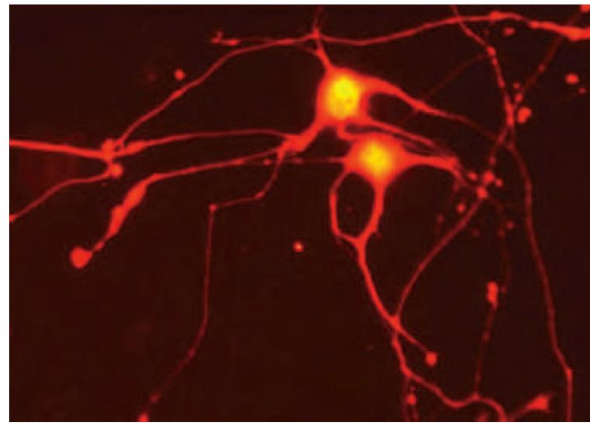
Figure 7-33 *Molecular Biology of the Cell* (© Garland Science 2015)

Transcription Regulators For The Specification of Cell Types



(A)

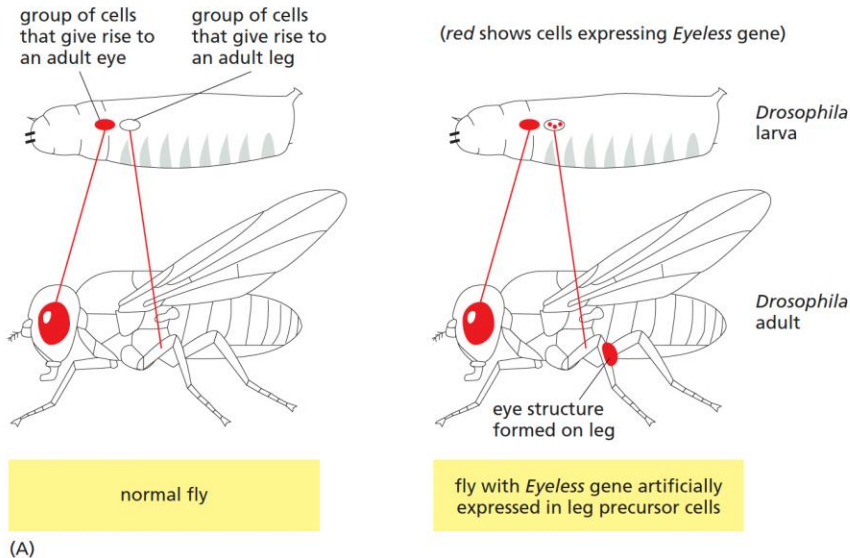
50 μ m



(B)

50 μ m

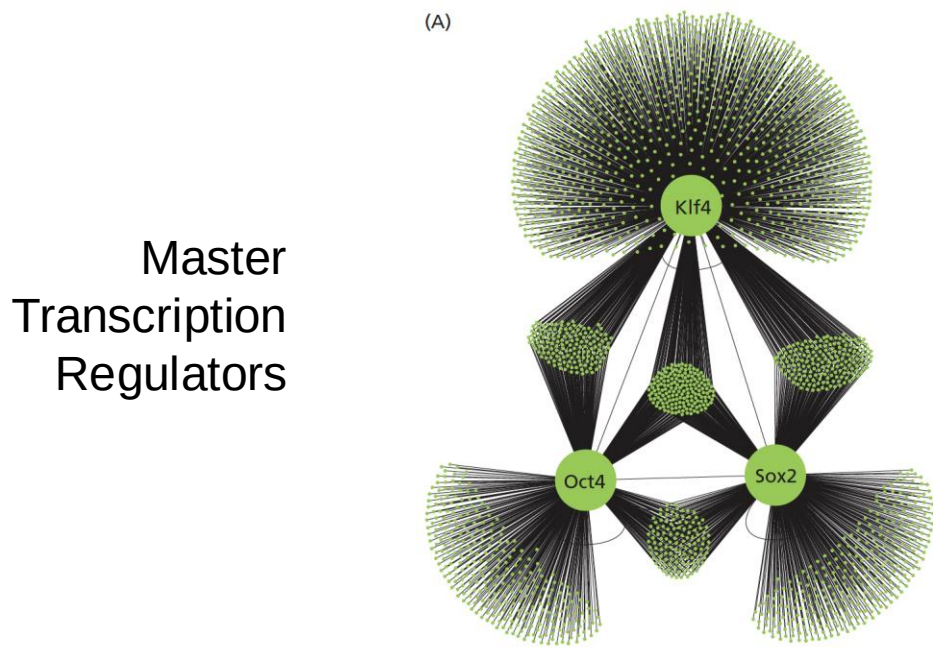
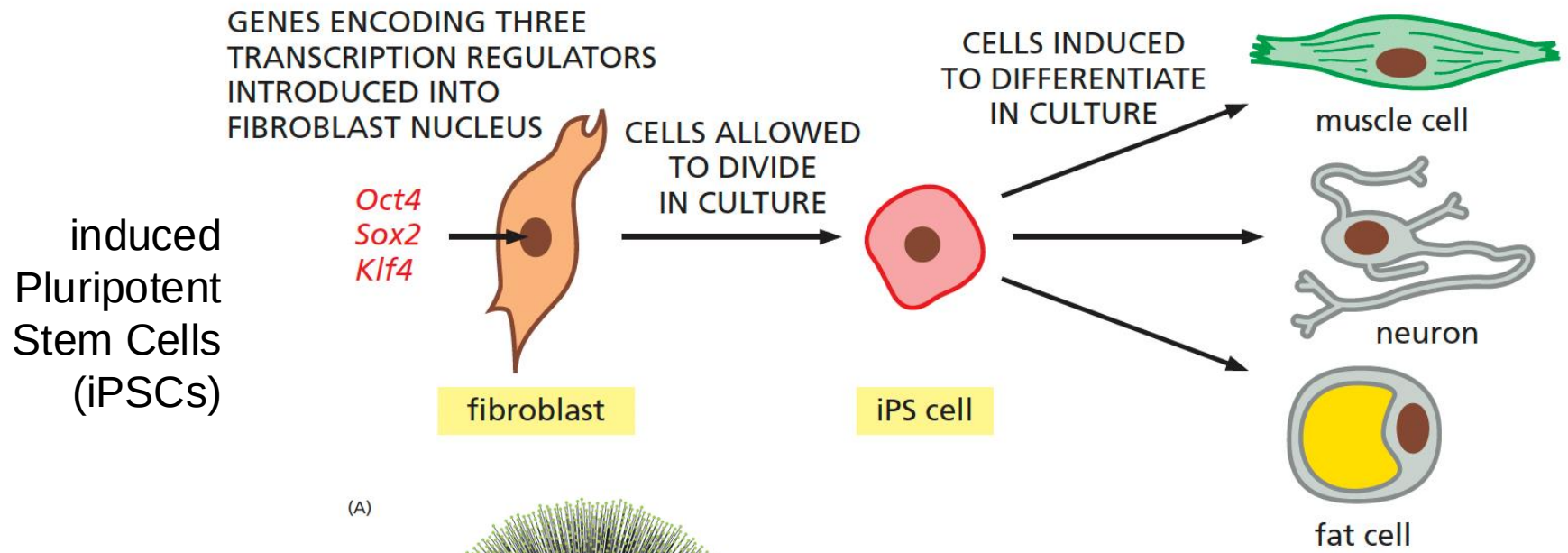
Liver cells grown in culture were converted into neuronal cells via the artificial expression of three nerve-specific transcription regulators.



(B)

Expression of the *Drosophila* *Eyeless* gene in precursor cells of the leg triggers the development of an eye on the leg.

Specialized Cell Types Can Be Experimentally Reprogrammed to Become Pluripotent Stem Cells



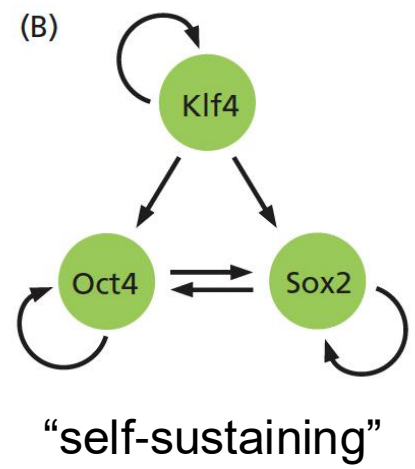
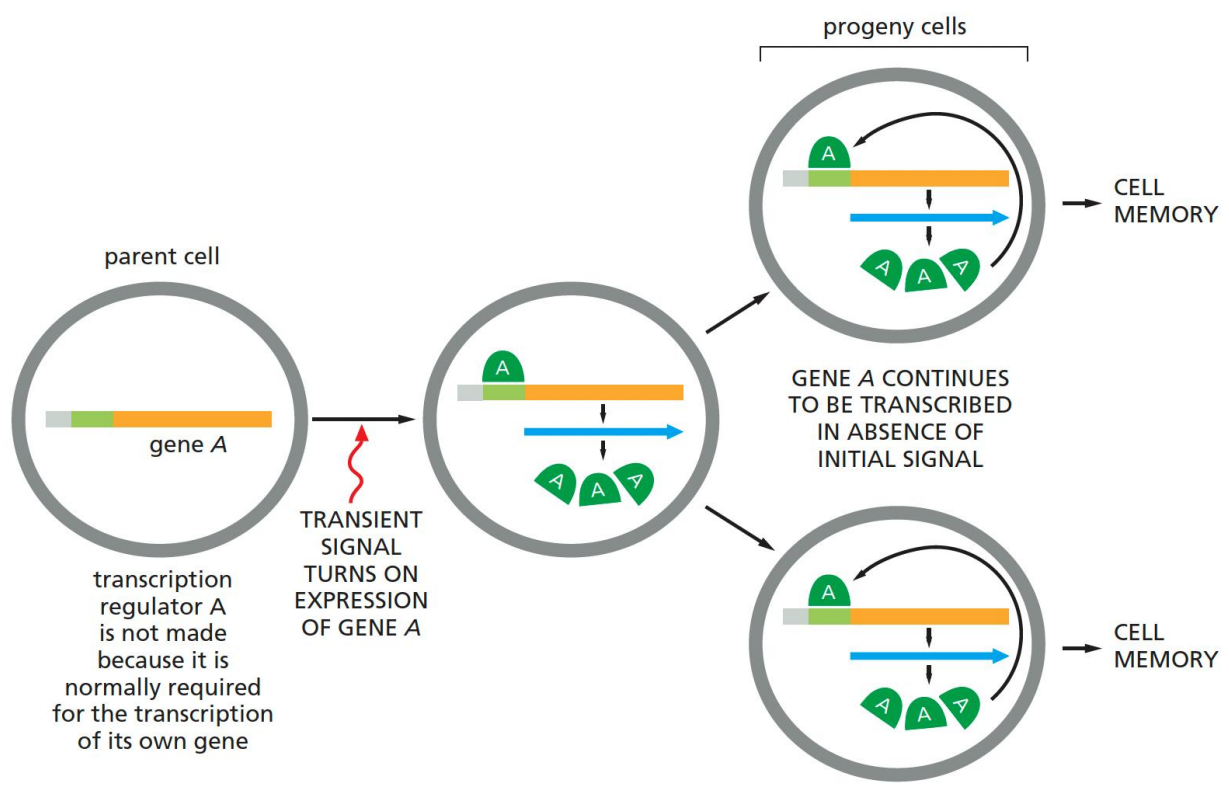
A portion of the transcription network specifying embryonic stem cells.



The Nobel Prize in Physiology or Medicine 2012

Figure 7-76, 7-37A *Molecular Biology of the Cell* (© Garland Science 2015)

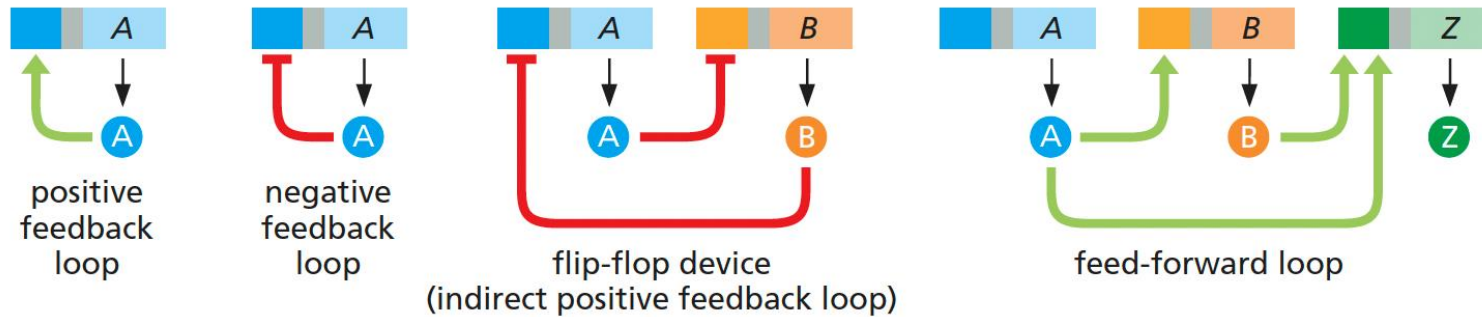
Differentiated Cells Maintain Their Identity



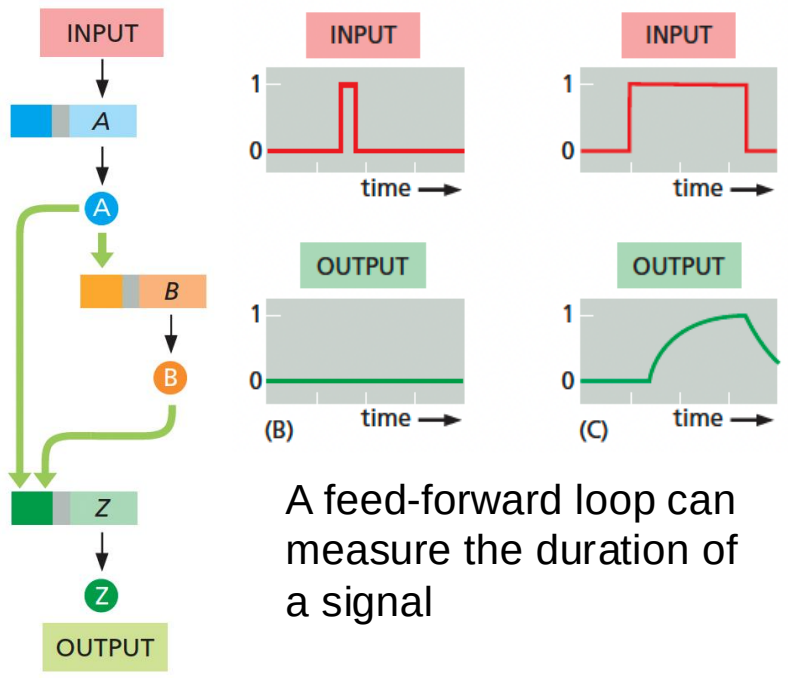
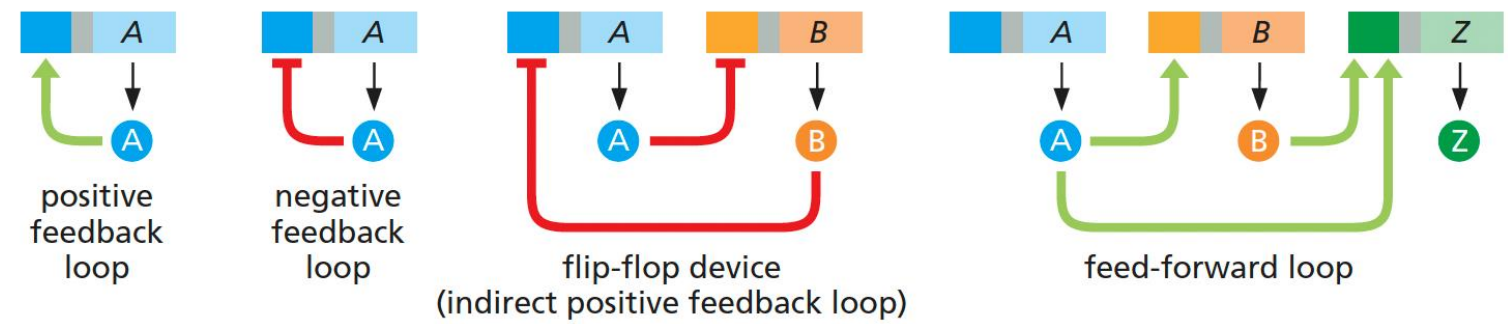
A **positive feedback** loop
can create **cell memory**

Figure 7-39, 7-37B *Molecular Biology of the Cell* (© Garland Science 2015)

Transcription Circuits Allow the Cell to Carry Out Logic Operations



Transcription Circuits Allow the Cell to Carry Out Logic Operations



A feed-forward loop can measure the duration of a signal

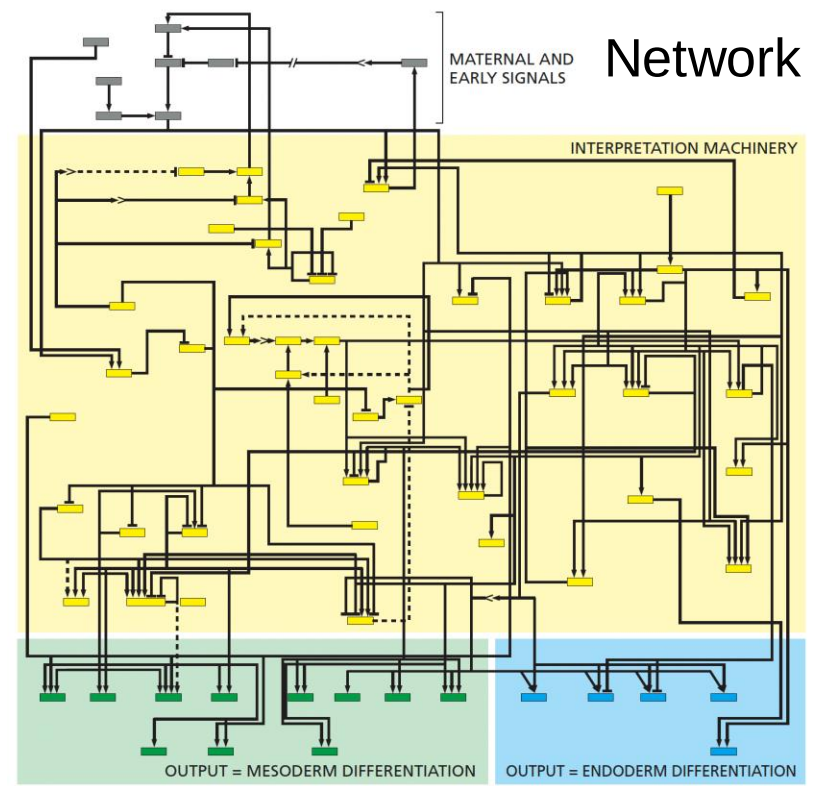


Figure 7-40, 7-41, 7-42 Molecular Biology of the Cell (© Garland Science 2015)

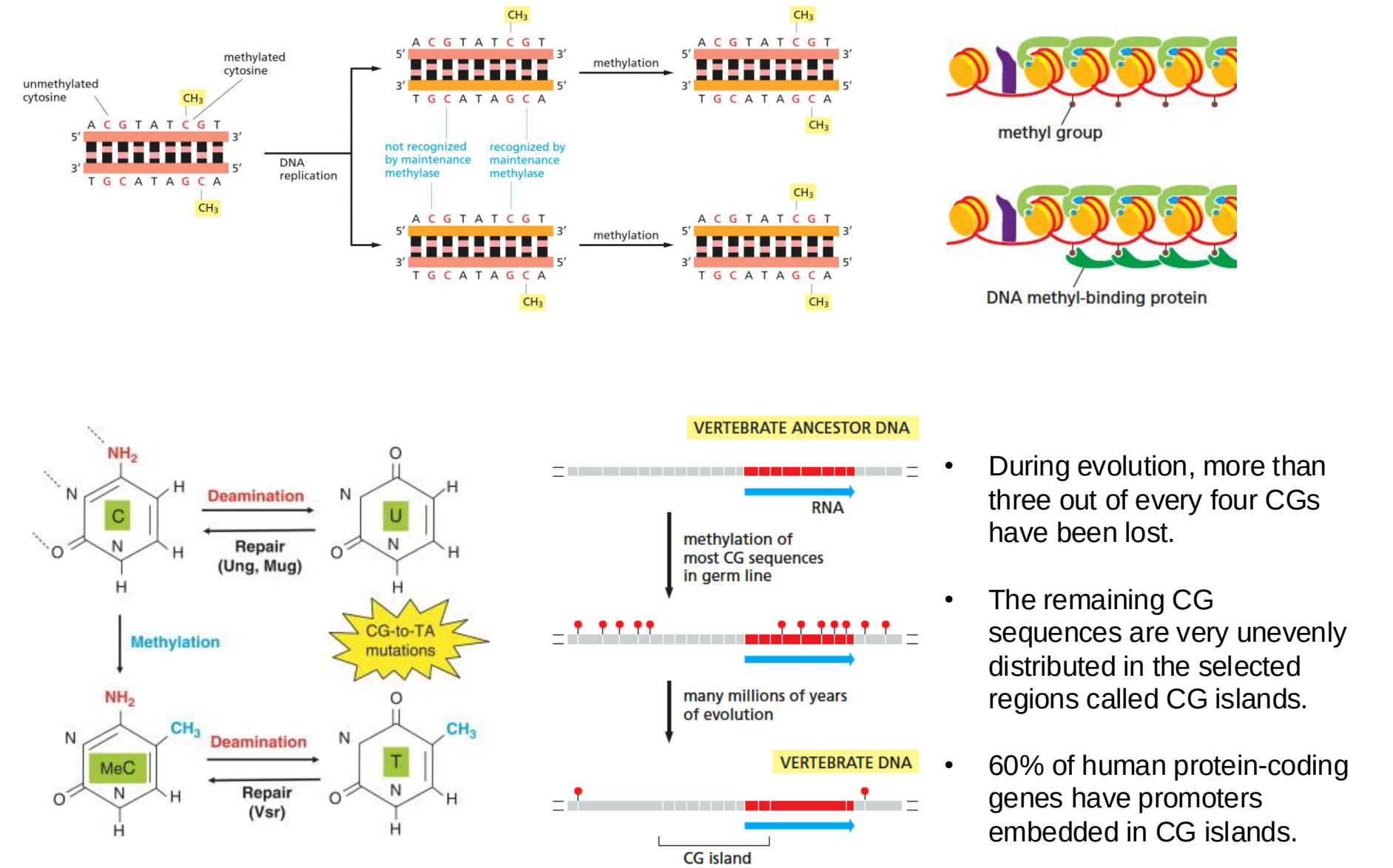
MECHANISMS THAT REINFORCE CELL MEMORY

DNA Methylation

Genomic Imprinting

X Chromosome Inactivation

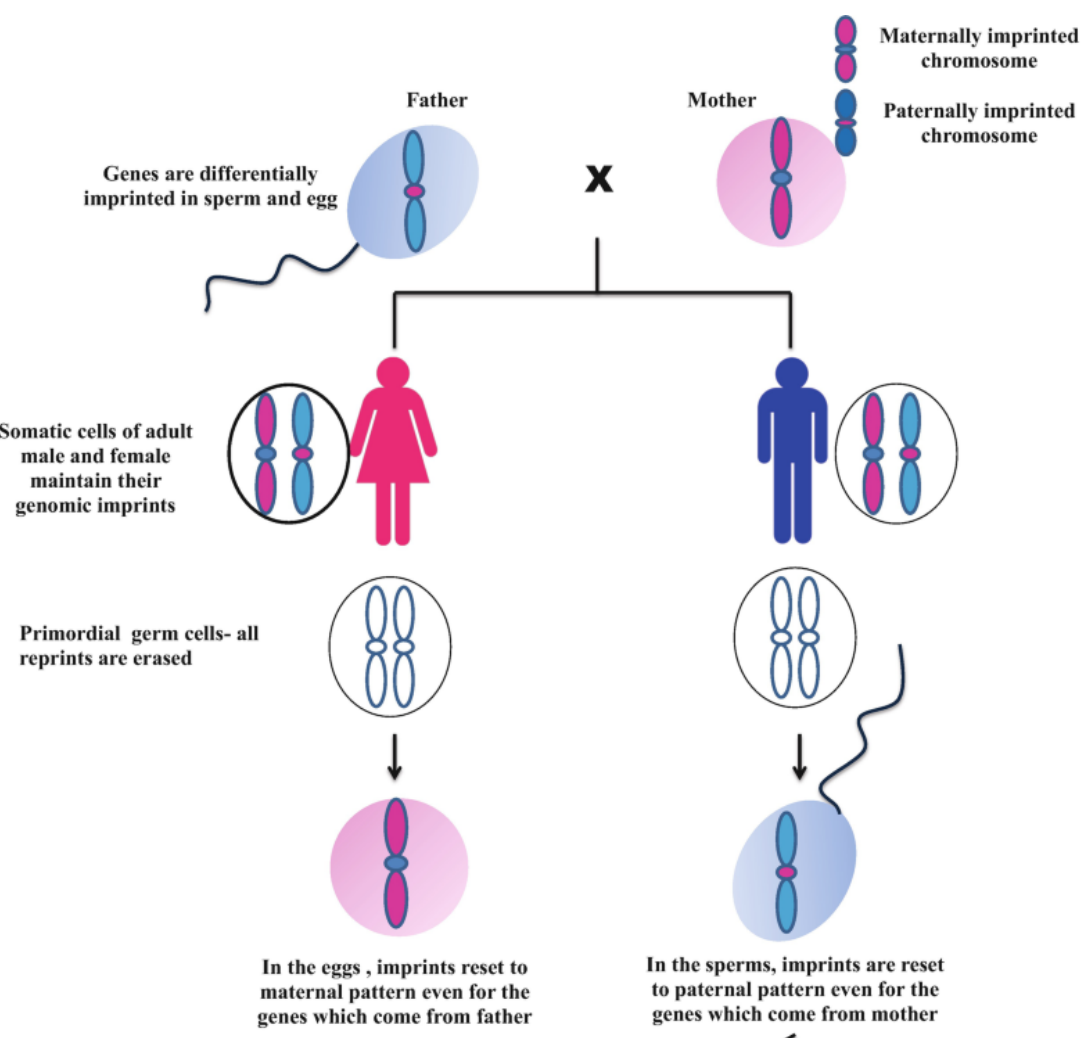
Patterns of DNA Methylation Can Be Inherited When Vertebrate Cells Divide



- During evolution, more than three out of every four CGs have been lost.
- The remaining CG sequences are very unevenly distributed in the selected regions called CG islands.
- 60% of human protein-coding genes have promoters embedded in CG islands.

Figure 7-43, 7-44, 7-47 *Molecular Biology of the Cell* (© Garland Science 2015)

Genomic Imprinting



Genomic imprinting is an epigenetic phenomenon which results in the **monoallelic expression** of a gene depending on the parental origin.

Roughly 300 genes are imprinted in humans.

Igf2 is paternally expressed

		Present	Absent
<i>Igf2r</i> is maternally expressed	Present	 Normally sized	 Dwarf
	Absent	 Overly large	 Normally sized

The Genetic Conflict Hypothesis

Mechanisms of Imprinting

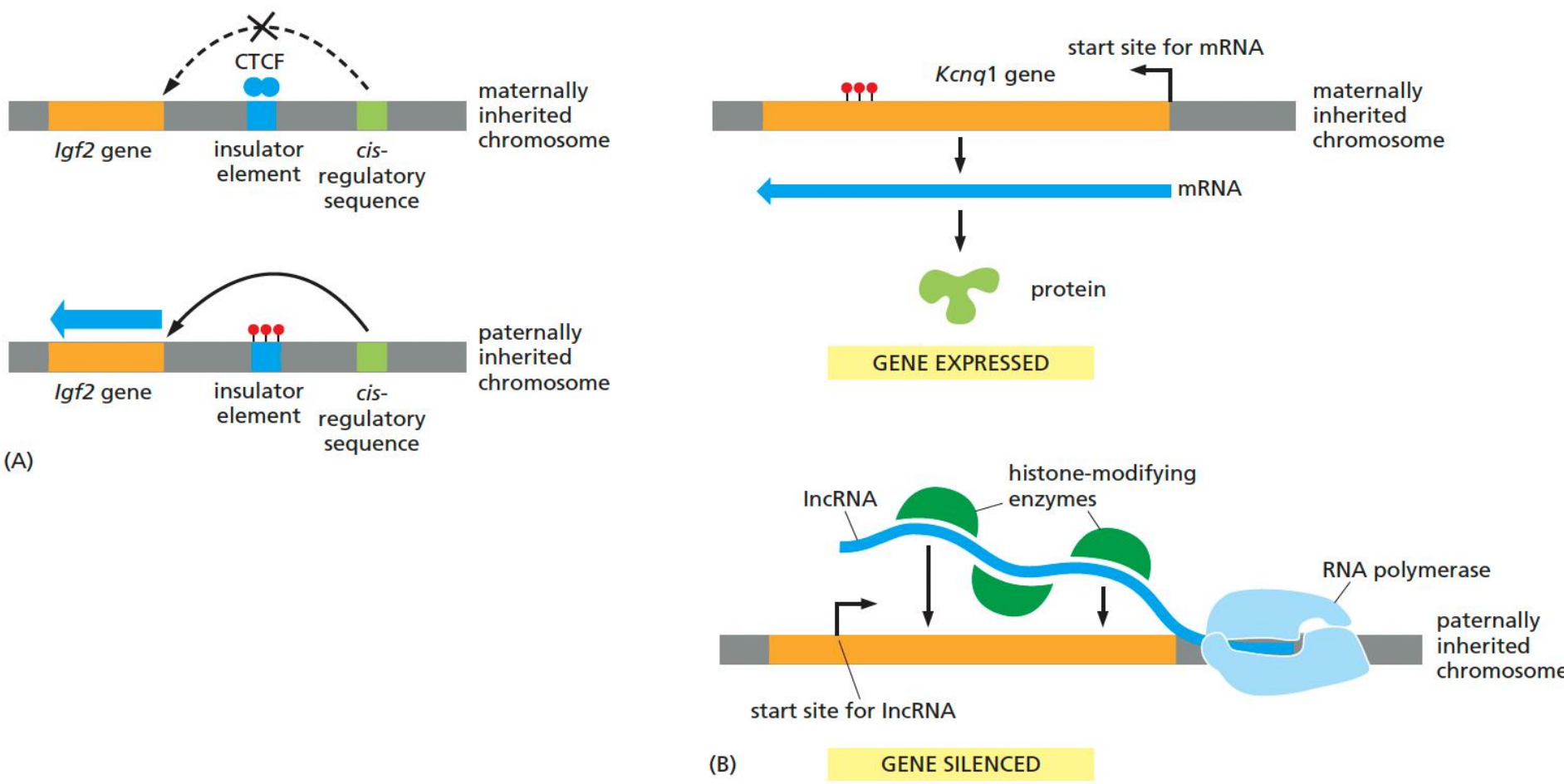
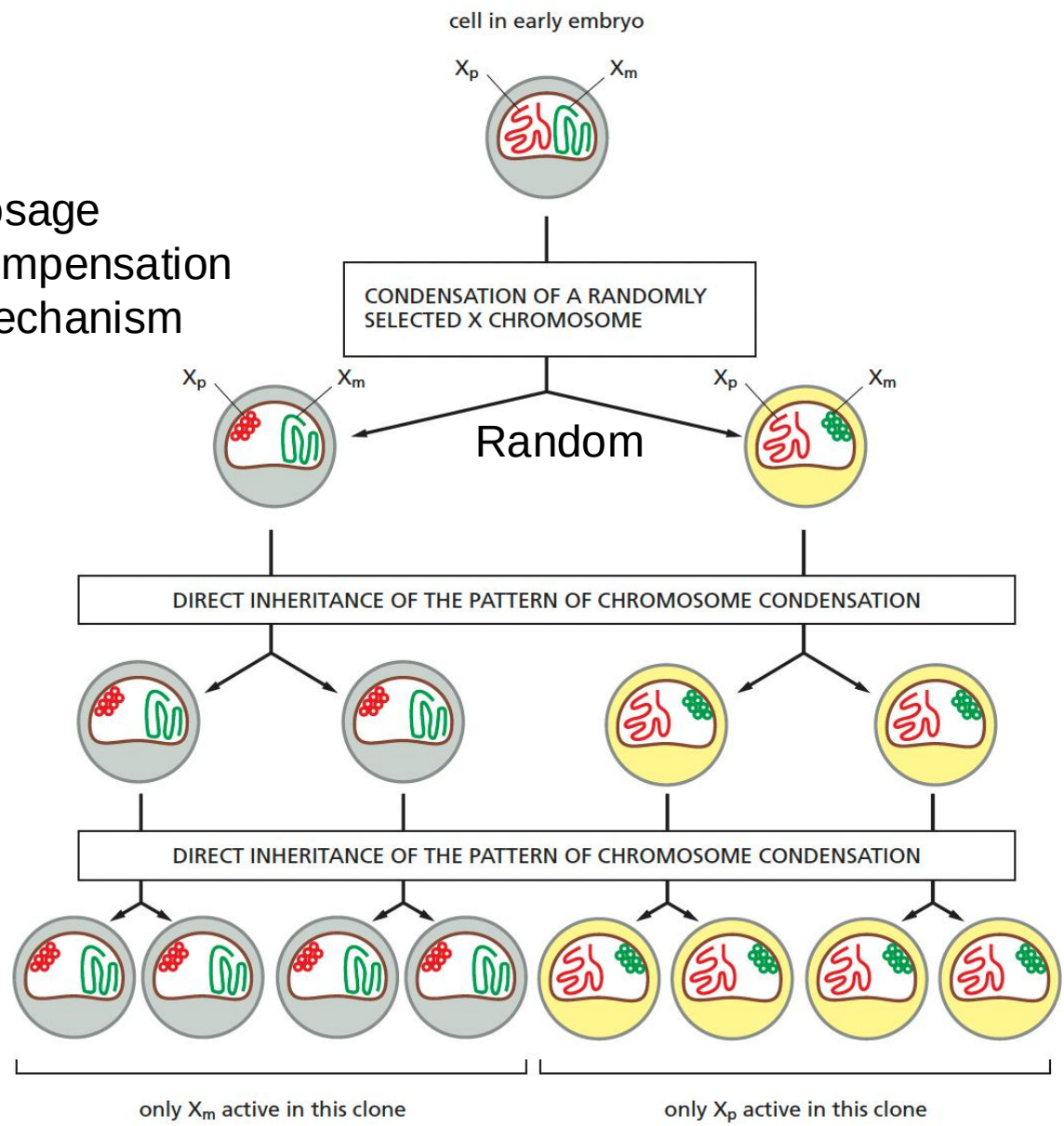


Figure 7-49 Molecular Biology of the Cell (© Garland Science 2015)

Chromosome-Wide Alterations in Chromatin Structure Can Be Inherited

dosage
compensation
mechanism



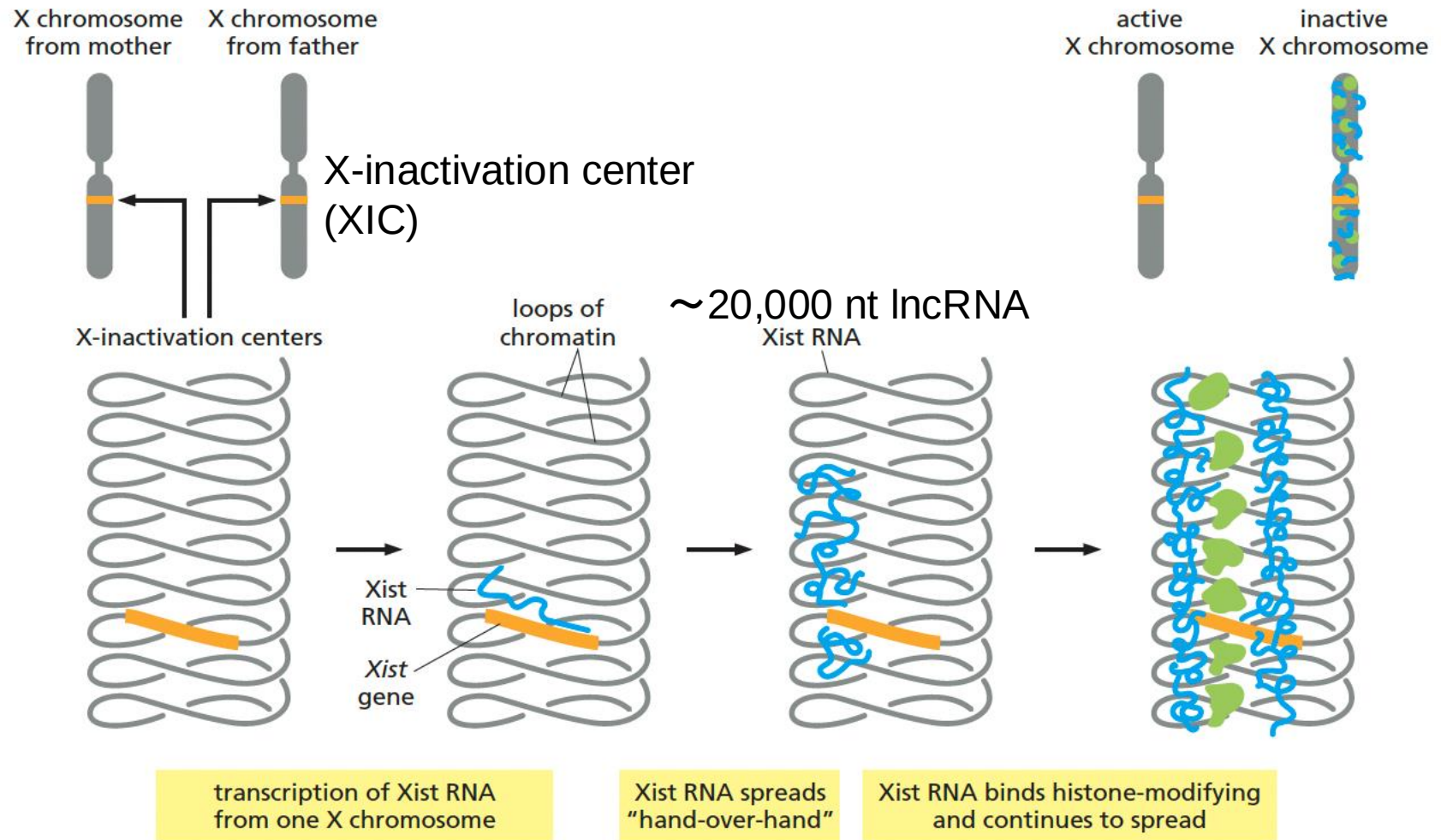
mosaics phenotype



Calico female cat

Figure 7-50 Molecular Biology of the Cell (© Garland Science 2015)

Mammalian X-chromosome Inactivation



About 10% of the genes on the X chromosome (including Xist itself) escape this silencing and remain active.

POST-TRANSCRIPTIONAL CONTROLS

Alternative RNA Splicing

3' Cleavage

RNA Editing

UTR Control

mRNA Decay

Post-transcriptional Controls of Gene Expression

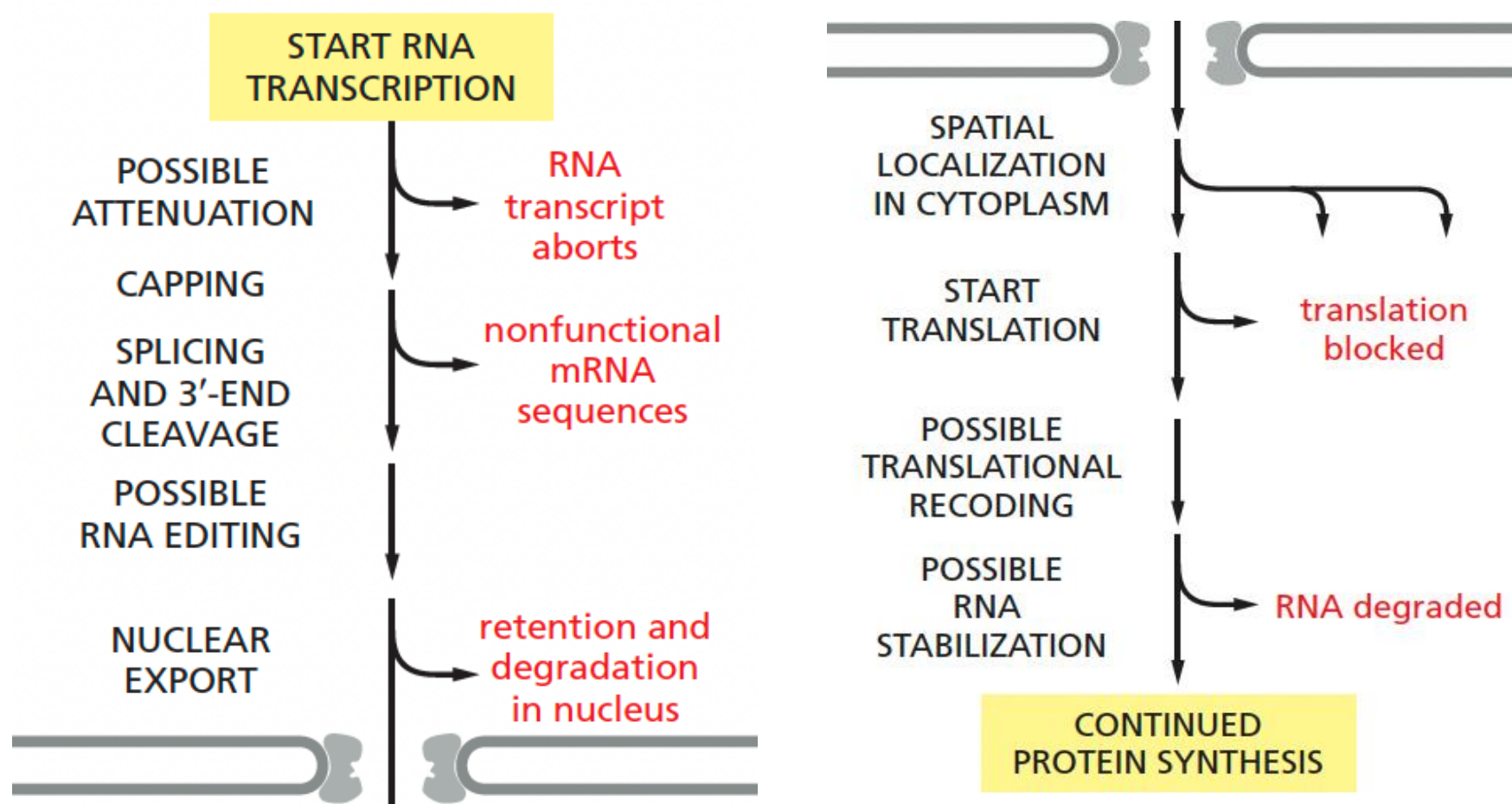
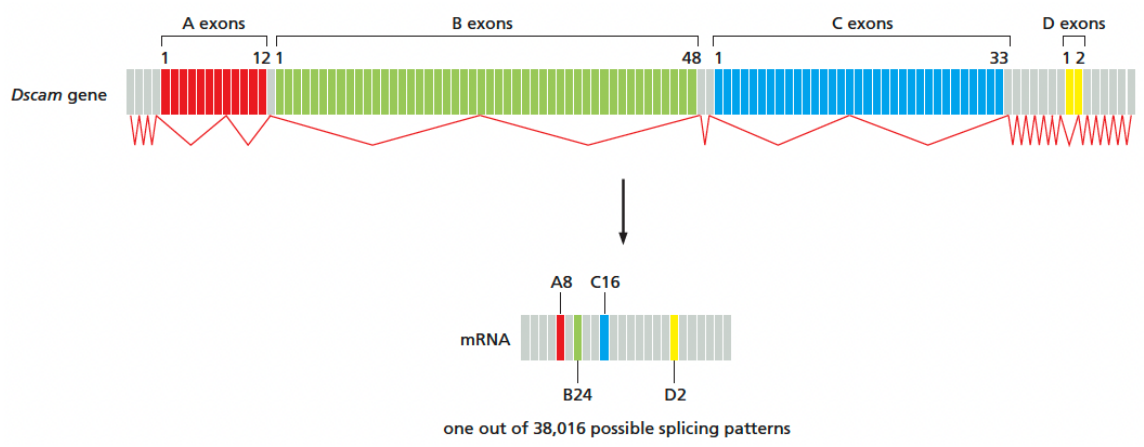
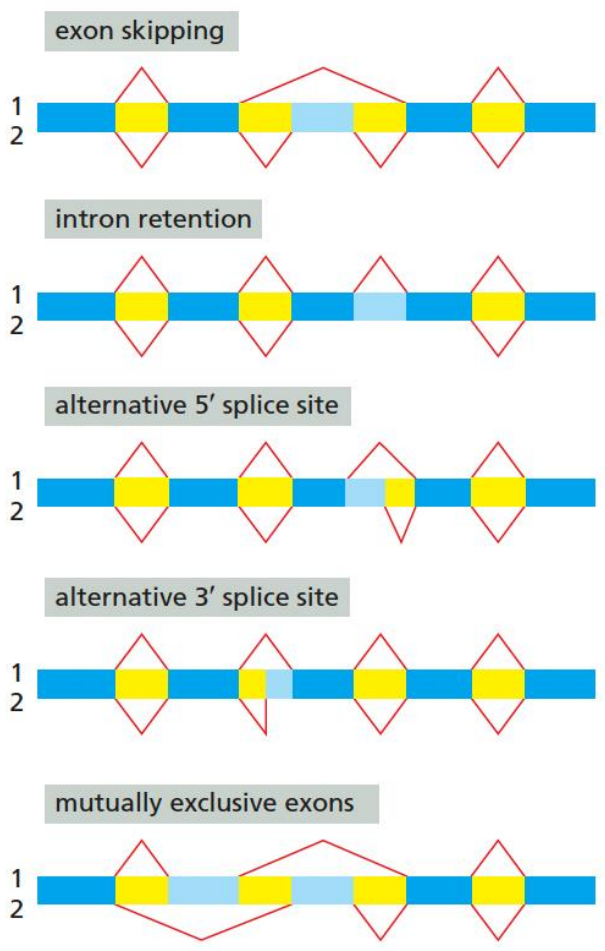


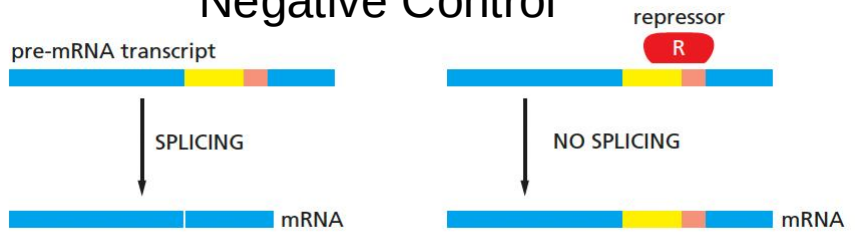
Figure 7-54 *Molecular Biology of the Cell* (© Garland Science 2015)

Negative And Positive Control of Alternative RNA Splicing

Alternative RNA Splicing



Negative Control



Positive Control

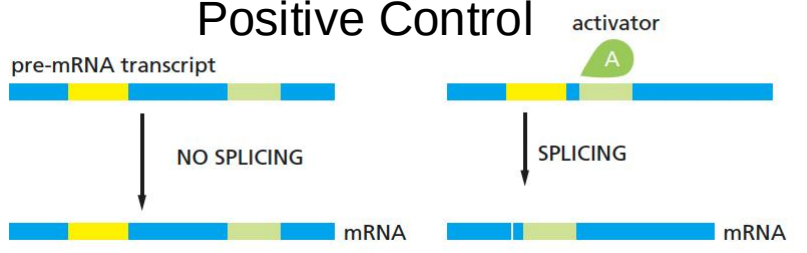
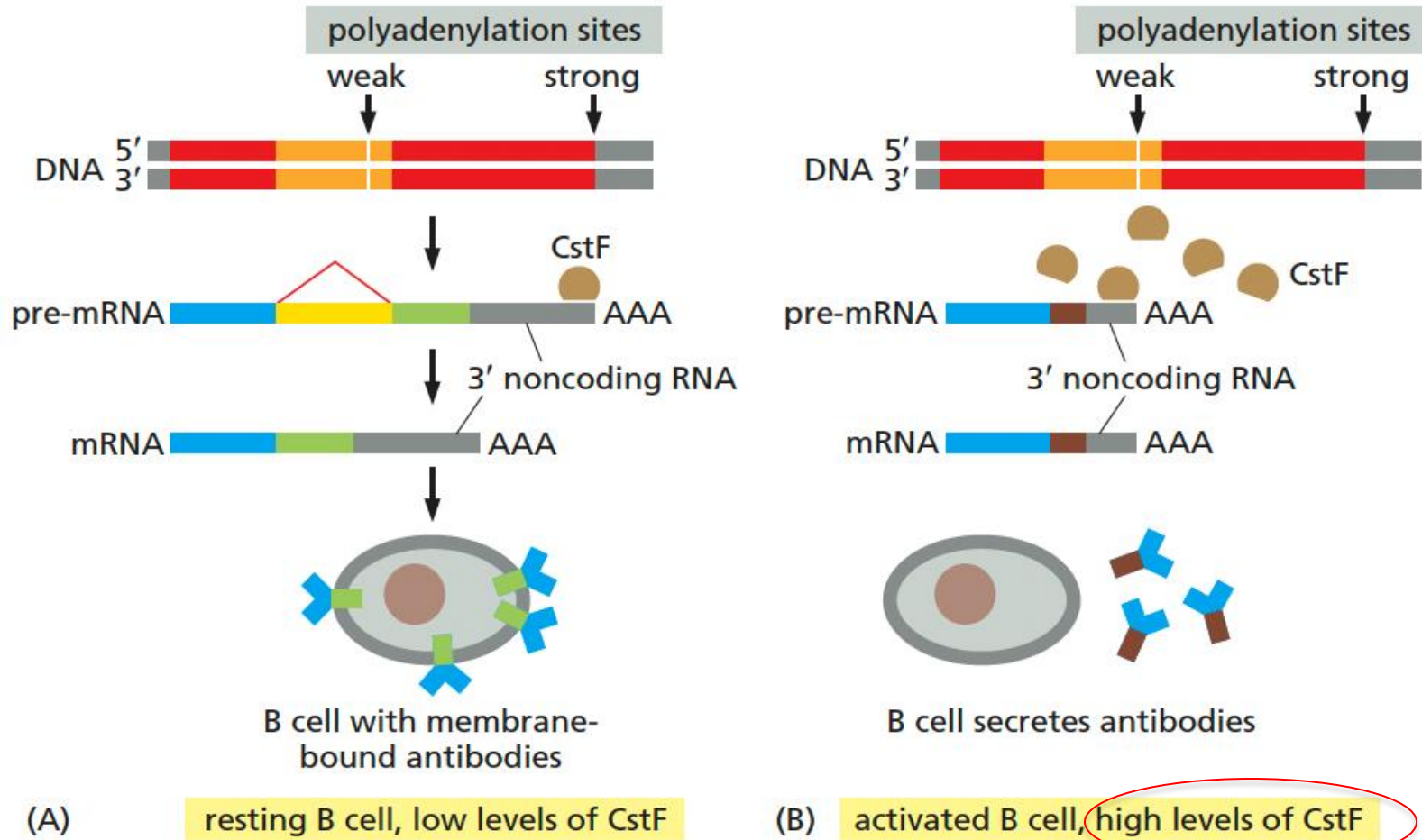


Figure 7-56, 7-58 *Molecular Biology of the Cell* (© Garland Science 2015)

A Change in the Site of RNA Transcript Cleavage and Poly-A Addition Can Change the C-terminus of a Protein

alternative cleavage and polyadenylation sites



RNA Editing Can Change the Meaning of the RNA Message

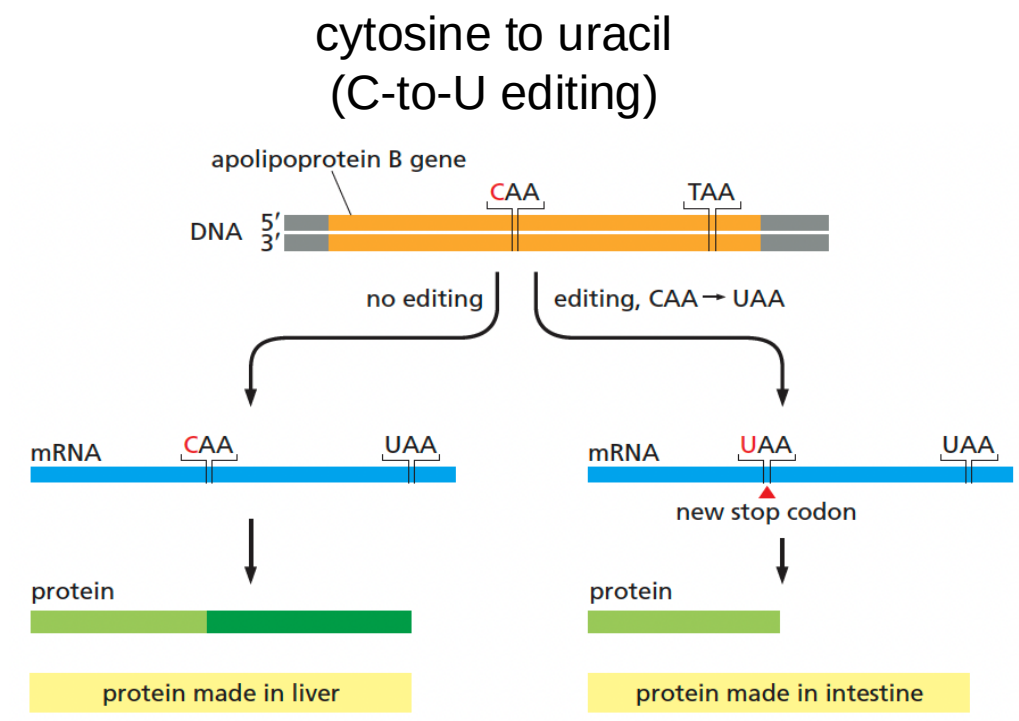
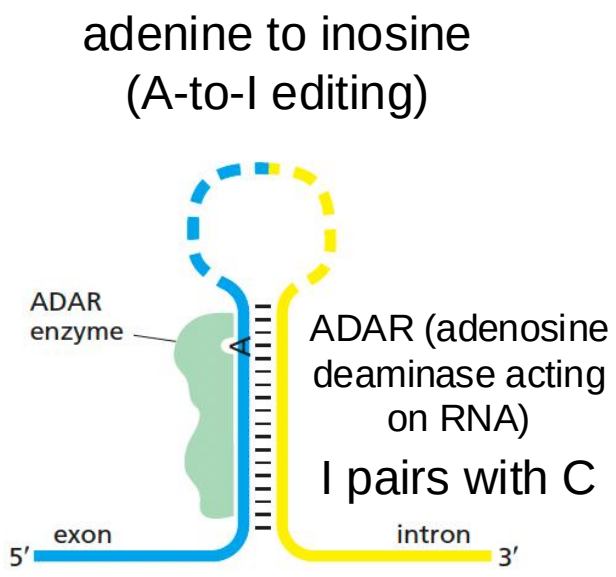
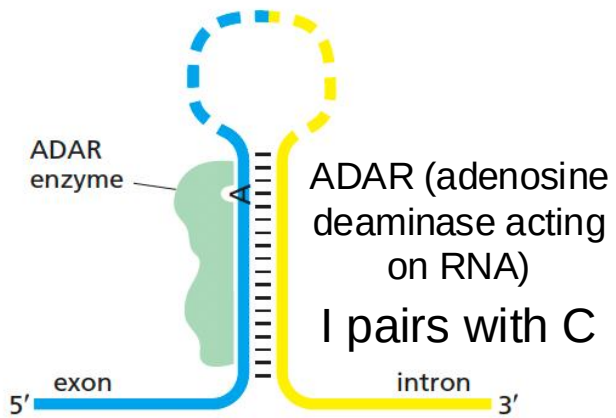


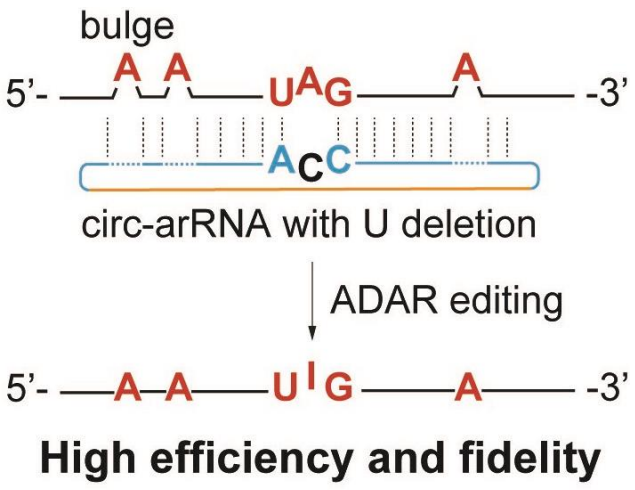
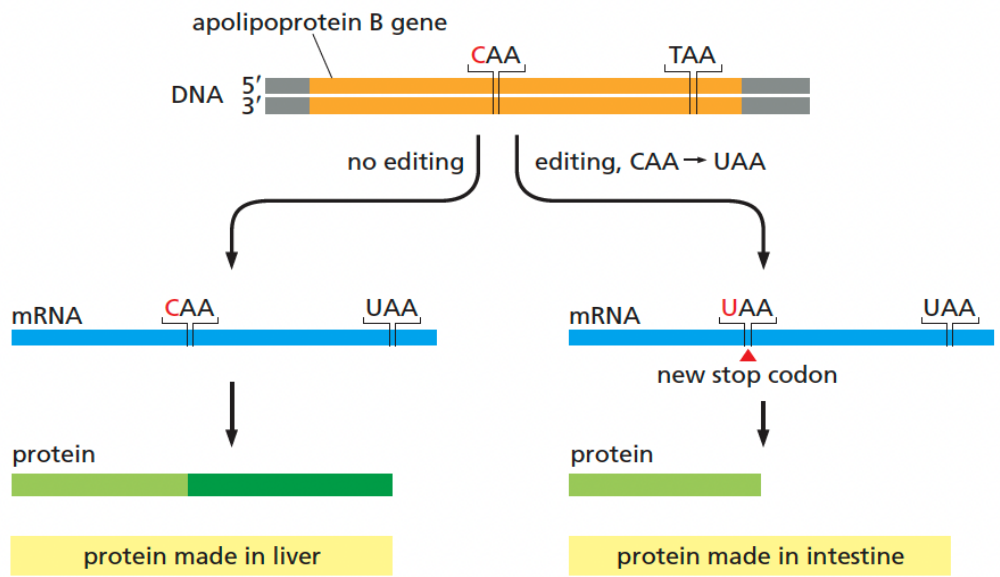
Figure 7-60, 7-61 *Molecular Biology of the Cell* (© Garland Science 2015)

RNA Editing Can Change the Meaning of the RNA Message

adenine to inosine
(A-to-I editing)



cytosine to uracil
(C-to-U editing)



Leveraging Endogenous ADAR
Protein For Targeted RNA Editing

Untranslated Regions of mRNAs Control Their Translation

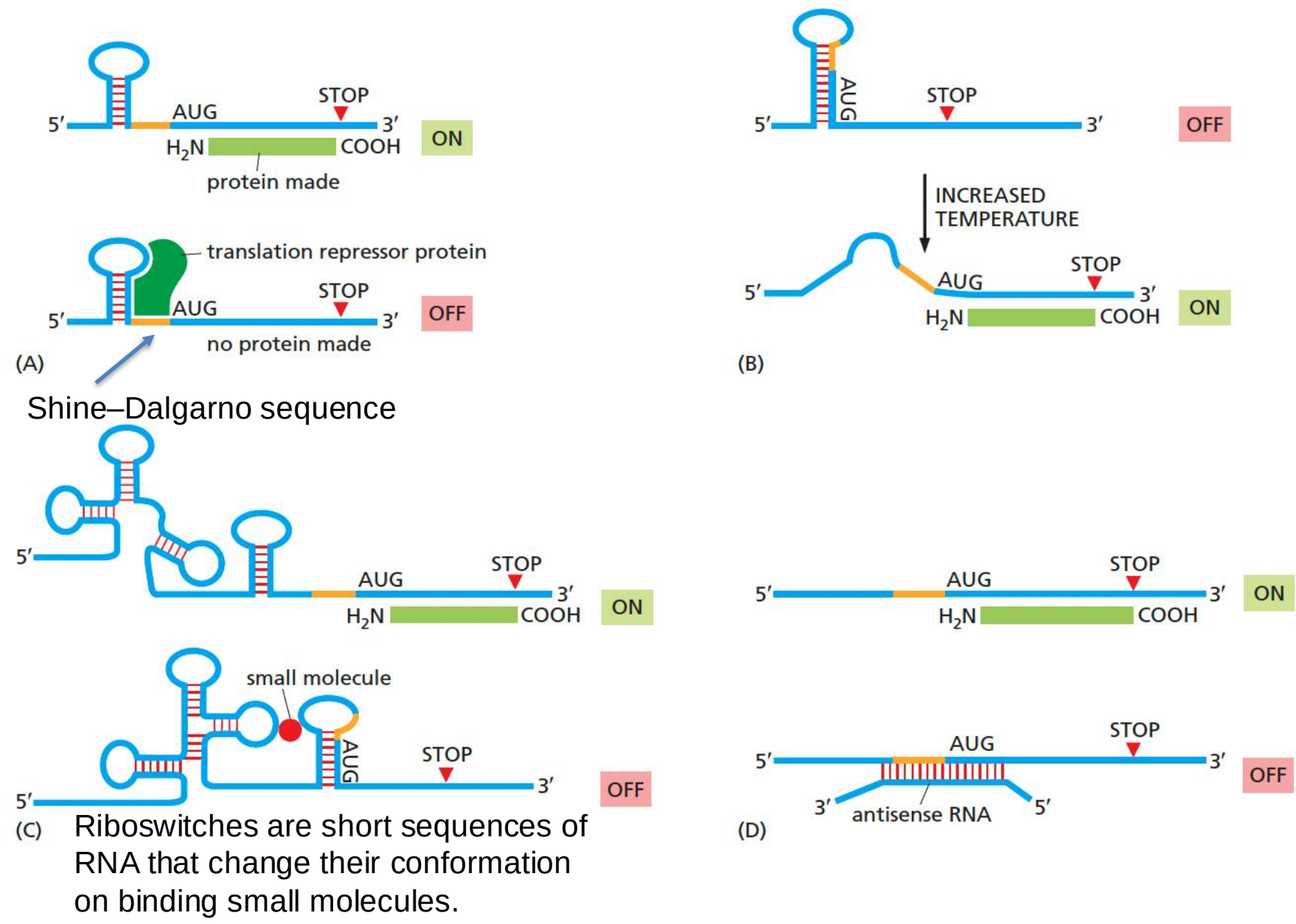
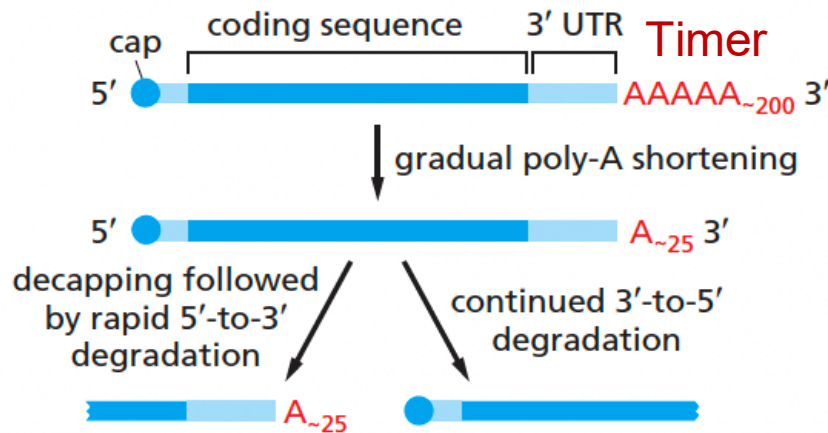


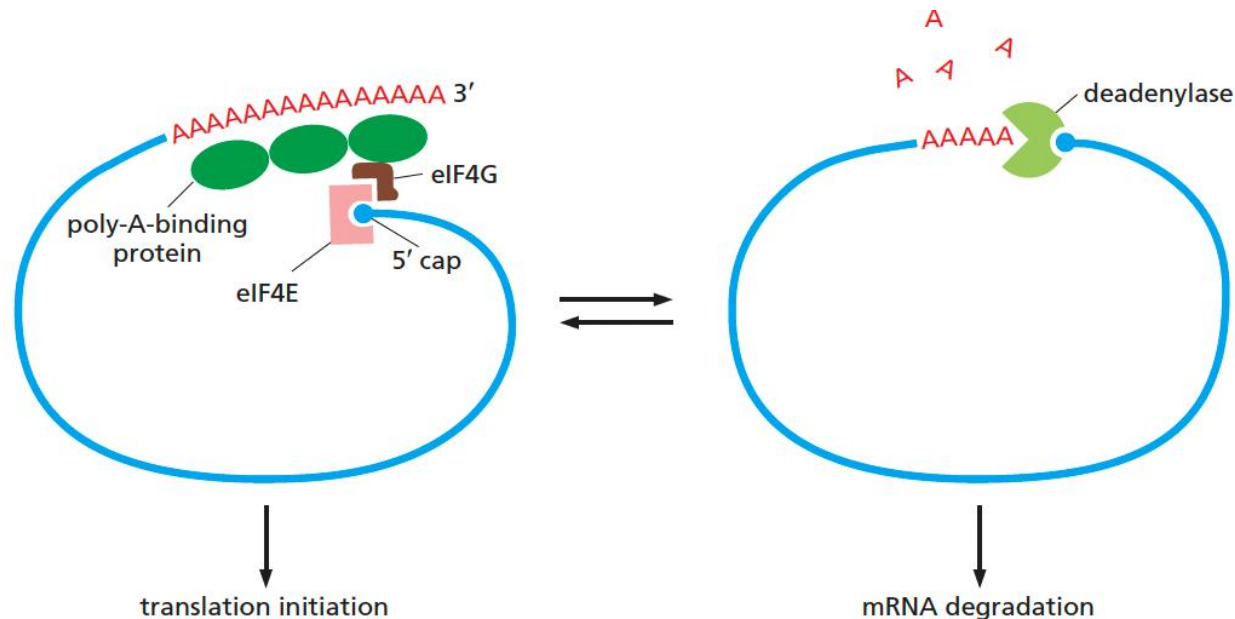
Figure 7-66 *Molecular Biology of the Cell* (© Garland Science 2015)

Changes in mRNA Stability Can Regulate Gene Expression



Eukaryotic mRNA decay, most have considerably shorter half-lives, typically less than 30 minutes.

The competition between mRNA translation and mRNA decay.



REGULATION OF GENE EXPRESSION BY NONCODING RNAs

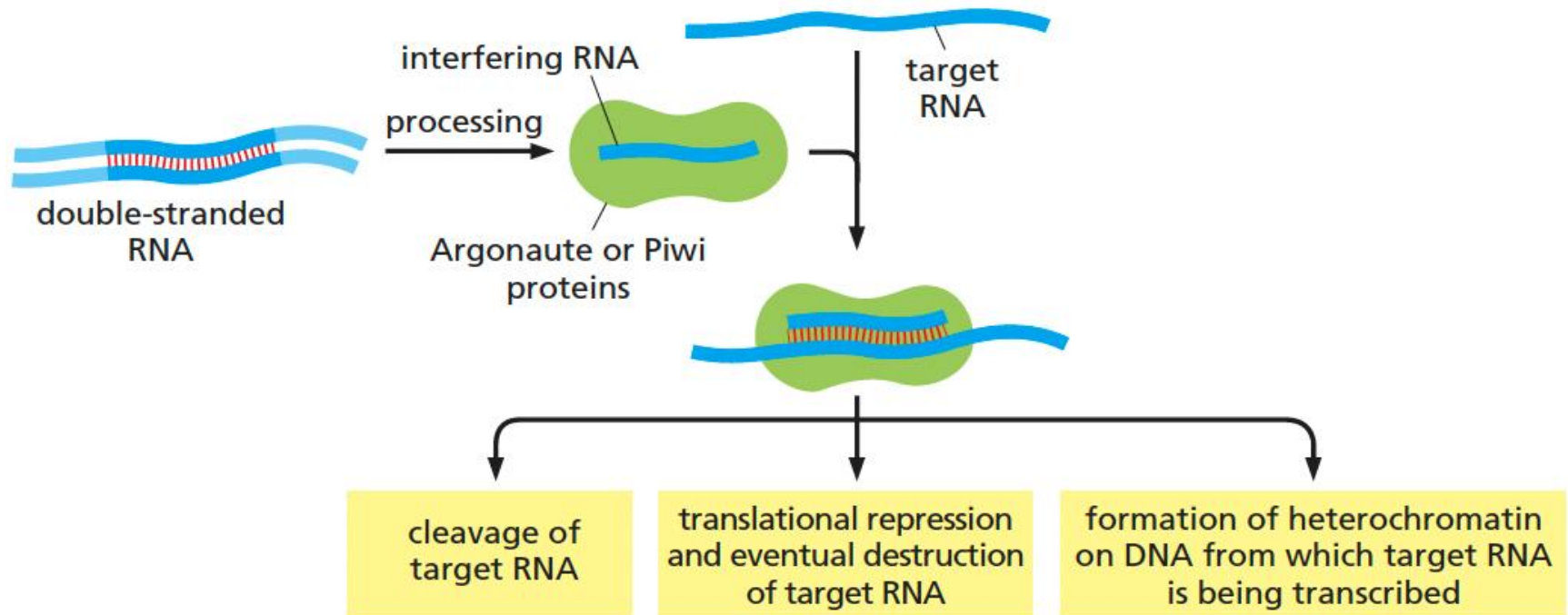
Micro RNAs (miRNAs)

Small interfering RNAs (siRNAs)

CRISPR

Long non-coding RNAs (lncRNAs)

RNA Interference In Eukaryotes



RNA Interference: **short single-stranded RNAs** (20–30 nucleotides) serve as guide RNAs that selectively reorganize and bind—through **base-pairing**—other RNAs in the cell. When the target is a mature mRNA, the small noncoding RNAs can inhibit its translation or even catalyze its destruction.

miRNAs Regulate mRNA Translation and Stability

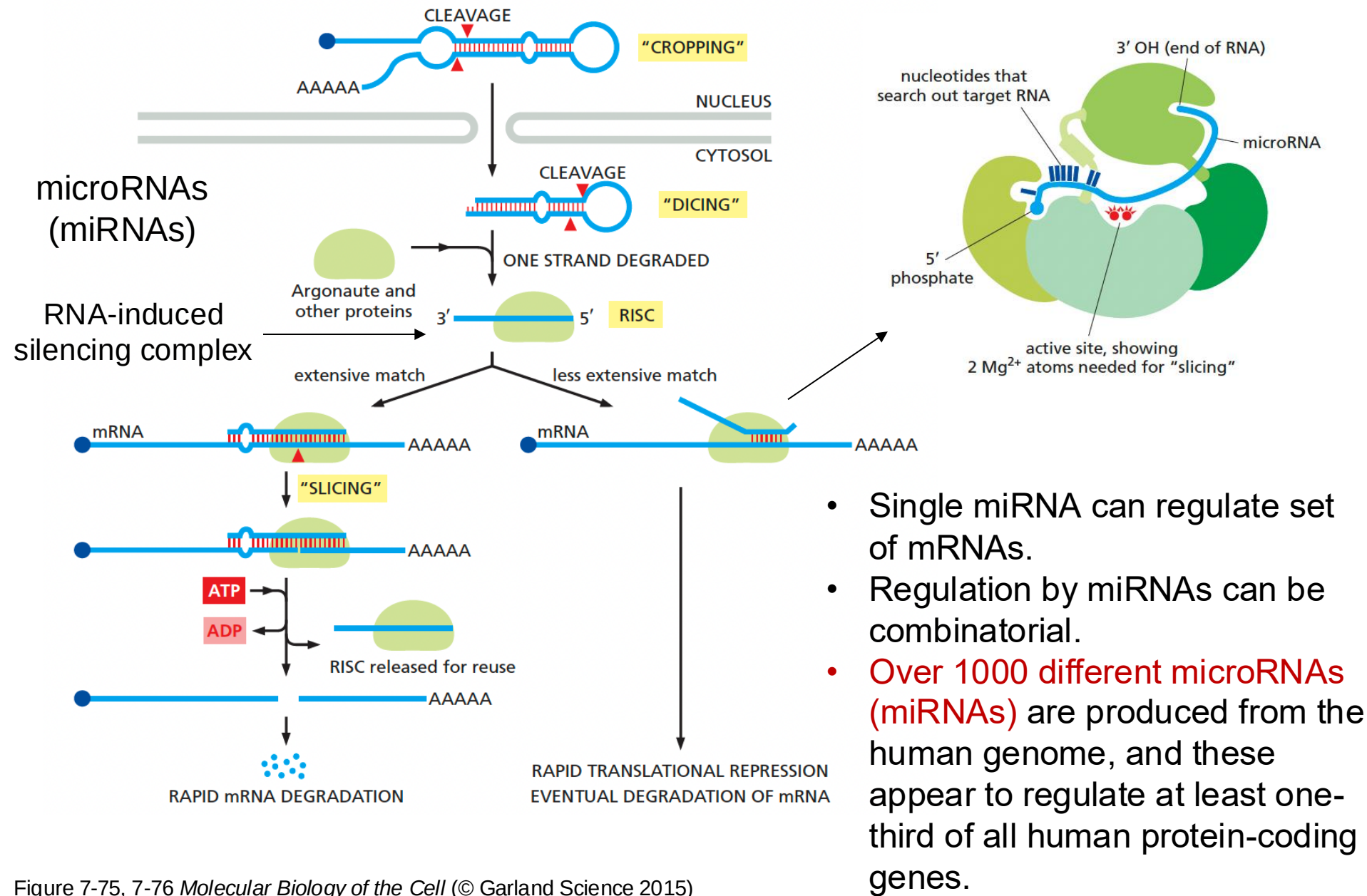
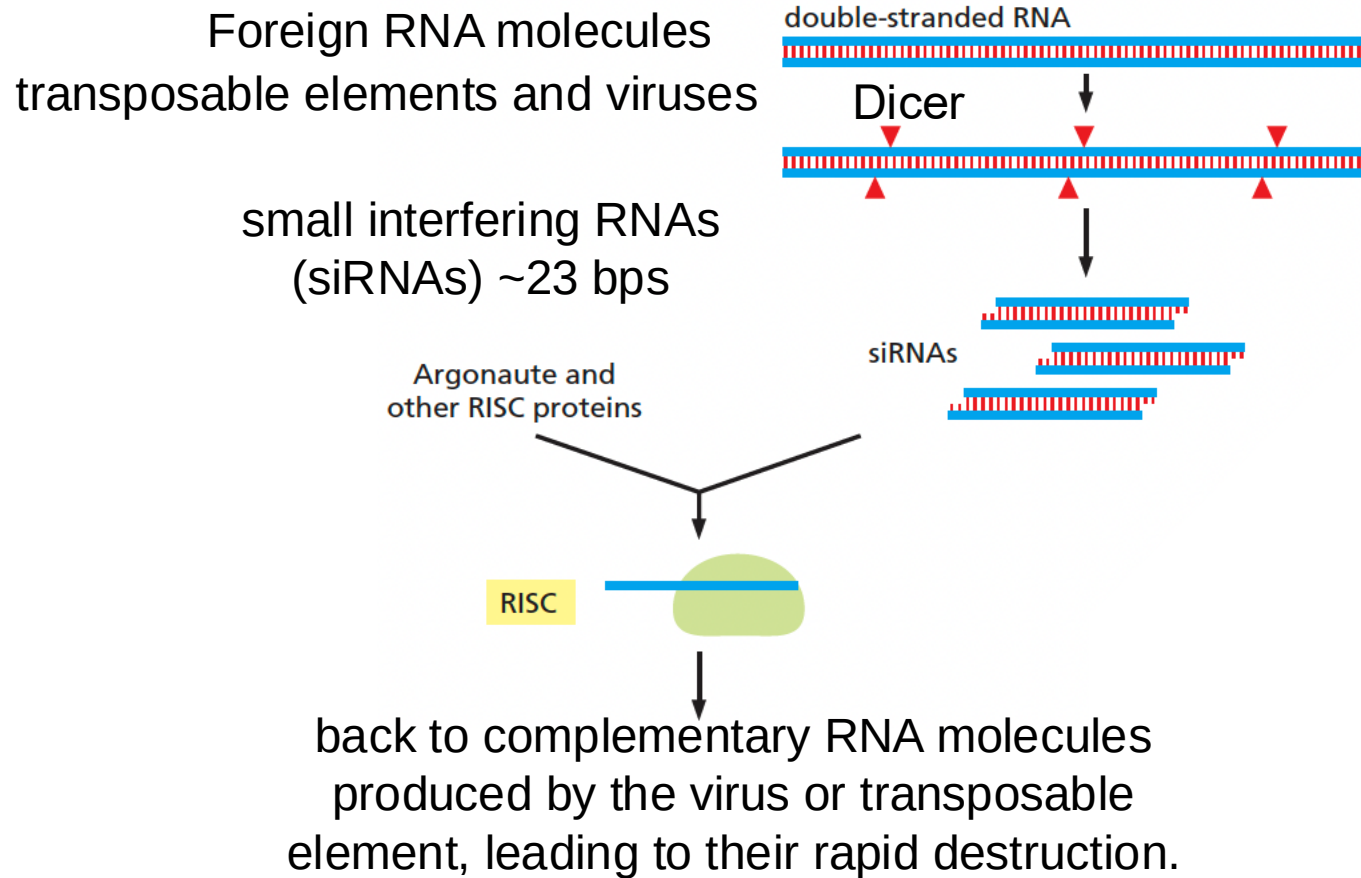


Figure 7-75, 7-76 *Molecular Biology of the Cell* (© Garland Science 2015)

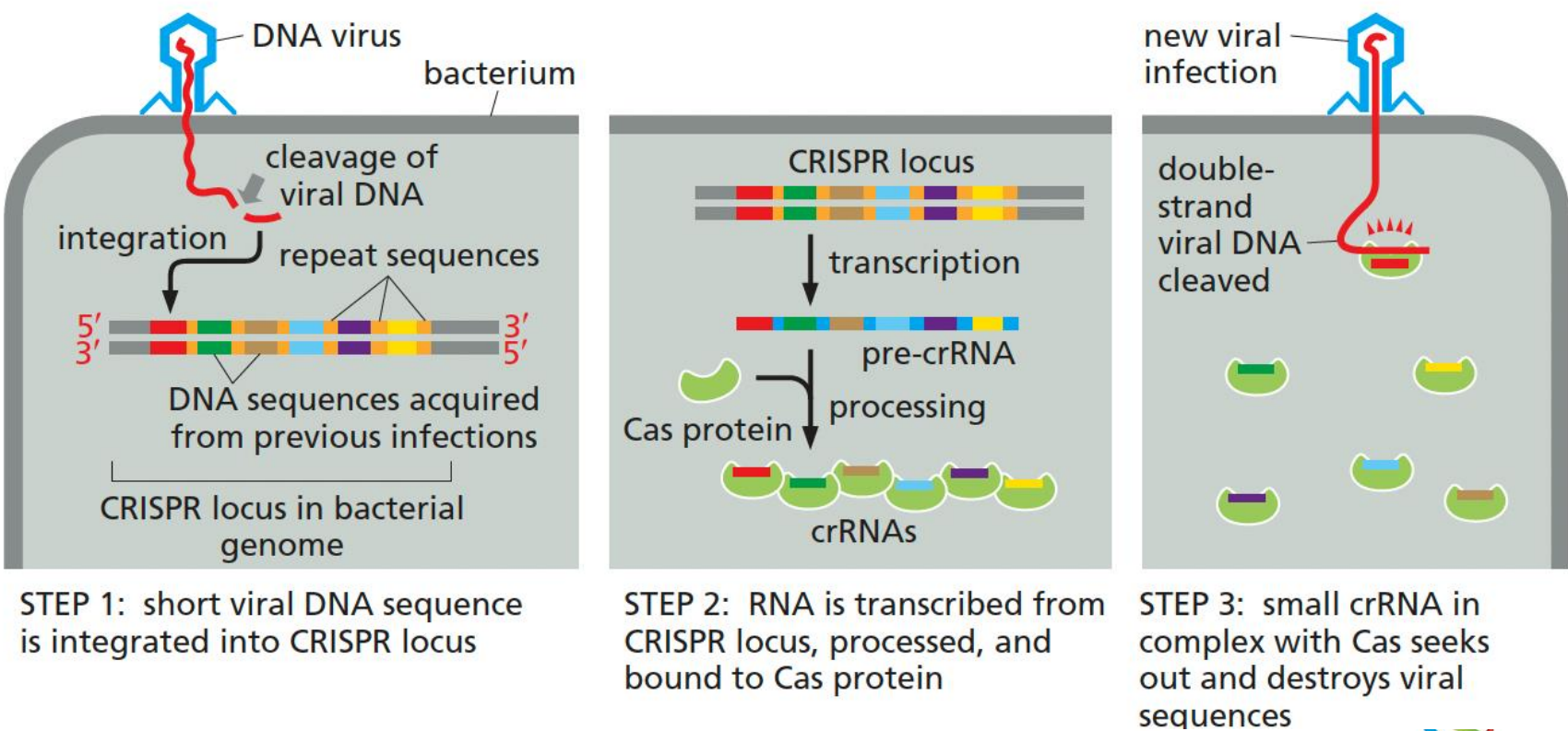
RNA Interference Directed By siRNAs



- RNA interference is used as a cell defense mechanism.
- In some organisms, the RNAi activity can be spread by transferring the siRNAs from cell to cell.
- RNAi has become a powerful experimental tool.

CRISPR-mediated Immunity In Bacteria And Archaeobacteria

CRISPR (clustered regularly interspersed short palindromic repeat)



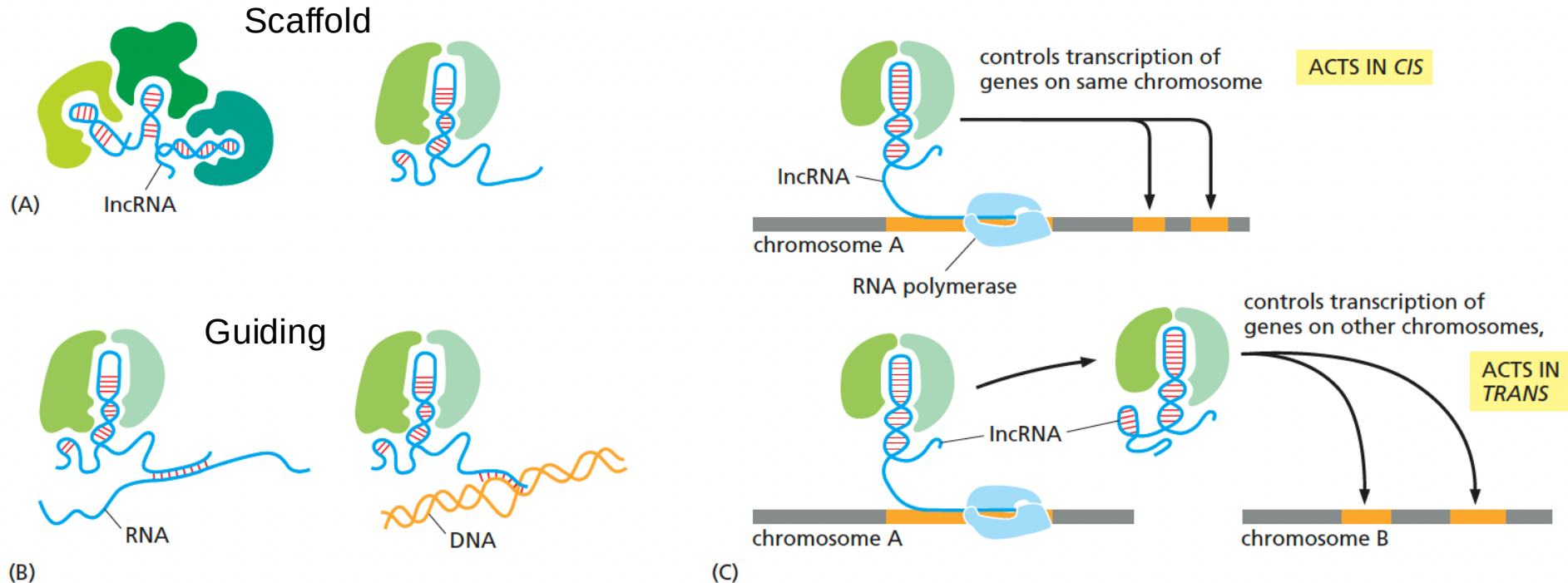
Two key differences comparing to RNAi:

- Bacteria and archaea have a mechanism to integrate the viral DNA into their genomes——“vaccination”.
- The CRISPR system destroys double-stranded DNA molecules, rather than single-stranded RNA molecules.



Long Non-coding RNAs Have Diverse Functions in the Cell

- RNAs longer than 200 nucleotides (arbitrarily) that do not code for protein.
- Most are background “noise” of transcription and RNA processing.



RNA molecules are well suited to act as scaffolds and guiding.